Discovering Physics (In Progress...)

Part I: The Calculus

"What I cannot create, I do not understand." — Richard Feynman

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October 15, 2026

Preface

This book is not a transcription of a calculus class. It is an attempt to hand you the same problem that sat in front of Newton, Leibniz, and every physicist since: *how do we describe change, precisely, when the thing that is changing never sits still long enough to be measured?* Everything here – limits, derivatives, integrals – exists because someone needed it to answer that question about *motion*. We will rebuild it in roughly the order it was needed, motivating each new idea with a physical question before we ever write down a formal definition.

A few notes on how this book is put together, since the format is unusual for a first course.

- **Boxes.** Every **Theorem**, **Proposition**, **Lemma**, **Corollary**, **Definition**, and **Example** is set off in its own numbered box. The numbering runs *continuously through each chapter*, regardless of type – so Definition 1.4 might be followed by Example 1.5 and Theorem 1.6. This is deliberate: it tells you at a glance how the ideas of a chapter are meant to flow, rather than sorting them into separate piles after the fact.
- **Gray boxes** pose a question and ask you to think before reading on. Please actually stop and think – even a wrong guess sharpens the question. Inside an Example, the work is introduced by a green **Solution** heading.
- **Side notes**, set in italic commentary boxes, are commentary rather than content: they say why a step is done the way it is, or warn about the place people usually go wrong. Some also spell out assumptions or limitations, so read them when you first meet an argument.
- **Proofs** are included whenever they teach you something about *why* a fact is true, not just *that* it is true. Where a fully rigorous proof would be long and not very illuminating at this stage, we give a sketch and say so honestly – rigor is a tool for removing doubt, not a hoop to jump through.

One ground rule, since this book claims to rebuild calculus rather than recite it. The rule is about *logical* order, not about dates. The core ideas begin with problems about motion, measurement and geometry. Later chapters also take physical models as starting points: we state a force law or circuit equation and ask what mathematics will solve it. Those models are supplied assumptions, not results proved here. The order is a reconstruction for learning, not a literal chronology; we will distinguish historical evidence, assumptions and deductions.

Two consequences of that rule are worth stating plainly, because a reader who takes the rule seriously will notice both. First, one piece of machinery here is openly anachronistic: the ε–δ language of Chapter 1 belongs to Cauchy and Weierstrass, two centuries after Newton. We use it anyway, and say so where it appears, because the seventeenth century never did settle the argument it was invented to settle. Second, once a tool *has* been built, we point it at whatever is worth pointing it at, and several of those things are modern — radioactive decay, molecular potentials, relativistic energy, electric circuits, a spacecraft's trajectory. Those are applications downstream of the mathematics rather than foundations underneath it, so using them costs the argument nothing. The rule forbids borrowing a result before we have earned it; it does not forbid enjoying one afterwards.

Read with a pencil. Try the gray boxes before you read past them. And welcome to the beginning of physics.

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Part I

Calculus

Chapter 1

Limits

Fire an arrow across a courtyard and photograph it with an impossibly fast camera. In the photograph the arrow is frozen: it occupies one definite position and covers no distance at all. Zeno of Elea drew the obvious conclusion twenty-four centuries ago. At every instant the arrow is motionless; motionless instants, however many you stack up, cannot add up to motion; therefore motion is impossible.

Something is clearly wrong with that argument. What exactly is wrong is not clear at all, and it took two thousand years to say it precisely. The repair is the idea of a *limit*, and it is the first real tool we need. So this chapter is not a formality to survive on the way to physics. It is the chapter where we learn how to say something about what happens *at* an instant using only information from the instants *around* it — which is, in the end, the only kind of information about motion that anyone has ever had.

1.1 The Problem of Instantaneous Speed

Drop a ball from rest. How far has it fallen after t seconds?

We are not going to answer that with theory, because Galileo answered it with a ramp, a ball, and a water clock. Free fall is far too quick to time by hand, so he slowed it down: he rolled balls down a gently inclined plane and measured how far they travelled in successive equal intervals of time. The distances came out in the ratio $1 : 3 : 5 : 7 : \dots$ — the odd numbers. Since $1 + 3 + 5 + \cdots + (2n - 1) = n^2$, the total distance after n intervals goes like n^2: distance grows as the *square* of the time. Converted to modern units, for a body falling freely from rest,

$$s(t) = 4.9\,t^2 \qquad \text{metres.}$$

> *Galileo's law is a* measurement*, made two generations before calculus existed. What is his is the* shape *of the law — distance grows as the square of the time — and we are taking that experimental fact and asking a question about it that the mathematics of his day could not answer.*

Two details of the history are worth having straight, since this book cares about not borrowing results it has not earned. Galileo stated the square-of-the-time law in a letter of 1604 and published his inclined-plane work in 1638; the exact route he took between the two is still argued over by historians. And the number 4.9 is our modern coefficient for free fall near the Earth's surface, not something he read off a ramp — a ramp gives a smaller constant, depending on its slope. Everything in this chapter grows out of that one measured law and a single question about it. Galileo could say how far the ball goes in any stretch of time. What nobody could yet say — not with any precision about what the words meant — was how fast it is going at a particular instant.

Start with a question we can already answer. How fast was the ball moving *between* $t = 1$ and $t = 2$? In that one second it fell $s(2) - s(1) = 19.6 - 4.9 = 14.7$ meters, so on average it was moving at 14.7 m/s. Nothing subtle happened there: we divided a distance by a time, exactly as you would to find the average speed of a car on a highway.

Before going any further we should give those two quantities a name, because everything in this book is built out of them. Notice that we did not use the ball's position, and we did not use the reading on the clock. We used the *change* in each.

1.1 Definition *the symbol* Δ

For a quantity q that moves from one value to another,

$$\Delta q \;=\; (\text{new value of } q) \;-\; (\text{old value of } q)$$

denotes the **change** in q. It is read "delta q".

Δq is one symbol, not a product. The Δ is not a number, so it cannot be cancelled, divided out, or moved across an equals sign on its own — writing $\Delta s/\Delta t = s/t$ is exactly as wrong as cancelling the letters in "cat/cot". Read Δ as a prefix meaning "the change in"; like the "un" in "unhappy", it means nothing at all standing by itself.

Two remarks about this symbol, the first of which trips up almost everybody at least once. A change is a difference of two numbers, so Δq carries a sign: if the ball is rising, Δs is negative, and that is information we want to keep, not an error. And Δq is always attached to a specific pair of moments — "the change in s" is a meaningless phrase until you say from when to when.

Why introduce a symbol for something so simple? Because the question physics asks about everything has exactly one shape: *if I change this, what happens to that, and by how much?* Push a spring by Δx and the force changes by ΔF. Let a time Δt pass and the position changes by Δs. The pairing of an input change with its output change is the fundamental object of the whole subject, and the ratio of the two — how much output you get per unit of input — is what we usually mean by a rate.

1.2 Definition *average velocity*

Let an object's position at time t be $s(t)$. If the time changes by Δt starting from $t = a$, the position changes by

$$\Delta s \;=\; s(a + \Delta t) - s(a),$$

and the object's **average velocity** over that interval is the ratio

$$\bar{v} \;=\; \frac{\Delta s}{\Delta t} \;=\; \frac{s(a + \Delta t) - s(a)}{\Delta t}.$$

Question. How fast was the ball falling at the *exact instant* $t = 1$ second — not on average across some stretch of time, but *at* that moment, the number a radar gun would report? Before reading on, try to answer it with Definition 1.2. What value of Δt describes a single instant, and what goes wrong when you use it?

An instant has no duration, so the honest choice is $\Delta t = 0$ — and that choice destroys the formula:

$$\frac{\Delta s}{\Delta t} = \frac{s(1) - s(1)}{0} = \frac{0}{0},$$

which is not a number. This is Zeno's arrow in algebraic dress. Over any stretch of time, however short, the ball moves a definite distance and we can divide. At a single instant it moves nothing in no time, and the division jams.

And yet the ball plainly *has* a speed at $t = 1$; a radar gun would report one. So the fault is not in the physics but in the question we are asking of the formula. Instead of demanding its value *at* $\Delta t = 0$, let us watch what it does as Δt gets small, and see whether it is heading somewhere definite.

For any $\Delta t \neq 0$,

$$\frac{\Delta s}{\Delta t} = \frac{4.9\,(1+\Delta t)^2 - 4.9}{\Delta t} = \frac{4.9\big(1 + 2\,\Delta t + (\Delta t)^2\big) - 4.9}{\Delta t} = \frac{4.9\big(2\,\Delta t + (\Delta t)^2\big)}{\Delta t} = 9.8 + 4.9\,\Delta t.$$

Notice where the trouble went. We never divided by zero: the Δt in the denominator was cancelled while Δt was still a perfectly good nonzero number, and only afterwards did we ask what happens as it shrinks. Every honest limit computation in this chapter is a version of that same two-step dance — simplify first, shrink second.

The expression $9.8 + 4.9\,\Delta t$ is ordinary algebra with no division left in it, and it makes sense for every Δt, including $\Delta t = 0$. That is the whole trick. The original quotient was undefined at $\Delta t = 0$; this equivalent expression is not, and the two agree at every Δt that both of them can handle.

It is worth pausing on *which* term dies. The numerator had two pieces, one carrying Δt and one carrying $(\Delta t)^2$. After dividing, the first became the constant 9.8 and the second became $4.9\,\Delta t$, which vanishes. The square of a small number is *very* small — a hundredth squared is a ten-thousandth — and that single observation, which looks like a triviality here, is the seed of the entire theory of derivatives in Chapter 2. Terms of size Δt survive division by Δt; terms of size $(\Delta t)^2$ do not.

Watch the numbers close in:

Δt (s)	interval	$\bar{v} = \Delta s/\Delta t = 9.8 + 4.9\,\Delta t$ (m/s)
1	[1, 2]	14.7
0.5	[1, 1.5]	12.25
0.1	[1, 1.1]	10.29
0.01	[1, 1.01]	9.849
0.001	[1, 1.001]	9.8049
0.0001	[1, 1.0001]	9.80049

The crucial thing to see is that 9.8 never appears in the table. No interval, however short, produces it. It is not a value the average velocity ever takes; it is the number all those values are aiming at. That distinction — between the values a quantity takes and the value it is approaching — is the subject of this chapter.

The geometry says the same thing more vividly. The ratio $\Delta s/\Delta t$ is a rise over a run: it is the slope of the straight line through the two points on the graph of s at the ends of the interval, a line called a **secant**. Shrinking Δt slides the second point down the curve toward the first, and the secants pivot as it goes — toward one particular line, the one that grazes the curve at $t = 1$ instead of cutting across it. That line is the **tangent**, and its slope is 9.8.

So we would like to write

$$\lim_{\Delta t \to 0} \frac{\Delta s}{\Delta t} = \lim_{\Delta t \to 0} \big(9.8 + 4.9\,\Delta t\big) = 9.8$$

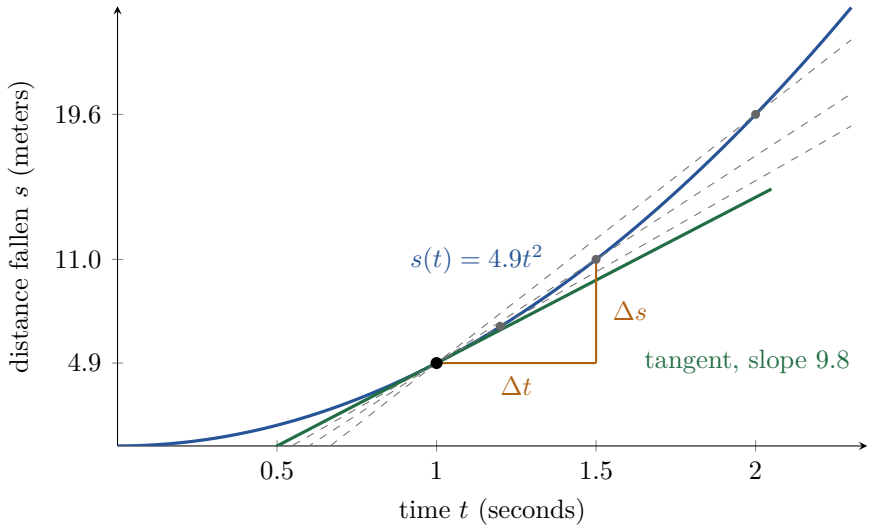


Figure 1.1: Each dashed secant line joins (1, 4.9) to a second point on the curve, and its slope is the ratio $\Delta s/\Delta t$ of the two orange sides — here drawn for $\Delta t = 0.5$, giving slope $6.125/0.5 = 12.25$. The three secants shown use $\Delta t = 1$, 0.5, and 0.2; the first two are rows of the table, and the third is an extra interval drawn because $\Delta t = 0.1$ would be too thin to see. As Δt shrinks, the secants pivot toward the solid green tangent line, whose slope is 9.8. The tangent is never a secant, just as 9.8 is never an entry in the table.

and call that number the instantaneous velocity at $t = 1$. The rest of this chapter earns the right to write it. In Chapter 2 this same limit — the limit of a ratio of changes, as the change in the input shrinks to nothing — will get a name of its own, the *derivative*, and a symbol, ds/dt, designed to remind you that it came from $\Delta s/\Delta t$.

1.2 What "Approaching" Really Means

Forget falling balls for a moment. We have a function f, a number a, and a recurring situation: f misbehaves, or is undefined, or is simply unknown *at* a, while behaving perfectly sensibly at every input *near* a. We want a way to extract the sensible information.

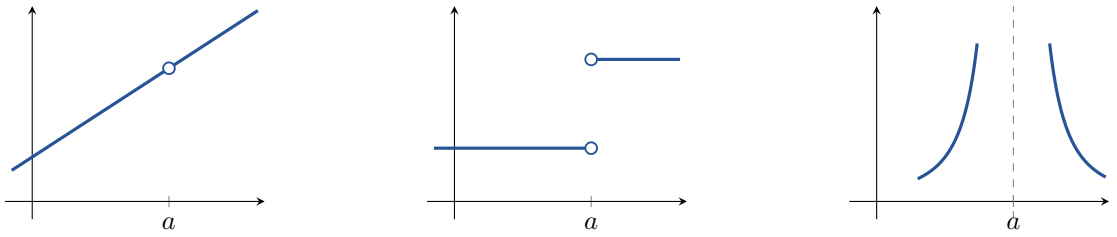
Here is the mental picture to carry through the whole chapter. Put your thumb over the graph at $x = a$, hiding that single point completely. Everything you can still see — the curve coming in from the left, the curve coming in from the right — is what the limit is allowed to depend on. The limit asks one question: *if you had to paint in the hidden point so that the graph looked seamless, what height would you paint it at?* Sometimes there is exactly one sensible answer, and that answer is the limit. Sometimes there is none.

1.3 Example *a limit that exists even though the function does not*

Let $f(x) = \dfrac{x^2 - 4}{x - 2}$.

Solution At $x = 2$ this formula reads $0/0$, so $f(2)$ does not exist — there is a hole in the graph at $x = 2$, exactly as in Figure 1.2(a). But for every $x \neq 2$ we may cancel:

$$f(x) = \frac{(x-2)(x+2)}{x-2} = x + 2.$$



(a) a hole (b) a jump (c) a blow-up

Figure 1.2: Three functions, three fates at $x = a$. In (a) the point is missing but both sides agree on where it should go, so there is exactly one sensible height to paint in. In (b) the two sides disagree, and no single height makes the graph seamless. In (c) the graph runs off to infinity on both sides and there is no finite height at all. Only in (a) does the limit exist.

So away from the single missing point, f *is* the line $y = x + 2$. Cover $x = 2$ with your thumb and the answer is obvious: both sides are marching toward height 4. Painting the hole at height 4 would make the graph seamless, and no other height would. We write $\lim_{x \to 2} f(x) = 4$, even though $f(2)$ is undefined.

That example is worth dwelling on, because it contains the whole philosophy: *a limit never asks what the function does at the point.* It asks only what the function's behavior near the point is pointing toward. This is not a technicality we tolerate; it is precisely the feature we needed in Section 1.1, where the quotient $\Delta s/\Delta t$ was undefined at $\Delta t = 0$ and useful everywhere else.

1.4 Example *a limit that fails to exist*

Let $g(x) = \dfrac{|x|}{x}$ for $x \neq 0$.

Solution For every $x > 0$ we have $|x| = x$, so $g(x) = 1$. For every $x < 0$ we have $|x| = -x$, so $g(x) = -1$. Its graph has the same jump shape as Figure 1.2(b), but here the flat line is at height 1 to the right of the origin and at height -1 to the left. Approaching 0 from the right, g is pinned at 1; approaching from the left, it is pinned at -1. No single height paints the hole seamlessly, so $\lim_{x \to 0} g(x)$ does not exist.

These examples suggest a working definition:

> $\lim_{x \to a} f(x) = L$ should mean that we can force $f(x)$ to be as close to L as we like, by forcing x to be close enough to a — while never setting $x = a$.

That sentence is a good mental picture and it will carry you a long way. But look hard at the words doing the work: *as close as we like* and *close enough*. Neither is a number, neither can be checked, and a definition assembled out of adjustable adjectives is a definition that cannot settle an argument. Newton's contemporaries noticed. Bishop Berkeley, in 1734, ridiculed the vanishing quantities of the new calculus as "ghosts of departed quantities" — things treated as nonzero when it was convenient to divide by them and as zero when it was convenient to discard them. He was not being unfair. It took another century to answer him, and the answer is the next section.

1.3 Making "Approaches" Precise

Suppose someone claims $\lim_{x\to 2}(3x+1) = 7$, and a skeptic refuses to accept it. The skeptic says: "Show me that $3x+1$ is within 0.001 of 7 whenever x is close enough to 2. Now show me within 0.000001. Now within any tolerance I care to name."
You cannot check infinitely many tolerances one at a time. You need a single argument covering all of them at once. So: given a tolerance, how close must x be to 2 to guarantee $3x+1$ lands inside it? Find the pattern before reading on.

Here is a way to think about it that engineers meet every day. You are machining a rod on a lathe, and the specification is on the *output*: the finished diameter must be within some tolerance of the target. What you actually control is the *input*: the setting on the machine. The practical question is never "is the setting exactly right" — no setting is ever exactly right — but "how tightly must I control the setting to guarantee the output meets spec?" A limit statement is exactly that guarantee, promised for every possible spec.

Give the output tolerance a name, ε (epsilon), any positive number the skeptic likes. Our job is to produce an input tolerance, δ (delta), that does the job. For $f(x) = 3x+1$ near $a = 2$,

$$|f(x) - 7| = |3x + 1 - 7| = |3x - 6| = 3\,|x - 2|,$$

so $|f(x) - 7| < \varepsilon$ holds exactly when $|x - 2| < \varepsilon/3$. There is the pattern: whatever ε the skeptic names, answer $\delta = \varepsilon/3$. The claim is now checkable, and the checking never ends in a shrug.

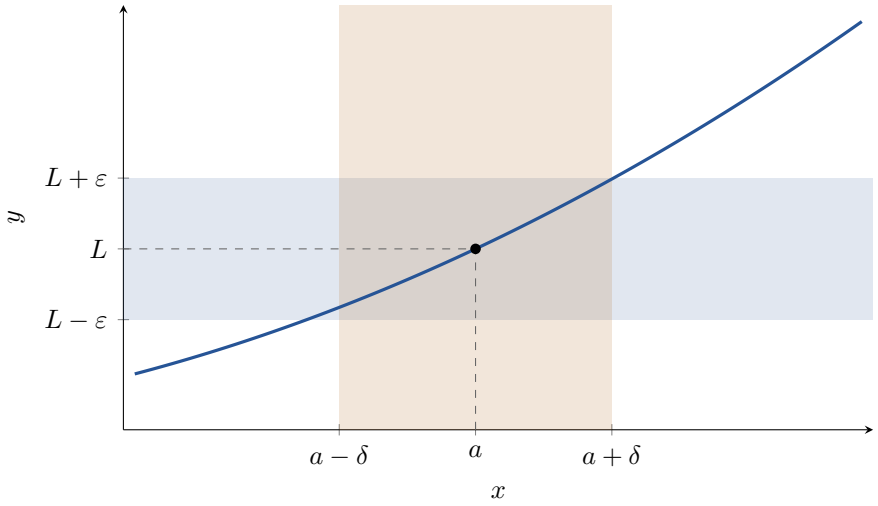


Figure 1.3: The ε–δ bargain. The skeptic chooses the horizontal band of half-width ε around L; we answer with the vertical strip of half-width δ around a. We win this round because every x inside the vertical strip has its output $f(x)$ inside the horizontal band — the curve does not escape the band until it has already left the strip. A smaller ε would flatten the band and force us to answer with a narrower strip, which is exactly what we must always be able to do.

1.5 Definition *the limit of a function*

Let f be defined near a, though not necessarily at a itself. We say

$$\lim_{x \to a} f(x) = L$$

if for every number $\varepsilon > 0$ there exists a number $\delta > 0$ such that

$$0 < |x - a| < \delta \quad \Longrightarrow \quad |f(x) - L| < \varepsilon.$$

Read the quantifiers as the order of play in a game. The skeptic moves first, naming ε; we move second, naming δ. So δ is allowed to depend on ε, and almost always must. What we may never do is wait to see which particular x will be tested — our δ has to work for all of them at once.

Unpacking that statement against Figure 1.3 is worth the five minutes it takes. The inequality $|f(x) - L| < \varepsilon$ says “the output lies in the horizontal band”; the inequality $|x - a| < \delta$ says “the input lies in the vertical strip”; and the implication says the strip is narrow enough that nothing inside it escapes the band. The extra condition $0 < |x - a|$ excludes $x = a$ itself, which is not fussiness but the entire point: it is what let us evaluate the limit in Example 1.3, where the function had no value at a at all, and what will let us differentiate in Chapter 2.

1.6 Example *verifying a limit straight from the definition*

Confirm that $\lim_{x \to 2} (3x + 1) = 7$.

Solution Let $\varepsilon > 0$ be given — and note that we are given it, we do not choose it. Choose $\delta = \varepsilon/3$. Then whenever $0 < |x - 2| < \delta$,

$$|(3x + 1) - 7| = 3|x - 2| < 3\delta = \varepsilon.$$

Since we produced a working δ for an arbitrary ε, the definition is satisfied and the limit is proved. The skeptic has nothing left to ask.

1.7 Example *a limit where δ takes more care*

Confirm that $\lim_{x \to 3} x^2 = 9$.

Solution We need $|x^2 - 9| < \varepsilon$, and $|x^2 - 9| = |x - 3|\,|x + 3|$. The factor $|x - 3|$ is the one we control. The factor $|x + 3|$ is a nuisance: it is not a constant, so we cannot simply divide by it.

The fix is to notice that we only ever care about x near 3, so we may *decide in advance* to keep x within 1 of 3, that is, to insist $\delta \le 1$. Under that restriction $2 < x < 4$, hence $|x + 3| < 7$, and

$$|x^2 - 9| = |x - 3|\,|x + 3| < 7\,|x - 3|.$$

So it is enough to have $7|x - 3| < \varepsilon$, i.e. $|x - 3| < \varepsilon/7$. Both requirements hold at once if we

choose

$$\delta = \min\Big(1,\ \tfrac{\varepsilon}{7}\Big),$$

and then $0 < |x-3| < \delta$ forces $|x^2 - 9| < \varepsilon$, as required.

The move in Example 1.7 — confine the input first, use that confinement to bound the nuisance factor, then take the smaller of the two requirements — is the standard technique for any polynomial. We are not going to repeat it by hand for the rest of the book. Instead we prove, once, which combinations of limits are safe, and then compute with algebra.

1.4 Limit Laws

1.8 Theorem *algebra of limits*

Suppose $\lim_{x\to a} f(x) = L$ and $\lim_{x\to a} g(x) = M$, and let c be a constant. Then

(a) $\lim_{x\to a}\big[f(x) + g(x)\big] = L + M$;

(b) $\lim_{x\to a}\big[c\,f(x)\big] = cL$;

(c) $\lim_{x\to a}\big[f(x)\,g(x)\big] = LM$;

(d) $\lim_{x\to a} \dfrac{f(x)}{g(x)} = \dfrac{L}{M}$, provided $M \neq 0$.

Proof sketch We do (a); the rest follow the same pattern with more bookkeeping. The idea in one sentence: *if each of two quantities is off by less than half your error budget, their sum is off by less than the whole budget.*

Formally, let $\varepsilon > 0$. Since $f(x) \to L$, we can force $|f(x) - L| < \varepsilon/2$ by taking x within some δ_1 of a; since $g(x) \to M$, we can force $|g(x) - M| < \varepsilon/2$ by taking x within some δ_2 of a. Both demands hold at once if we take $\delta = \min(\delta_1, \delta_2)$, and then for $0 < |x-a| < \delta$ the triangle inequality gives

$$\big|(f(x)+g(x)) - (L+M)\big| \le |f(x)-L| + |g(x)-M| < \tfrac{\varepsilon}{2} + \tfrac{\varepsilon}{2} = \varepsilon. \qquad \blacksquare$$

Theorem 1.8 is why we will hardly ever touch ε and δ again. Two limits are immediate from Definition 1.5: $\lim_{x\to a} x = a$ (take $\delta = \varepsilon$) and $\lim_{x\to a} c = c$ for a constant c (any δ works, since $|c - c| = 0$). Every polynomial is built from those two by adding and multiplying, so the limit of any polynomial can be computed by substitution — and every rational function too, as long as the denominator does not vanish.

1.9 Example *limits by substitution*

$$\lim_{x\to 1}\big(2x^3 - 5x + 4\big) = 2(1)^3 - 5(1) + 4 = 1,$$

by repeated use of Theorem 1.8(a)–(c) on the two basic limits above. No ε, no δ, no table of values.

1.10 Example *when substitution fails, factor first*

Compute $\lim_{x\to 1} \dfrac{x^2-1}{x-1}$.

Solution Substituting $x = 1$ gives $0/0$, so Theorem 1.8(d) does not apply — the denominator's limit is 0. But the limit does not care what happens *at* $x = 1$, and for every $x \neq 1$,

$$\frac{x^2-1}{x-1} = \frac{(x-1)(x+1)}{x-1} = x+1.$$

Therefore

$$\lim_{x\to 1} \frac{x^2-1}{x-1} = \lim_{x\to 1}(x+1) = 2.$$

This is the same manoeuvre that rescued the falling ball in Section 1.1, and it is no accident: that difference quotient was a $0/0$ expression too.

1.5 One-Sided Limits and Continuity

Example 1.4 failed because the two sides disagreed. That is common enough — and physically meaningful enough — to deserve its own vocabulary.

1.11 Definition *one-sided limits*

$\lim_{x\to a^-} f(x) = L$ means that $f(x) \to L$ as x approaches a through values *less* than a; likewise $\lim_{x\to a^+} f(x) = L$ for values *greater* than a. The two-sided limit $\lim_{x\to a} f(x)$ exists and equals L if and only if both one-sided limits exist and both equal L.

Most functions we meet in physics do something stronger than merely having a limit: their limit at a point agrees with their actual value there. The graph has no hole to paint in, because the point is already sitting where the two sides say it should. That property has a name, and an enormous amount of calculus depends on it.

1.12 Definition *continuity*

f is **continuous** at a if

$$\lim_{x\to a} f(x) = f(a).$$

f is continuous on an interval if it is continuous at every point of that interval, with one adjustment at the ends: at an endpoint we ask only about the side of the interval that exists, using the right-hand limit at a left endpoint and the left-hand limit at a right endpoint.

Read that equation carefully: it quietly makes three demands at once. The limit must exist, the value $f(a)$ must exist, and the two must be equal. Figure 1.2 is a catalogue of the ways to fail. Panel (a) has a perfectly good limit but no value — a removable failure, fixed by painting in one point. Panel (b) has both one-sided limits but they disagree, so no repair is possible. Panel (c) has nothing finite to work with at all. Informally, continuity is the licence to trace the graph without lifting your pencil.

Continuity is not optional for the position of a physical object: a ball cannot be at one place and, an instant later, somewhere else without having been in between, so $s(t)$ is always continuous. Velocity is a different matter. A ball bouncing off a wall reverses direction in what we usually idealize as no time at all, and its velocity graph looks like Figure 1.2(b) — a genuine jump. Idealizations like that are exactly where continuity fails, and knowing that will matter when we study collisions.

1.13 Theorem *intermediate value theorem*

If f is continuous on $[a, b]$ and N is any number between $f(a)$ and $f(b)$, then there exists at least one c in $[a, b]$ with $f(c) = N$.

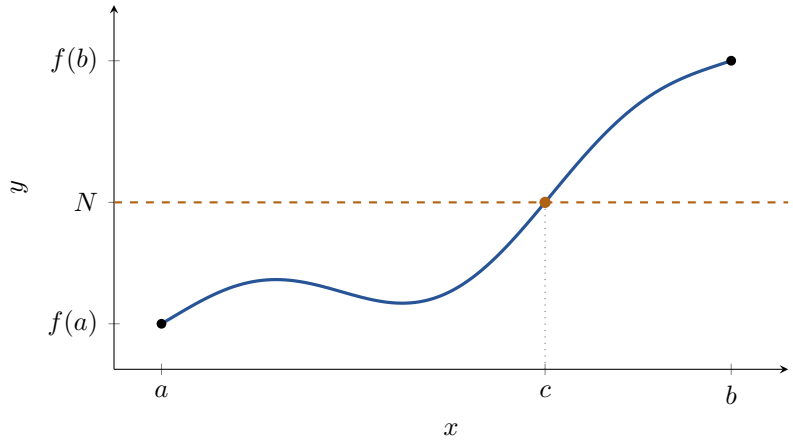


Figure 1.4: The intermediate value theorem. The curve starts below the dashed level N and ends above it, and being continuous it has no way to get from one side to the other without touching it. The theorem promises at least one such crossing c — and says nothing about where it is, or how many there are.

Proof sketch Physically the statement is almost too obvious to argue with. If a car's position is continuous in time, and it is at mile marker $f(a)$ at time a and at mile marker $f(b)$ at time b, then it cannot have skipped mile marker N in between; the road has no gaps to jump across. Making that rigorous requires knowing that the real numbers themselves have no gaps – the completeness property – which is a foundational matter we set aside for this course. The picture in Figure 1.4 is the honest content of the theorem. ■

This unassuming result does real work. It is what guarantees that a continuously varying quantity which starts negative and ends positive *must* pass through zero somewhere — which is how we will know, later on, that equilibrium positions and turning points actually exist rather than merely seeming to.

1.6 Limits at Infinity, and Limits That Are Infinite

Two more situations use the word "limit", and they are easy to confuse with each other because both involve the symbol ∞. They are different questions, and we will need both.

The first asks about the long run. Drop a coffee filter instead of a cannonball and air resistance matters: the filter speeds up, but the faster it goes the harder the air pushes back, and it settles

into a steady fall. There is a number its speed is approaching — its **terminal speed** — but it never quite reaches it and there is no particular instant at which it arrives. That is a limit, taken not as the input closes in on some finite a but as the input runs away forever.

1.14 Definition *limit at infinity*

$\lim_{x\to\infty} f(x) = L$ means: for every $\varepsilon > 0$ there is a number M such that

$$x > M \quad \Longrightarrow \quad |f(x) - L| < \varepsilon.$$

The definition of $\lim_{x\to-\infty} f(x) = L$ is the same with $x < M$.

Compare this with Definition 1.5 and notice that only one phrase changed. "Close enough to a" became "far enough out"; the skeptic still names ε first and we still have to answer. The ∞ is not a place the input arrives at, and it is not a number — it is shorthand for "eventually, and from then on".

The second situation is not a limit at all, strictly speaking, and the notation is a small abuse that everyone commits. Look again at Figure 1.2(c): as $x \to a$ the function $1/(x-a)^2$ does not approach any number. But it does not merely fail to settle down, the way $\sin(1/x)$ fails by oscillating; it fails in an orderly way, by growing without bound. That orderly failure is worth naming.

1.15 Definition *infinite limits*

$\lim_{x\to a} f(x) = \infty$ means: for every number N there is a $\delta > 0$ such that

$$0 < |x - a| < \delta \quad \Longrightarrow \quad f(x) > N.$$

Similarly $f(x) \to -\infty$ means $f(x) < N$ can be forced for every N. One-sided versions are defined the same way.

Writing $\lim f(x) = \infty$ does *not* say the limit exists. It says the limit fails to exist, and specifies how. Keeping that straight matters later: when we ask in Chapter 6 whether an infinitely long region has a finite area, the whole question will be which of these two definitions applies.

1.16 Example *both kinds at once*

Evaluate $\lim_{x\to\infty} \dfrac{3x^2+1}{x^2-5}$ and $\lim_{x\to0^+} \dfrac{1}{x}$.

Solution For the first, divide numerator and denominator by the highest power present. For $x \neq 0$,

$$\frac{3x^2+1}{x^2-5} = \frac{3 + 1/x^2}{1 - 5/x^2}.$$

Now $1/x^2 \to 0$ as $x \to \infty$ (given $\varepsilon > 0$, take $M = 1/\sqrt{\varepsilon}$), so the limit laws of Theorem 1.8 — which hold verbatim for limits at infinity — give $3/1 = 3$. The trick is always the same:

the largest power wins, and everything else is eventually negligible beside it.
For the second, let N be any number and put $B = \max\{1, N\}$. If $0 < x < 1/B$, then $1/x > B \geq N$. So $1/x$ exceeds every bound as $x \to 0^+$, and we write $\lim_{x\to 0^+} 1/x = \infty$. Approaching from the left gives $-\infty$ instead, so the two-sided limit does not exist in any sense at all.

1.7 The Squeeze Theorem and the Small-Angle Limit

One particular limit deserves its own section, because every later chapter on oscillation, rotation, and waves depends on it. Getting it also gives us an excuse to meet a tool that is useful far beyond this one application.

1.17 Theorem *squeeze theorem*

Suppose $\ell(x) \leq f(x) \leq u(x)$ for all x near a (except possibly at a itself), and

$$\lim_{x\to a} \ell(x) = \lim_{x\to a} u(x) = L.$$

Then $\lim_{x\to a} f(x) = L$ as well.

Proof sketch If f is trapped between two functions that are both closing in on L, then f has nowhere else to go. Formally: given $\varepsilon > 0$, both ℓ and u eventually lie within ε of L, and since $\ell(x) \leq f(x) \leq u(x)$, so must f. ■

The application we need concerns $\sin\theta$ for small θ. The ratio $\sin\theta/\theta$ is $0/0$ at $\theta = 0$ — the same shape of problem as every other interesting limit so far — and everything hinges on a picture from plane geometry.

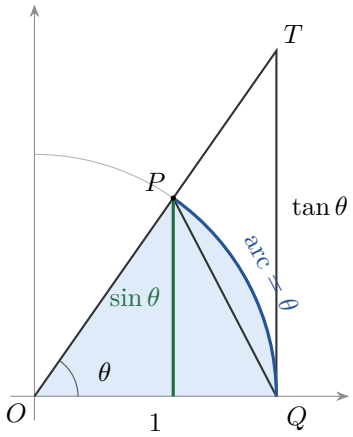


Figure 1.5: The picture behind the small-angle limit, drawn in a circle of radius 1. The green segment has length $\sin\theta$; the blue arc has length θ, which is exactly what it means to measure angles in radians; the outer vertical segment QT has length $\tan\theta$. The inequality $\sin\theta \leq \theta \leq \tan\theta$ comes from comparing the areas of three nested regions: the triangle OQP, the shaded sector, and the triangle OQT, each of which contains the one before it.

1.18 Example *the small-angle limit*

Show that $\lim_{\theta\to 0}\dfrac{\sin\theta}{\theta}=1$, with θ measured in radians.

Solution Take $0<\theta<\pi/2$ and look at the three nested regions in Figure 1.5. The triangle OQP sits inside the shaded sector, which sits inside the larger triangle OQT. Their areas are

$$\underbrace{\tfrac{1}{2}\sin\theta}_{\text{triangle } OQP} \;\le\; \underbrace{\tfrac{1}{2}\theta}_{\text{sector}} \;\le\; \underbrace{\tfrac{1}{2}\tan\theta}_{\text{triangle } OQT},$$

since triangle OQP has base OQ of length 1 and height $\sin\theta$; a sector of angle θ in a unit circle has area $\theta/2$; and triangle OQT has base 1 and height $\tan\theta$. Multiplying by 2,

$$\sin\theta\le\theta\le\tan\theta.$$

Divide through by $\sin\theta$, which is positive here:

$$1\le\frac{\theta}{\sin\theta}\le\frac{1}{\cos\theta}.$$

Taking reciprocals reverses the inequalities:

$$\cos\theta\le\frac{\sin\theta}{\theta}\le 1.$$

We still owe ourselves the claim that $\cos\theta\to 1$, and the same picture pays it. The left half of the area inequality says $0\le\sin\theta\le\theta$, so $\sin\theta\to 0$; and since $\cos\theta>0$ on this range,

$$0\;\le\;1-\cos\theta\;=\;\frac{1-\cos^2\theta}{1+\cos\theta}\;=\;\frac{\sin^2\theta}{1+\cos\theta}\;\le\;\sin^2\theta\;\le\;\theta^2,$$

which squeezes $\cos\theta$ up to 1 without using anything we have not already proved. So the ratio is trapped between two quantities both heading for 1, and Theorem 1.17 finishes the job. The function $\sin\theta/\theta$ is unchanged when θ is replaced by $-\theta$, so the limit from the left is the same, and the two-sided limit is 1.

This limit is the reason physics measures angles in radians and not degrees. In degrees the arc in Figure 1.5 has length $\pi\theta/180$ rather than θ, and the limit comes out to $\pi/180\approx 0.01745$ instead of 1 — carrying an ugly constant into every formula that follows. Radians are not a convention; they are the choice that makes this limit come out clean.

Figure 1.6 shows the squeeze happening, and the numbers show how fast it closes in:

θ (rad)	$\sin\theta$	$\sin\theta/\theta$
0.5	0.479426	0.958851
0.1	0.099833	0.998334
0.05	0.049979	0.999583
0.01	0.0099998	0.999983

At $\theta=0.1$ radians the agreement between $\sin\theta$ and θ is already three decimal places, and it improves from there. This is the origin of the **small-angle approximation** $\sin\theta\approx\theta$, which you

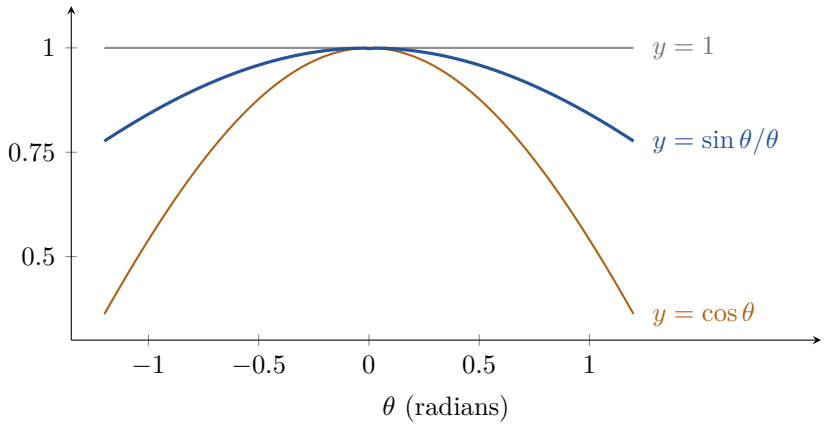


Figure 1.6: The squeeze in action. The ratio $\sin\theta/\theta$ is trapped between $\cos\theta$ below and 1 above. As $\theta \to 0$ the floor rises to meet the ceiling, leaving the ratio no room to be anything but 1.

will see used without apology the moment we study pendulums and springs in Part II. It is not a fudge and it is not guesswork: it is this limit, applied at a small but nonzero angle, with the table above as the record of how good the trade is.

1.8 Chapter Recap

We started with Zeno's arrow and a formula that jammed. Average rates of change are easy to write down, instantaneous ones apparently are not, and the gap between them is a $0/0$ that no amount of arithmetic will fill.

The way through was to stop asking for the value *at* the point and start asking what the values *near* the point are aiming at. The ε–δ statement (Definition 1.5) turned that into a promise that can be checked rather than a phrase that can be argued about: name any output tolerance, and we will name an input tolerance that guarantees it. The limit laws (Theorem 1.8) then did the real labor, so that in practice we compute limits with algebra and reserve ε and δ for the rare occasion when algebra is not enough. Continuity (Definition 1.12) named the functions whose limit and value agree — the ones a moving object's position is forced to be — and the squeeze theorem handed us $\sin\theta/\theta \to 1$, the single fact that all of oscillation will later rest on.

With all of that in place, the question that opened the chapter is finally answerable. What *is* the instantaneous velocity of the falling ball at $t = 1$? It is

$$\lim_{\Delta t \to 0} \frac{\Delta s}{\Delta t} = \lim_{\Delta t \to 0} (9.8 + 4.9\,\Delta t) = 9.8 \text{ m/s},$$

and every symbol in that sentence now means something specific.

Chapter 2 takes this one construction — the limit of $\Delta f/\Delta x$ as $\Delta x \to 0$ — strips the falling ball out of it, and finds that what remains is one of the two central objects in mathematics. It will also turn the statement around in a way that is easy to miss and impossible to forget once seen. Saying that $\Delta s/\Delta t$ approaches 9.8 is the same as saying that, for small Δt,

$$\Delta s \approx 9.8\,\Delta t,$$

with an error that shrinks faster than Δt itself. A limit statement about a ratio has become a *prediction*: tell me a small change in the input and I will tell you the resulting change in the output, by multiplying. That is what a derivative is really for, and it is why physics can get away with first-order approximations nearly everywhere.

Problem 1.1. Following Section 1.1, write $\Delta s/\Delta t$ for $s(t) = 4.9t^2$ starting from $t = 2$, simplify it until no division remains, and identify which term survives the limit and which dies. State the instantaneous velocity at $t = 2$. Does your answer fit the pattern $v = 9.8t$?

Problem 1.2. Use Definition 1.5 directly — an honest ε–δ argument — to show $\lim_{x\to 5}(2x-3) = 7$.

Problem 1.3. Compute $\displaystyle\lim_{x\to -2}\frac{x^2+5x+6}{x+2}$ by factoring first, as in Example 1.10. Then sketch the graph near $x = -2$ and say which panel of Figure 1.2 it resembles.

Problem 1.4. Let

$$f(x) = \begin{cases} x+1 & x < 0 \\ 2 & x = 0 \\ x^2+1 & x > 0. \end{cases}$$

Find $\lim_{x\to 0^-} f(x)$ and $\lim_{x\to 0^+} f(x)$. Does $\lim_{x\to 0} f(x)$ exist? Is f continuous at 0? Which of the three demands hidden in Definition 1.12 fails?

Problem 1.5. Using the squeeze theorem and the fact that $-1 \le \cos\left(\frac{1}{x}\right) \le 1$ for $x \neq 0$, show that $\displaystyle\lim_{x\to 0} x^2\cos\left(\frac{1}{x}\right) = 0$. Sketch what the graph must look like near the origin.

Problem 1.6. A ball is thrown so that its height is $s(t) = 20t - 4.9t^2$ meters. Use the intermediate value theorem to argue that the ball is at a height of exactly 10 meters at some time between $t = 0$ and $t = 1$. (You do not need to find that time.)

Chapter 2

Derivatives

Two questions were asked, by different people, for different reasons, in the century before Newton.

The geometers asked: given a curve, what is the straight line that touches it at a point? They wanted tangents for optics — light reflects off a curved mirror according to the tangent at the point of impact — and they had no general method, only a collection of clever tricks that worked one curve at a time.

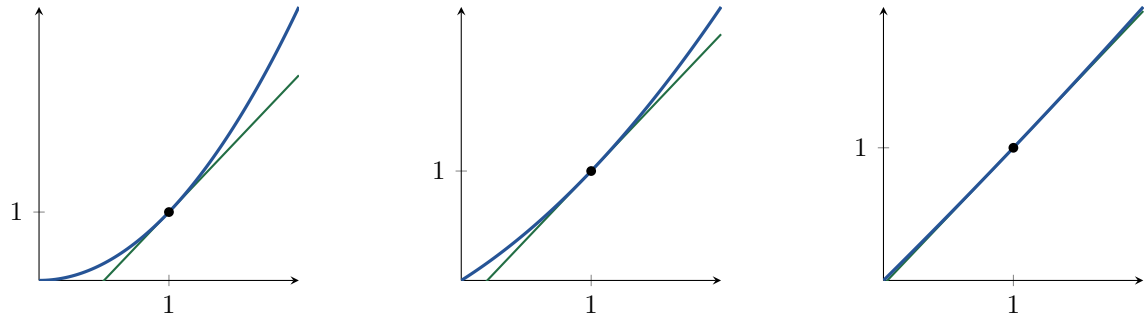
Newton asked the question of Chapter 1: given where a thing is at every instant, how fast is it moving right now?

These look like different subjects. One is about frozen geometry, the other about motion. They turn out to be the same question with different words, and the object that answers both is the *derivative*. This is the chapter where calculus actually starts, so we are going to take our time with why it looks the way it does.

2.1 Zooming In

Here is an experiment you can run on any graphing calculator, and it is worth running, because it is the single most important picture in differential calculus.

Take a smooth curve — $y = x^2$ will do — and pick a point on it, say $(1, 1)$. Now zoom in. Not a little: a lot. Shrink the window around that point and let the curve redraw itself each time.



(a) Δx up to 1 (b) Δx up to 0.4 (c) Δx up to 0.06

Figure 2.1: The parabola $y = x^2$ near the point $(1, 1)$, at three magnifications, with the same green straight line drawn in each. Far out, the curve and the line are obviously different objects. Close in, they become impossible to tell apart. The curve has not stopped being curved — we have simply stopped being able to see it.

The curve flattens. Keep zooming and it becomes visually indistinguishable from a straight line. This is not a defect of the screen; it is the defining property of the curves we care about, and it has a name: such a curve is **locally straight**.

This is why the Earth feels flat. A sphere is not flat, but you are looking at a patch of it a few kilometres wide out of forty thousand, and at that magnification the curvature has nowhere to show itself. Local straightness is the same phenomenon, and a derivative is the number that says which straight line you are standing on.

Everything in this chapter follows from taking that observation seriously. If a curve looks like a line when you get close enough to a point, then near that point the curve is described, to very good accuracy, by something we completely understand: a straight line, with a slope. The whole business of differential calculus is finding that slope, and then exploiting it.

So: which line is it? We would like to say "the line that touches the curve at the point", but that phrase does not survive contact with examples. Infinitely many lines pass through $(1, 1)$. "Touches at exactly one point" is no better — the tangent to a wave crosses the wave again further along, and the tangent to a straight line touches it everywhere. We cannot define the tangent using only the one point, because one point does not know which way the curve is going.

The way out is the one Chapter 1 spent its whole length preparing. Use two points, which *do* determine a line, and then let the second one slide into the first.

2.2 The Derivative

Fix a point a and move the input by Δx. The output moves by

$$\Delta f \;=\; f(a + \Delta x) - f(a),$$

and the ratio $\Delta f / \Delta x$ is the slope of the secant line through the two points — the average rate of change of f across that interval, exactly as $\Delta s / \Delta t$ was an average velocity in Chapter 1. Shrinking Δx slides the second point toward the first, and the secant pivots toward the tangent.

2.1 Definition *the derivative*

Let f be defined near a. The **derivative** of f at a, written $f'(a)$, is

$$f'(a) \;=\; \lim_{\Delta x \to 0} \frac{\Delta f}{\Delta x} \;=\; \lim_{\Delta x \to 0} \frac{f(a + \Delta x) - f(a)}{\Delta x},$$

provided this limit exists. When it does, f is called **differentiable** at a, and $f'(a)$ is both the slope of the tangent line to the graph of f at $(a, f(a))$ and the instantaneous rate of change of f with respect to x at a.

Notice what kind of object this is. The quotient $\Delta f / \Delta x$ is undefined when $\Delta x = 0$ — that is the $0/0$ that jammed us in Chapter 1 — so the limit is doing genuine work: it extracts a definite number from an expression that has no value at the point of interest. That is precisely the manoeuvre Definition 1.5 was built to license.

2.2 Example *the falling ball, finished*

For $s(t) = 4.9t^2$, find $s'(t)$ at an arbitrary time t.

Solution Move the time by Δt and compute the change in position:

$$\Delta s = 4.9\,(t + \Delta t)^2 - 4.9\,t^2 = 4.9(2t\,\Delta t + (\Delta t)^2).$$

Divide by Δt — legitimate, since $\Delta t \neq 0$ — and let it shrink:

$$s'(t) = \lim_{\Delta t \to 0} \frac{\Delta s}{\Delta t} = \lim_{\Delta t \to 0} \big(9.8\,t + 4.9\,\Delta t\big) = 9.8\,t.$$

So the velocity of a ball dropped from rest is $v(t) = 9.8\,t$: it grows in direct proportion to the time elapsed. At $t = 1$ this gives 9.8 m/s, the number the table in Section 1.1 was creeping toward, now obtained in one line and for every t at once.

Look at which term died. The change Δs had a piece proportional to Δt and a piece proportional to $(\Delta t)^2$; after dividing, the first survived and the second went to zero. Chapter 1 flagged this as a triviality worth remembering. Section 2.3 turns it into the main theorem.

Notation, and why there is more than one

Since $f'(a)$ exists at every a where the limit does, the derivative is itself a function, and gets written several ways:

$$f'(x), \qquad \frac{df}{dx}, \qquad \frac{dy}{dx}, \qquad \dot{s}(t).$$

The first is Lagrange's and is compact. The second and third are Leibniz's, and they are the ones to use when you want to remember where the object came from: df/dx is built to look like $\Delta f/\Delta x$ with the Δ's replaced, because that is exactly what it is — the limit of that ratio. The last is Newton's, a dot meaning "derivative with respect to time", and physics still uses it constantly: $\dot{s}$ is velocity, $\ddot{s}$ is acceleration.

Leibniz's df/dx is not, at this stage, a fraction — it is one symbol denoting a limit, and df and dx have not been given separate meanings. Hold that thought for exactly one section. In Section 2.3 we will give them separate meanings, after which df/dx really is a ratio of two honest quantities, and the way physicists shove dx around the page stops being an abuse of notation.

It is worth being slightly suspicious of the Leibniz symbol at first. We have defined df/dx as a single indivisible piece of notation standing for a limit, and yet everyone — physicists especially — writes things like $df = f'(x)\,dx$ and multiplies through by dx as if it were a number. Either that is sloppiness that happens to work, or there is a real idea underneath it. There is a real idea underneath it, and it is the most useful thing in this chapter.

2.3 What Differentiability Actually Promises

Let us take Definition 2.1 and turn it inside out.

Saying that $\Delta f/\Delta x \to f'(a)$ means that for small Δx the ratio is *close* to $f'(a)$ — close, but not equal. Give the discrepancy a name: define

$$\varepsilon(\Delta x) \;=\; \frac{\Delta f}{\Delta x} - f'(a),$$

so that $\varepsilon(\Delta x) \to 0$ as $\Delta x \to 0$; that is all the derivative's existence says. Now multiply through by Δx:

$$\Delta f \;=\; f'(a)\,\Delta x \;+\; \varepsilon(\Delta x)\,\Delta x.$$

Read that equation carefully, because it is the whole chapter. It says the change in the output splits into two pieces. The first is $f'(a)\,\Delta x$: exactly proportional to the change in the input, with the derivative as the constant of proportionality. The second is a product of two things that both go to zero — so it is not merely small, it is small *compared to* Δx *itself*. The first term is exactly proportional to Δx: halve the input change and it halves too. The second term is not proportional to anything — what we know about it is that its ratio to Δx tends to zero, so once Δx is small enough it is as small a fraction of Δx as we care to demand. If $f'(a) \neq 0$ that also makes it negligible beside the first term, which is how we shall usually picture it. If $f'(a) = 0$ the first term is simply gone and the remainder is the entire change — not a defect in the statement, but precisely the situation that Section 2.8 is about.

That last distinction is important enough to name.

2.3 Definition *little-o*

A quantity $E(\Delta x)$ is said to be $o(\Delta x)$, read "little-o of Δx", if it vanishes faster than Δx does:

$$\lim_{\Delta x \to 0} \frac{E(\Delta x)}{\Delta x} = 0.$$

So $(\Delta x)^2$ is $o(\Delta x)$, and so is $(\Delta x)^3$, and so is $\varepsilon(\Delta x)\,\Delta x$ for any ε tending to zero. But Δx itself is not $o(\Delta x)$, and neither is $5\,\Delta x$: those shrink at exactly the same rate as Δx, not faster. The notation is a way of saying "negligible at this order" without having to specify the junk in question, and it is the reason physicists can write "to first order" and be making a precise statement.

2.4 Theorem *differentiability is local linearity*

f is differentiable at a with derivative $f'(a)$ if and only if the change in f can be written

$$\Delta f \;=\; f'(a)\,\Delta x \;+\; o(\Delta x).$$

That is: f is differentiable exactly when a small change in the input produces a change in the output that is proportional to it, up to an error that is negligible compared with the change itself.

Proof Both directions are the algebra above, read in the two directions. If $f'(a)$ exists, set $\varepsilon(\Delta x) = \Delta f/\Delta x - f'(a)$; then $\varepsilon \to 0$, so $\varepsilon\,\Delta x$ is $o(\Delta x)$ and the displayed equation holds. Conversely, if $\Delta f = m\,\Delta x + E(\Delta x)$ with E of size $o(\Delta x)$, divide by Δx to get

$$\frac{\Delta f}{\Delta x} = m + \frac{E(\Delta x)}{\Delta x},$$

and the last term tends to 0 by Definition 2.3, so the limit exists and equals m. ∎

This reformulation is worth more than the original definition, for three reasons.

It explains *why differentiability is the property worth caring about.* A differentiable function is one whose response to a small nudge is predictable by multiplication. You do not need to know the function; you need one number, the derivative, and then $\Delta f \approx f'(a)\Delta x$. Nothing in mathematics is easier to work with than multiplication by a constant, and differentiability is precisely the licence to replace a complicated function by that, locally.

It explains the zooming picture. Figure 2.1 showed a curve becoming a line; Theorem 2.4 says which line, and how fast the discrepancy disappears.

And it is the form the rest of physics actually uses — almost never as a limit, almost always as an approximation with a controlled error.

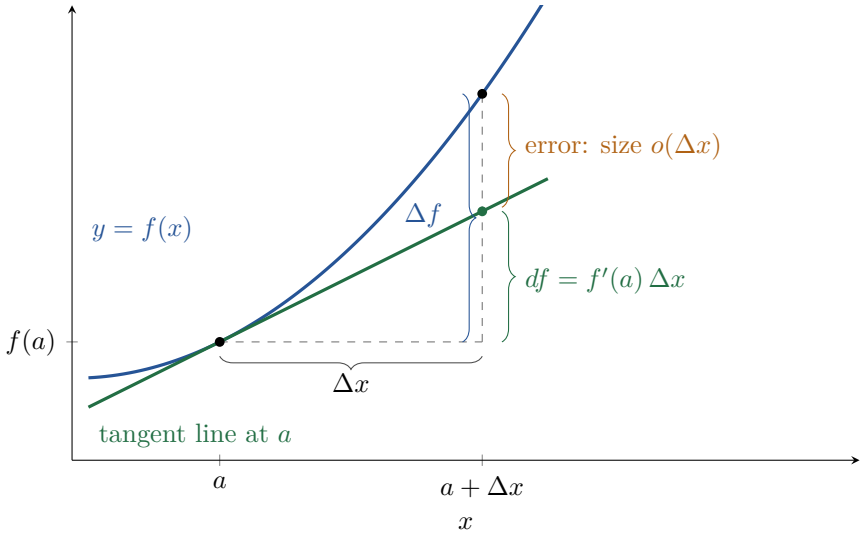


Figure 2.2: The anatomy of a small change. Moving the input from a to $a+\Delta x$ moves the output by Δf, the blue bracket. The tangent line predicts a change of $f'(a)\Delta x$, the green bracket. The orange bracket is what the prediction missed. Shrink Δx and all three shrink — but the orange one shrinks faster than the other two, which is the entire content of Theorem 2.4.

Differentials

We can now give df and dx meanings of their own, which is what Leibniz intended all along.

2.5 Definition *differentials*

Let f be differentiable at x. For a change in the input, written dx and chosen freely, the **differential** of f is

$$df \;=\; f'(x)\,dx.$$

The differential df is not the true change in f. It is the change the *tangent line* predicts: the green bracket in Figure 2.2, whereas Δf is the blue one. The two differ by the orange bracket, which is $o(\Delta x)$. So

$$\Delta f = df + o(\Delta x),$$

and if we divide Definition 2.5 by dx we get $df/dx = f'(x)$ — the Leibniz symbol, now genuinely a quotient of two quantities rather than an indivisible squiggle. This is the licence physicists have been using all along when they write $dV = 4\pi r^2\,dr$ and treat it as ordinary algebra. It is not sloppiness. It is Theorem 2.4, with the error term left politely unmentioned because it is $o(dr)$ and everybody knows it.

2.6 Example *the differential of x^2, geometrically*

Where does the error term actually live? Take $f(x) = x^2$, which is the area of a square of side x, and grow the side by Δx.

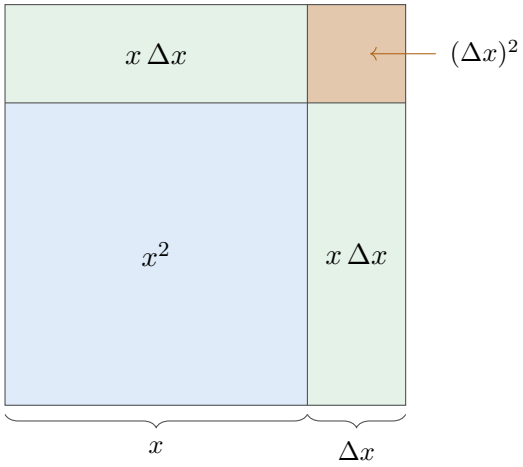


Figure 2.3: Growing a square. The new area exceeds the old by two strips of area $x\,\Delta x$ and one corner of area $(\Delta x)^2$. The strips are the differential $df = 2x\,\Delta x$; the corner is the entire error. Its area shrinks like the *square* of Δx, so as the growth gets small the corner becomes negligible compared with the strips — which is $o(\Delta x)$ made visible.

Figure 2.3 says it all: $\Delta f = 2x\,\Delta x + (\Delta x)^2$, so $df = 2x\,dx$ and $f'(x) = 2x$. The derivative of x^2 is the combined length of the two sides that sweep out the new strips, and the corner you are throwing away is the price of using a straight-line approximation on a curved function. Grow a circle of radius r by Δr instead and the same argument gives a thin ring of area $2\pi r\,\Delta r$ plus a negligible sliver: $\dfrac{d}{dr}(\pi r^2) = 2\pi r$, the circumference. Differentials turn several formulas you memorized in geometry into obvious statements about growth.

2.4 Differentiability and Continuity

With Theorem 2.4 in hand, one standard result becomes immediate.

2.7 Theorem *differentiable implies continuous*

If f is differentiable at a, then f is continuous at a.

Proof By Theorem 2.4, $\Delta f = f'(a)\Delta x + o(\Delta x)$. As $\Delta x \to 0$ both terms on the right go to 0, so $\Delta f \to 0$, which says $f(a + \Delta x) \to f(a)$ — exactly continuity at a (Definition 1.12). ■

The converse fails, and the failure is worth seeing, because it is the zooming picture run in reverse.

2.8 Example *continuous but not differentiable*

Let $f(x) = |x|$. This is continuous everywhere. Is it differentiable at 0?

Solution Compute the ratio from each side:

$$\frac{\Delta f}{\Delta x} = \frac{|0 + \Delta x| - 0}{\Delta x} = \begin{cases} +1 & \Delta x > 0, \\ -1 & \Delta x < 0. \end{cases}$$

The one-sided limits are 1 and -1, so by Definition 1.11 the limit does not exist and f is not differentiable at 0. There is no number that deserves to be called the slope there, because the two sides never agree on one.



(a) $y = |x|$ near the origin (b) the same corner, magnified 100×

Figure 2.4: Compare with Figure 2.1. Zooming in on a corner does nothing at all: it is a corner at every magnification. A function is differentiable at a point precisely when it has a finite-slope linear approximation whose error divided by Δx tends to zero. The line must not be vertical. The two figures illustrate a smooth point and a corner, one passing that test and the other failing.

This is the mathematical version of a remark from Chapter 1. A ball bouncing off a wall has a continuous position — it never teleports — but if we idealize the bounce as instantaneous, its *position* graph has a corner and its velocity jumps, exactly like Figure 1.2(b). At that instant the position has no derivative, so "the velocity" — and with it the acceleration — does not exist. Idealizations are where differentiability fails, and it is always worth knowing which of your idealizations has a corner hidden in it.

2.5 The Rules

Computing every derivative from Definition 2.1 would be unbearable. As with limits, we prove a handful of rules once and then compute by algebra. Every proof below is the same move: write down Δf, split it into a part proportional to Δx and a part that is $o(\Delta x)$, and read off the derivative from the first part.

2.9 Theorem *power, constant multiple, and sum rules*

Let f, g be differentiable, c a constant, and n a positive integer. Then

(a) $\dfrac{d}{dx}[x^n] = n\,x^{n-1}$;

(b) $\dfrac{d}{dx}[c\,f(x)] = c\,f'(x)$;

(c) $\dfrac{d}{dx}[f(x)+g(x)] = f'(x)+g'(x)$.

Proof of (a) By the binomial theorem,

$$(x+\Delta x)^n = x^n + n\,x^{n-1}\,\Delta x + \binom{n}{2}x^{n-2}(\Delta x)^2 + \cdots + (\Delta x)^n,$$

so

$$\Delta f = n\,x^{n-1}\,\Delta x + \underbrace{\binom{n}{2}x^{n-2}(\Delta x)^2 + \cdots + (\Delta x)^n}_{\text{every term carries } (\Delta x)^2 \text{ or higher}}.$$

The underbraced part is $o(\Delta x)$, so by Theorem 2.4 the derivative is the coefficient of Δx, namely $n\,x^{n-1}$. For $n=2$ this is exactly the square of Figure 2.3: two strips, one negligible corner. ■

Parts (b) and (c) come straight from the corresponding limit laws (Theorem 1.8) applied to the difference quotients of cf and $f+g$; we omit the bookkeeping.

One honest caveat about part (a). The binomial theorem needs n to be a positive integer, so that is all we have proved. The formula $\dfrac{d}{dx}[x^n] = n\,x^{n-1}$ is in fact true for *every* real exponent — negative, fractional, irrational — and we will use it in that generality, most importantly for $n=\frac{1}{2}$ and $n=-\frac{1}{2}$ in Section 2.10. Our proof for all real exponents uses the logarithm, which we have not built yet, so we flag the debt and pay it when we build that function. Together with the fact that a constant has derivative 0 — a quantity that never changes has $\Delta f = 0$ for every Δx — they differentiate any polynomial on sight.

2.10 Example *differentiating a polynomial*

$$\frac{d}{dx}[3x^4 - 5x^2 + 7x - 2] = 3(4x^3) - 5(2x) + 7 - 0 = 12x^3 - 10x + 7.$$

2.6 Products and Quotients

Let $f(x) = x$ and $g(x) = x$, so that $f(x)g(x) = x^2$. We know the derivative of x^2 is $2x$. Is that the same as $f'(x)\,g'(x) = 1\cdot 1 = 1$? It is not. So the derivative of a product is not the product of the derivatives — and before reading on, look at Figure 2.3 again and see whether you can guess what the missing terms should be.

The square picture already answers this, once we let the two sides be different.

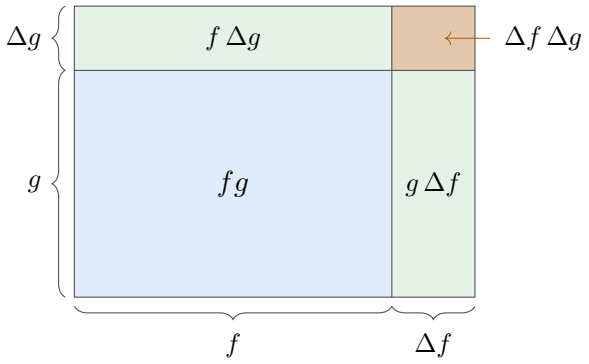


Figure 2.5: A rectangle with sides f and g has area fg. Nudge both sides and the area grows by two strips, $g\,\Delta f$ and $f\,\Delta g$, plus the corner $\Delta f\,\Delta g$. The corner is a product of two small quantities, so it is $o(\Delta x)$ and contributes nothing to the derivative — the two strips are the product rule.

2.11 Theorem *product rule*

If f and g are differentiable, then

$$\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x).$$

Proof Read the area off Figure 2.5:

$$\Delta(fg) = g\,\Delta f + f\,\Delta g + \Delta f\,\Delta g.$$

Divide by Δx:

$$\frac{\Delta(fg)}{\Delta x} = g\,\frac{\Delta f}{\Delta x} + f\,\frac{\Delta g}{\Delta x} + \Delta f\,\frac{\Delta g}{\Delta x}.$$

As $\Delta x \to 0$ the first two terms tend to gf' and fg'. In the third, $\Delta g/\Delta x \to g'$ while $\Delta f \to 0$ (by Theorem 2.7), so the product of the two tends to 0: the corner of the rectangle is negligible, as promised. ■

2.12 Theorem *quotient rule*

If f and g are differentiable and $g(x) \neq 0$, then

$$\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}.$$

Proof sketch Write $q = f/g$, so $f = qg$, and differentiate that with the product rule: $f' = q'g + qg'$. Solve for q' and substitute $q = f/g$:

$$q' = \frac{f' - (f/g)g'}{g} = \frac{f'g - fg'}{g^2}.$$

(This assumes q is differentiable, which it is; the honest proof establishes that along the way.) ■

2.13 Example *using the quotient rule*

$$\frac{d}{dx}\left[\frac{x^2+1}{x-3}\right] = \frac{(2x)(x-3)-(x^2+1)(1)}{(x-3)^2} = \frac{x^2-6x-1}{(x-3)^2}.$$

2.7 The Chain Rule

A bicycle's pedals turn a chainring, which drives a sprocket, which turns the wheel. Suppose the sprocket turns 3 times for every turn of the pedals, and the wheel turns 1 time for every turn of the sprocket. How many times does the wheel turn per turn of the pedals? You did not need calculus for that. Now say it in the language of changes: what is $\Delta(\text{wheel})/\Delta(\text{pedal})$ in terms of the two ratios you were given?

Rates compose by multiplying, and in Δ notation the reason is embarrassingly visible. If y depends on u and u depends on x, then as long as $\Delta u \neq 0$ we may write

$$\frac{\Delta y}{\Delta x} = \frac{\Delta y}{\Delta u}\cdot\frac{\Delta u}{\Delta x},$$

which is just cancellation of a common factor. Let $\Delta x \to 0$; then $\Delta u \to 0$ too (Theorem 2.7), and each ratio tends to its derivative.

2.14 Theorem *chain rule*

If g is differentiable at x and f is differentiable at $g(x)$, then $f \circ g$ is differentiable at x and

$$(f\circ g)'(x) = f'(g(x))\,g'(x).$$

In Leibniz notation, with $y = f(u)$ and $u = g(x)$,

$$\frac{dy}{dx} = \frac{dy}{du}\cdot\frac{du}{dx}.$$

The Leibniz form is the reason this rule is impossible to forget: it looks like cancelling du. With differentials in hand that is very nearly what it is — $dy = (dy/du)\,du$ and $du = (du/dx)\,dx$, so substituting one into the other gives the result. The one thing the slogan hides is the case where $\Delta u = 0$ for arbitrarily small Δx, which is why the paragraph after the theorem re-proves the rule without ever dividing by Δu.

The cancellation argument above is honest wherever $\Delta u \neq 0$, and the awkward case can be disposed of with the tool we already have. Local linearity lets us write $\Delta y = f'(u)\,\Delta u + \eta(\Delta u)\,\Delta u$, where $\eta(\Delta u) \to 0$ and where we are free to set $\eta(0) = 0$, so that the formula holds even when Δu vanishes — notice that no division by Δu has been performed anywhere in it. Substituting $\Delta u = g'(x)\,\Delta x + o(\Delta x)$ and dividing by Δx leaves $f'(u)\,g'(x)$ plus terms carrying the factor $\eta(\Delta u)$, which tends to zero while $\Delta u/\Delta x$ stays bounded. So the rule holds with no exceptional case at all. That is twice now that local linearity has done work the ratio definition could not, and it will not be the last time.

The rule also tells you what it *means*: sensitivities multiply along a chain of dependence. If the volume of a balloon is 4 times as sensitive to its radius as you would naively guess, and its radius is changing slowly with time, the volume's sensitivity to time is the product of the two. Physics is full of such chains, because physical quantities almost always depend on each other through intermediate variables.

2.15 Example *a power inside a function*

For differentiable $u(x)$ and integer n, taking $f(u) = u^n$ gives

$$\frac{d}{dx}\big[u(x)^n\big] = n\,u(x)^{n-1}\,u'(x).$$

For instance $\dfrac{d}{dx}\big[(3x^2+1)^5\big] = 5(3x^2+1)^4(6x) = 30x\,(3x^2+1)^4$.

2.16 Example *inflating a balloon*

A spherical balloon's radius grows at a constant 0.1 cm/s from an initial 2 cm, so $r(t) = 2 + 0.1t$. How fast is its volume growing at $t = 0$?

Solution With $V = \frac{4}{3}\pi r^3$ we have $\dfrac{dV}{dr} = 4\pi r^2$ — the surface area, which is exactly the differentials picture again: growing a sphere's radius by dr adds a shell of area $4\pi r^2$ and thickness dr. Since $\dfrac{dr}{dt} = 0.1$,

$$\frac{dV}{dt} = \frac{dV}{dr}\cdot\frac{dr}{dt} = 4\pi r^2(0.1).$$

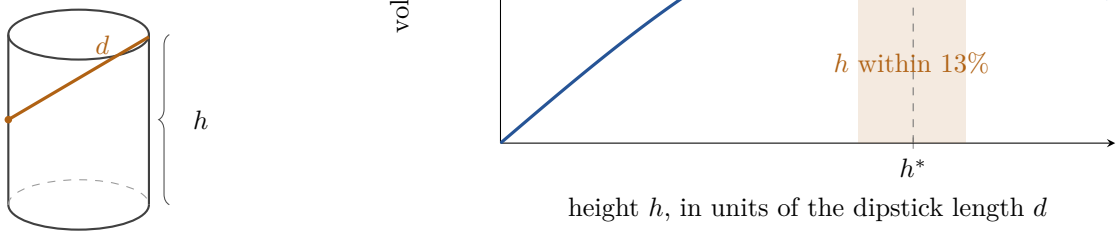
At $t = 0$, $r = 2$, so $\dfrac{dV}{dt} = 4\pi(4)(0.1) = 1.6\pi \approx 5.0$ cm^3/s.

2.8 Maxima, Minima, and Kepler's Barrels

In 1613 Johannes Kepler bought wine for his wedding, and could not leave the merchant's way of measuring it alone. What he worked out in response went into print two years later.[3]

The merchant measured barrels the way merchants did: he pushed a dipstick in through the bung hole in the middle of the side, slid it up to the far edge of the lid, pulled it out, and read off the wetted length d. One measurement, one number, one price. Kepler — who had spent years computing the areas swept out by planets — immediately objected that this is absurd. The same dipstick reading fits a tall narrow barrel and a short fat one, and those do not hold the same wine.

Model the barrel as a cylinder of height h. The dipstick runs from the middle of the side to the opposite edge of the lid, so its length d satisfies $d^2 = (2r)^2 + (h/2)^2$, where r is the radius.
For a *fixed* dipstick reading d, which shape holds the most wine? And — this is the question Kepler actually cared about — how badly is a merchant cheated if the shape is a little off?



(a) the merchant's dipstick

(b) why the maximum is forgiving

Figure 2.6: Kepler's problem. For a fixed dipstick length d, the volume depends on the barrel's proportions, and peaks at h^*. The striking feature is not where the peak is but how *flat* it is. The graph is horizontal at h^*, so getting the height wrong by 10% costs only about 1.5% of the wine, and getting it wrong by 13% costs about 2.5%. The two directions are not quite symmetric.

Kepler's answer to the first question is a calculation we can now do in two lines, and his answer to the second is the reason this section exists at all.

2.17 Example *the best barrel*

Maximise V over h for a fixed dipstick length d.

Solution From $d^2 = 4r^2 + h^2/4$ we get $r^2 = \frac{1}{4}d^2 - \frac{1}{16}h^2$, so the volume $V = \pi r^2 h$ is a function of h alone:

$$V(h) = \frac{\pi}{4}d^2 h - \frac{\pi}{16}h^3.$$

Differentiate with Theorem 2.9:

$$\frac{dV}{dh} = \frac{\pi}{4}d^2 - \frac{3\pi}{16}h^2.$$

This is zero when $h^2 = \frac{4}{3}d^2$, that is

$$h^* = \frac{2d}{\sqrt{3}} \approx 1.155\,d,$$

and since dV/dh is positive for smaller h and negative for larger, this is the maximum for our cylindrical model. Near this shape, modest changes of proportion hardly change the volume, which helps explain why one dipstick reading can be useful for similarly shaped barrels. It cannot determine volume reliably when the shapes vary widely. Real barrels bulge, so this model illustrates the idea rather than reproducing a measurement of Kepler's actual barrels.

Now the part that matters more than the answer. Put $d = 1$ and compare:

$$V(1.00) = 0.589, \qquad V(h^*) = 0.6046, \qquad V(1.30) = 0.590.$$

Changing the height by *thirteen percent* in either direction costs about two and a half percent of the wine. The maximum is flat, and this is not special to barrels.

Why should a maximum be flat? Suppose the graph were *not* level at its highest point — suppose it had some definite nonzero slope there. What could you then do to get a bigger value, and why does that contradict the point being highest?

Theorem 2.4 answers this immediately, and the argument is worth having in your bones. Near c,

$$\Delta f = f'(c)\,\Delta x + o(\Delta x).$$

If $f'(c) > 0$, then taking a small step to the right makes Δf positive, because the first term dominates the second once Δx is small enough — so $f(c)$ was not the largest value after all. If $f'(c) < 0$, step left instead. The only escape is $f'(c) = 0$.

2.18 Theorem *Fermat's rule: extrema are flat*

If f is differentiable at an interior point c of its domain and f has a local maximum or a local minimum at c, then

$$f'(c) = 0.$$

Pierre de Fermat had this rule by about 1636, thirty years before Newton. His method, which he called adequality*, was to replace x by $x + E$, set the two expressions almost-equal, cancel, divide by E, and then throw away every term still carrying an E. That is precisely the habit of this chapter — keep the part proportional to the step, discard the corner — performed a generation before anyone had a definition of a limit to justify it.*

The converse is false, and forgetting that is the classic error. $f'(c) = 0$ says only that the graph is momentarily level, which a maximum must be, and a minimum must be, and so must the flat spot of $f(x) = x^3$ at the origin — which is neither.

2.19 Definition *critical point*

A point c in the domain of f where $f'(c) = 0$, or where $f'(c)$ fails to exist, is called a **critical point** of f. Every interior maximum or minimum occurs at a critical point, but not every critical point is a maximum or a minimum.

To tell them apart, restrict attention to a *stationary* point — one where $f'(c) = 0$, rather than one where f' fails to exist, which has to be inspected on its own terms — and look at the next term. The first-order term is gone there, so if f has a second derivative the leading behaviour is

$$\Delta f \approx \tfrac{1}{2} f''(c)\,(\Delta x)^2.$$

Since $(\Delta x)^2 > 0$ on both sides, the sign of $f''(c)$ decides everything.

2.20 Theorem *second derivative test*

Suppose $f'(c) = 0$ and f'' exists and is continuous near c. If $f''(c) > 0$ then f has a local minimum at c; if $f''(c) < 0$, a local maximum. If $f''(c) = 0$ the test is silent and you must look further.

Two debts have just been run up, and it is only fair to say so. The symbol f'' is not defined until Section 2.11, a few pages on; and the quadratic expansion is Chapter 5's Taylor remainder (Theorem 5.2). Neither is circular — nothing in Chapter 5 leans on this test — but you are entitled to know when you are being asked to accept something on credit. There is also a shorter route, which we could take once Chapter 3 is in hand: $f'(c) = 0$ with $f''(c) > 0$ forces f' to pass from negative to positive at c, and the mean value theorem (Theorem 3.6) turns that into "f falls, then rises".

For Kepler's barrel, $V''(h) = -\frac{3\pi}{8}h$, which is negative for every positive h — a maximum, as the picture showed. And notice that the same second derivative controls the *flatness*: the bigger $|f''|$, the sharper the peak and the more a mistake costs. Kepler's peak is gentle, which is exactly why generations of coopers could find it by trial and error without knowing any mathematics.

This is the single most reused idea in physics, so it is worth naming now. A mechanical system sits at equilibrium where the potential energy U has a stationary point, $U' = 0$, because force is $-U'$. If $U'' > 0$ there — a valley rather than a hilltop — the equilibrium is stable, the valley floor is approximately a parabola, and a small disturbance oscillates. Read the implication in that direction only: $U = x^4$ has a perfectly stable minimum at the origin with $U''(0) = 0$, and its floor is flatter than any parabola. Stability is the general notion; $U'' > 0$ is the well-behaved case, and it is the case most of Part II lives in.

Before leaving the subject we should be honest about two things the apparatus above does *not* do, because mechanics will lean on both.

The first is reading off where a function rises and falls. It is entirely believable that $f' > 0$ throughout an interval means f is increasing there — "positive velocity means moving forward" — but believing it is not proving it, and the derivative is a statement about single points while "increasing" is a statement about a whole interval. The bridge between the two is the mean value theorem (Theorem 3.6), which we will not have until Chapter 3. We will use the sign of f' before then, and this is the notice that we are doing so on credit.

2.21 Proposition *finding the largest and smallest values on a closed interval*

Let f be continuous on a closed interval $[a, b]$. Then f attains a largest and a smallest value there, and each is attained at one of: a stationary point inside the interval, a point inside where f' does not exist, or one of the two endpoints a and b. So: list the candidates, evaluate f at each, and compare.

The endpoints are the part everybody forgets, and physics punishes it. A projectile's range as a function of launch angle is largest in the interior, at $45°$, when launch and landing heights agree and air resistance is neglected. But the speed $v(t) = gt$ of a freely falling object, released from rest and observed over $0 \le t \le T$, is greatest at the endpoint T, even though $v'(T) = g \neq 0$. The existence is half of the proposition — that a continuous function on a closed bounded interval really does attain its extremes — is the extreme value theorem, which we assume, as we assumed the intermediate value theorem in Chapter 1.

Fermat's rule finds interior stationary points and nothing else. That is the right tool for an open-ended question like Kepler's, where the answer is clearly somewhere in the middle. It is the wrong tool on its own when the variable is confined — an angle between 0 and $\pi/2$, a length between 0 and L, a speed that cannot exceed c — and confined variables are the normal case in mechanics.

Optimization was one of the four problems that forced calculus into existence, and it was not an academic exercise. Kepler was pricing wine. Fermat was doing optics — and the most beautiful application of his rule is a law of physics you already know, obtained from a principle rather than from measurement.

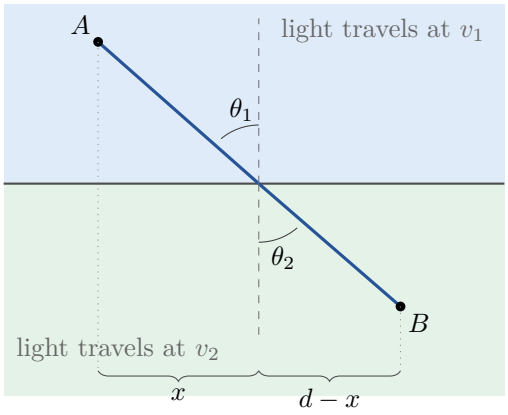


Figure 2.7: Fermat's principle. Light leaves A in one medium and arrives at B in another, where it travels at a different speed. Of all the possible crossing points, nature picks the one that makes the total travel time least — and asking which one that is turns the law of refraction into an exercise in finding a minimum.

2.22 Example *the law of refraction, from least time*

Light travels at speed v_1 above an interface and v_2 below it. It leaves A at height a above the surface and arrives at B at depth b below it, the two being a horizontal distance d apart. Fermat's principle says the path taken is the one of *least time*. Find it.

Solution Let x be the horizontal distance from A to the crossing point. The two straight legs have lengths $\sqrt{a^2 + x^2}$ and $\sqrt{b^2 + (d - x)^2}$, so the total time is

$$T(x) = \frac{\sqrt{a^2 + x^2}}{v_1} + \frac{\sqrt{b^2 + (d - x)^2}}{v_2}.$$

By Theorem 2.18 the minimum occurs where $T'(x) = 0$. Differentiating with the chain rule (Theorem 2.14),

$$T'(x) = \frac{x}{v_1\sqrt{a^2 + x^2}} - \frac{d - x}{v_2\sqrt{b^2 + (d - x)^2}}.$$

Now read the two fractions off Figure 2.7. In the upper triangle, x is the side opposite θ_1 and $\sqrt{a^2 + x^2}$ is the hypotenuse, so that quotient is exactly $\sin\theta_1$; likewise the other is $\sin\theta_2$. So $T'(x) = 0$ says

$$\frac{\sin\theta_1}{v_1} = \frac{\sin\theta_2}{v_2},$$

which is Snell's law of refraction. (In the form usually quoted, multiply through by the speed of light and write $n = c/v$ for the refractive index: $n_1 \sin\theta_1 = n_2 \sin\theta_2$.)

Stop and notice what just happened, because it is a template for most of theoretical physics. We did not measure anything. We asserted a principle — light takes the quickest route — and

then a derivative turned that principle into a quantitative law that had previously been found only by experiment. Fermat did this in 1662.[4] In the century that followed, the same move applied to mechanics produced Lagrange's reformulation of Newton's laws — his *Mécanique analytique* appeared in 1788[5] — and it remains the way fundamental physics is written today.

2.9 Derivatives of Sine and Cosine

Here the limit we worked for at the end of Chapter 1 finally earns its keep. We take it now, rather than later, because the section that follows wants to approximate $\sin\theta$ and $\cos\theta$ and has no business doing so until these two facts are actually proved.

2.23 Theorem *derivative of sine*

$$\frac{d}{dx}[\sin x] = \cos x.$$

Proof Using the angle-addition identity $\sin(x+\Delta x) = \sin x\cos\Delta x + \cos x\sin\Delta x$,

$$\frac{\Delta f}{\Delta x} = \frac{\sin(x+\Delta x)-\sin x}{\Delta x} = \sin x\cdot\frac{\cos\Delta x - 1}{\Delta x} + \cos x\cdot\frac{\sin\Delta x}{\Delta x}.$$

The second fraction tends to 1 by Example 1.18. For the first, write $\cos\Delta x - 1 = -\dfrac{\sin^2\Delta x}{1+\cos\Delta x}$, so that

$$\frac{\cos\Delta x - 1}{\Delta x} = -\frac{\sin\Delta x}{\Delta x}\cdot\frac{\sin\Delta x}{1+\cos\Delta x} \longrightarrow -1\cdot\frac{0}{2} = 0.$$

Hence the whole expression tends to $\sin x\cdot 0 + \cos x\cdot 1 = \cos x$. ∎

2.24 Corollary *derivative of cosine*

$$\frac{d}{dx}[\cos x] = -\sin x.$$

Proof Since $\cos x = \sin(x+\frac{\pi}{2})$, the chain rule (Theorem 2.14) gives $\dfrac{d}{dx}[\cos x] = \cos(x+\frac{\pi}{2}) = -\sin x$. ∎

These two facts are the reason a pendulum, a mass on a spring, a vibrating string and an alternating current all obey the same equations. Sine and cosine are, essentially, the functions whose rate of change feeds back into themselves in exactly the way an oscillation requires: differentiate twice and you get back what you started with, times -1. We will build Part II's entire treatment of oscillation on that single observation.

2.10 First-Order Approximation in Practice

Theorem 2.4 is not a piece of theory to be admired and filed away. Rearranged as

$$f(a+\Delta x) \;\approx\; f(a) + f'(a)\,\Delta x,$$

it is probably the most-used equation in physics, and the rest of this section is about learning to reach for it.

2.25 Example *the workhorse approximations*

Apply the formula at $a = 0$ to three functions.

Solution For $f(x) = (1+x)^n$: $f(0) = 1$ and $f'(x) = n(1+x)^{n-1}$ so $f'(0) = n$, giving

$$(1+x)^n \approx 1 + nx \qquad \text{for small } x.$$

Taking $n = \frac{1}{2}$ gives $\sqrt{1+x} \approx 1 + \frac{x}{2}$; taking $n = -1$ gives $\dfrac{1}{1-x} \approx 1 + x$.
For $f(\theta) = \sin\theta$: using $f'(0) = \cos 0 = 1$, proved in the previous section, $\sin\theta \approx \theta$ — which is the small-angle approximation from Chapter 1, no longer a special geometric fact but one instance of the general rule.
For $f(\theta) = \cos\theta$: $f'(0) = -\sin 0 = 0$, so to first order $\cos\theta \approx 1$. The first-order term vanishing is itself information: it says cos is flat at the origin, and that the leading correction lives at second order, where it turns out to be $-\theta^2/2$.

The next example is deliberately out of period: it is from 1905. That is allowed by the ground rule of the preface — we are pointing a finished tool at something, not borrowing a result we have not yet earned — and later chapters will do the same with radioactive decay, molecular potentials and electric circuits. It is here because it is the best advertisement for first-order approximation ever written. Nothing later depends on it.

2.26 Example *where $\frac{1}{2}mv^2$ comes from*

Einstein's relativity says a particle of mass m moving at speed v has energy $E = \gamma mc^2$ with $\gamma = \left(1 - v^2/c^2\right)^{-1/2}$, where c is the speed of light. What does this become at everyday speeds, $v \ll c$?

Solution Write $x = -v^2/c^2$, which is tiny when $v \ll c$, and use $(1+x)^n \approx 1 + nx$ with $n = -\frac{1}{2}$:

$$\gamma = (1+x)^{-1/2} \approx 1 - \tfrac{1}{2}x = 1 + \frac{v^2}{2c^2}.$$

Therefore

$$E = \gamma mc^2 \approx mc^2 + \tfrac{1}{2}mv^2.$$

The famous rest energy, plus the kinetic energy you learned in high school — which is revealed to be nothing more than the first-order term of a more accurate formula. At $v = 30$ km/s (the Earth's orbital speed) the neglected terms are smaller than the kept one by a factor of about 10^{-8}, which is why nobody noticed for two centuries.

This is what "to first order" means whenever you hear it: keep the Δx term, discard the $o(\Delta x)$. It is not hand-waving, because Theorem 2.4 tells you exactly what you discarded and guarantees it shrinks faster than what you kept. When a first-order term vanishes — as it does for cos *at* 0 *— the discarded part is suddenly the whole story, and you go to second order.*

Differentials also answer a question every experimentalist has: if my measurement is a little off, how far off is the result? Divide $\Delta f \approx f'(x)\Delta x$ by f to get the *relative* error, which is usually

what you care about.

2.27 Example *propagating a measurement error*

You measure a sphere's radius to within 1% and compute its volume. How accurate is the volume?

Solution With $V = \frac{4}{3}\pi r^3$ we have $dV = 4\pi r^2\,dr$, so

$$\frac{dV}{V} = \frac{4\pi r^2\,dr}{\frac{4}{3}\pi r^3} = 3\,\frac{dr}{r}.$$

A 1% error in the radius becomes about a 3% error in the volume. The exponent became the multiplier — and the same computation shows that for any power law $f = kx^n$, signed relative changes obey $df/f = n\,dx/x$, so an uncertainty is magnified by $|n|$. Being a first-order statement it is an estimate, not an identity: the exact volume changes here are $+3.03\%$ and -2.97%, and the small asymmetry between them is exactly the $o(\Delta x)$ we agreed to throw away.

2.11 Higher Derivatives, and What the Second One Measures

2.28 Definition *higher derivatives*

The derivative of f' is the **second derivative** f'', also written $\dfrac{d^2f}{dx^2}$. In general $f^{(n)}$ denotes f differentiated n times.

If $s(t)$ is position, then $v = \dot{s}$ is velocity and $a = \dot{v} = \ddot{s}$ is acceleration: the rate at which the rate is changing.

2.29 Example *free fall, completed*

For $s(t) = 4.9t^2$ we found $v(t) = 9.8\,t$ in Example 2.2. Differentiating once more,

$$a(t) = v'(t) = 9.8,$$

a constant. The acceleration of a freely falling body never changes — this is the $g \approx 9.8$ m/s^2 that physics quotes, and we have now derived it rather than been told it, from nothing but the shape of $s(t)$ and two applications of Definition 2.1.

There is a second way to think about f'' that connects it to everything in Section 2.3. The first derivative tells you the best straight-line approximation; the second tells you how fast that approximation goes wrong. For a smooth f, the error discarded in Theorem 2.4 is approximately

$$\Delta f - f'(a)\,\Delta x \;\approx\; \tfrac{1}{2}f''(a)\,(\Delta x)^2,$$

which is the next term of the same expansion — exactly the corner of the square in Figure 2.3, where $f(x) = x^2$, $f'' = 2$, and the corner is $\frac{1}{2}(2)(\Delta x)^2 = (\Delta x)^2$. So f'' measures how far the

graph pulls away, vertically, from its own tangent line, and therefore how quickly a first-order approximation stops being good enough. It is close kin to *curvature* but not the same thing: a line of slope 1000 and a line of slope 0 are both perfectly straight, yet a given f'' bends the steep graph far less, and the geometric curvature $|f''|/(1+(f')^2)^{3/2}$ is what corrects for that. That is why $\cos\theta \approx 1-\theta^2/2$ had to wait for second order, and why a tightly curved orbit needs smaller time steps than a gentle one.

2.12 Chapter Recap

We started with two questions, one about tangent lines and one about instantaneous speed, and found that both are answered by the same limit of the same ratio of changes: $f'(a) = \lim_{\Delta x\to 0} \Delta f/\Delta x$ (Definition 2.1).

Then we turned that definition around, and this is the part to carry forward. Differentiability says $\Delta f = f'(a)\,\Delta x + o(\Delta x)$ (Theorem 2.4): a differentiable function responds to a small nudge by multiplication, with an error that vanishes faster than the nudge. That is why curves look straight under magnification, why the differential $df = f'(x)\,dx$ can be manipulated as ordinary algebra, and why physics can replace almost any function by a straight line and still get the right answer to the accuracy it needs. Every rule in this chapter — power, product, quotient, chain — came from the same three-step habit: write Δf, keep the part proportional to Δx, and throw away the corner.

What we cannot yet do is run the process backwards. Differentiating took us from position to velocity to acceleration. Physics hands us the acceleration — a force divided by a mass — and asks where the object ends up. That reverse trip is Chapter 3.

Problem 2.1. Use Definition 2.1 directly, in Δ notation, to compute $f'(x)$ for $f(x) = x^2 + 3x$. Identify which part of Δf is proportional to Δx and which part is $o(\Delta x)$, then check your answer against Theorem 2.9.

Problem 2.2. A cube's side is measured as 10.0 cm with an uncertainty of 0.1 cm. Use differentials to estimate the resulting uncertainty in the volume, both absolutely (in cm^3) and relatively (as a percentage). Then draw the three-dimensional version of Figure 2.3 and say what the discarded pieces are.

Problem 2.3. Differentiate $p(x) = (2x+1)^3(x-4)$ using the product rule together with the shortcut of Example 2.15.

Problem 2.4. A pendulum's period is $T = 2\pi\sqrt{L/g}$. If the length L is increased by 1%, by roughly what percentage does the period change? Answer using differentials, not by recomputing T twice.

Problem 2.5. A car's position is $s(t) = t^3 - 6t^2 + 9t$ (meters, t in seconds). Find $v(t)$ and $a(t)$, and find every time at which the car is momentarily at rest. At those instants, what is the sign of a, and what does that tell you physically?

Problem 2.6. Use the approximation $(1+x)^n \approx 1 + nx$ to estimate $\sqrt{1.02}$ and $1/0.98$ without a calculator. Then compute the true values and check that the errors are of size roughly $(0.02)^2$. One caution: local linearity by itself promises only that the error is $o(\Delta x)$, which is weaker than a bound of size $(\Delta x)^2$. The stronger estimate uses the smoothness of these particular functions, and Chapter 5 will justify it.

Chapter 3

Integrals

Chapter 2 gave us a machine that runs in one direction. Feed it a position function and it returns a velocity; feed it the velocity and it returns an acceleration. Every step throws information away and answers the question "how fast is this changing?"

We almost always need the machine to run backwards. A speedometer does not record where you went; an odometer has to build that up out of what the speedometer said. (Newton would later pose the same question in its sharpest form: his second law hands you a force, and therefore an acceleration, and asks where the object ends up. But that is 1687, and the mathematics we are about to build is 1665 — the reverse question was already pressing long before the *Principia* gave it its most famous shape.) So the question for this chapter is:

> If I know the *rate* at which something changes, at every instant, how do I recover the *total change*?

We are going to answer it with one idea, borrowed directly from Chapter 2 — *a small change is rate times step* — applied many times and added up. Along the way we will find that the answer is also the area under a graph, and that the two seemingly unrelated questions of this subject are the same question.

That the answer should turn out to be an area is not, historically, the surprise. It is the oldest part of the story. What the seventeenth century lacked was not the idea that areas mattered but any general method for computing them: *quadrature*, the problem of finding the area of a region with curved boundaries, was the great unsolved problem of the age. Archimedes had done extraordinary things with the methods he had — exhaustion, and the mechanical balancing argument of his *Method* — and his work remained a model for later geometers. The centuries between Archimedes and calculus were not empty: mathematicians extended methods for areas and volumes. What was still missing was a broadly usable calculus linking quadrature and tangents, so each new region still called for a new idea. Kepler had needed barrel volumes and the areas swept by planets. Everyone could see that areas were the key to something, and nobody could compute them except case by case. What was missing was a machine, and this chapter builds it.

3.1 Adding Up the Little Changes

Start with the easiest possible case. A car travels at a constant 20 m/s for 6 seconds; it goes 120 m. You did not need calculus: distance is rate times time. Draw the velocity graph — a horizontal line at height 20 from $t = 0$ to $t = 6$ — and notice that 120 is also the *area* of the rectangle underneath it, because area is height times width and we just multiplied height by width. Keep that coincidence in mind; it is about to stop being a coincidence.

Now let the velocity change — but change in the simplest possible way, increasing steadily from rest, as it does for a falling body. The velocity graph is now a sloped line rather than a flat one, and the region beneath it is a triangle. Is the distance still the area?

Scholars at Merton College, Oxford, answered this in the 1330s, and in the mid-fourteenth century Nicole Oresme gave a geometrical argument.[2] Draw the sloped velocity line over the interval, and draw the horizontal line at the *average* of the starting and ending speeds. Those two lines cut off two triangles, marked A and B in Figure 3.1 — and the triangles are congruent. Whatever the sloped graph has too little of on the left, it has exactly too much of on the right. So the triangle under the sloped line has the same area as the rectangle, and the rectangle is an honest constant-velocity trip at the average speed, whose distance we already know.

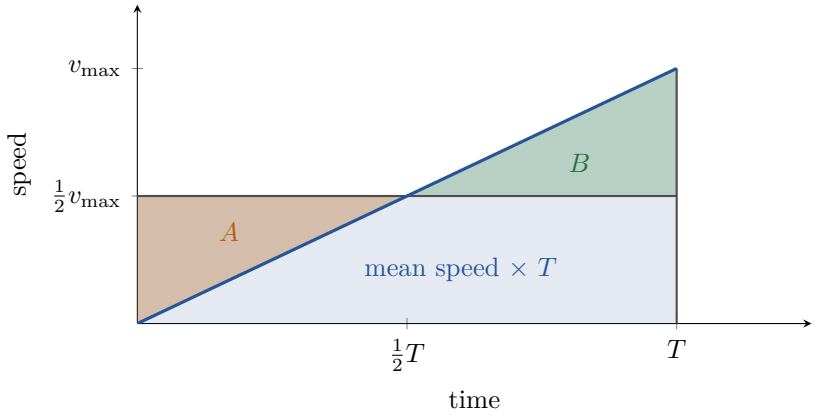


Figure 3.1: A reconstruction of Oresme's mid-fourteenth-century diagram — roughly three centuries before calculus. For a speed increasing steadily from rest, triangles A and B are congruent, so the area under the sloped line equals the area of the shaded rectangle, whose height is the average speed. The distance covered is therefore the average speed times the time. A half-turn about the point where the two lines cross carries B exactly onto A. Note what the picture does and does not settle: it proves the two *areas* are equal. That area is distance is still a claim, and the limiting sums later in this section are what will justify it.

Run Oresme's result forwards and you get Galileo's law. If the speed grows steadily from 0 *to* $v_{\max} = aT$*, the distance is the mean speed times the time,* $\frac{1}{2}(aT)\,T = \frac{1}{2}aT^2$ *— proportional to the square of the time, which is precisely the* $1 : 3 : 5 : 7$ *pattern Galileo described mathematically in his letter of 1604 and supported with an inclined-plane account published in 1638. The medieval result predates Galileo's work by roughly two and a half centuries; the exact route from his experiments to the falling-body law remains a historical question.[1]*

So for a steadily increasing speed, the distance is still the area, and it can be had with nothing more than the geometry of triangles. The trouble starts when the velocity graph is *curved* — when the acceleration itself changes, which is what happens to a falling raindrop, a planet, a charge in a field, essentially everything. Then no cutting and sliding of triangles will do it, because the region has a curved edge, and we are back to quadrature: the very problem nobody could solve in general.

Here is where Chapter 2 pays for itself. Over a long stretch we cannot multiply rate by time, because there is no single velocity to multiply by. But over a *short* stretch we almost can, and Chapter 2 says exactly how well. Applying Theorem 2.4 to the position function s, whose derivative is v,

$$\Delta s \;=\; v(t)\,\Delta t \;+\; o(\Delta t).$$

Over a short enough interval, distance really is rate times time, with an error we know how to control. That is the entire idea of this chapter.

This is not a metaphor for how measurement works — it is literally how it works. A car's trip computer samples the speed every fraction of a second, multiplies each reading by the sampling interval, and adds. Every numerical simulation ever run, from spacecraft trajectories to climate models, is doing the same thing: chop time into steps, use the rate to step forward, repeat.

So chop the whole interval $[a, b]$ into n short steps, each of width $\Delta t = (b-a)/n$, with endpoints $t_0 = a,\ t_1,\ \ldots,\ t_n = b$. Over the i-th step the position changes by some exact amount $\Delta s_i = s(t_i) - s(t_{i-1})$, and two things are true about those n little changes.

The first is exact and almost embarrassingly simple. Add them all up and everything cancels except the ends:

$$\sum_{i=1}^{n} \Delta s_i = \big(s(t_1) - s(t_0)\big) + \big(s(t_2) - s(t_1)\big) + \cdots + \big(s(t_n) - s(t_{n-1})\big) = s(b) - s(a).$$

Each interior value appears once with a plus and once with a minus. This is called **telescoping**, and it is the statement that the total change is the sum of the little changes — true for any function whatsoever, with no approximation anywhere. Remember it; it is half of the Fundamental Theorem.

It is worth knowing that this is not a convenient trick we have adopted for the occasion. Sums and differences helped Leibniz develop his route to calculus; Newton reached the inverse relationship independently. One of Leibniz's starting problems concerned whole numbers. In 1672 Christiaan Huygens, testing the young diplomat who kept pestering him for mathematics lessons, challenged Gottfried Leibniz to add up the reciprocals of the triangular numbers.[8]

3.1 Example *Huygens's challenge, and what Leibniz saw in it*

The triangular numbers are $1, 3, 6, 10, 15, \ldots$, with $T_n = \frac{1}{2}n(n+1)$. Find $1 + \frac{1}{3} + \frac{1}{6} + \frac{1}{10} + \cdots$.

Solution Leibniz noticed that each term is a *difference*:

$$\frac{1}{T_n} = \frac{2}{n(n+1)} = 2\left(\frac{1}{n} - \frac{1}{n+1}\right).$$

So the partial sums collapse, every interior term cancelling its neighbour:

$$\sum_{n=1}^{N} \frac{1}{T_n} = 2\left[\left(1 - \tfrac{1}{2}\right) + \left(\tfrac{1}{2} - \tfrac{1}{3}\right) + \cdots + \left(\tfrac{1}{N} - \tfrac{1}{N+1}\right)\right] = 2\left(1 - \frac{1}{N+1}\right),$$

which tends to 2 as N grows. The sum is exactly 2.

The answer mattered less to Leibniz than what he had done to get it. To add up a long list, he had found a second list whose *differences* were the terms he wanted, and then everything cancelled but the ends. Summing and differencing, he realised, are inverse operations — and if that is true for sequences of numbers, it ought to be true for the continuous versions: areas and slopes. His work on sums and differences supplied an important route toward that connection.

The finite cancellation is not yet a proof of the continuous theorem: passing from sums to areas still needs justification. Everything we do for the rest of this chapter is the continuous form of the cancellation you just watched happen in Example 3.1.

The second is the approximation: each $\Delta s_i \approx v(t_i)\,\Delta t$. Putting the two together,

$$s(b) - s(a) \;=\; \sum_{i=1}^{n} \Delta s_i \;\approx\; \sum_{i=1}^{n} v(t_i)\,\Delta t.$$

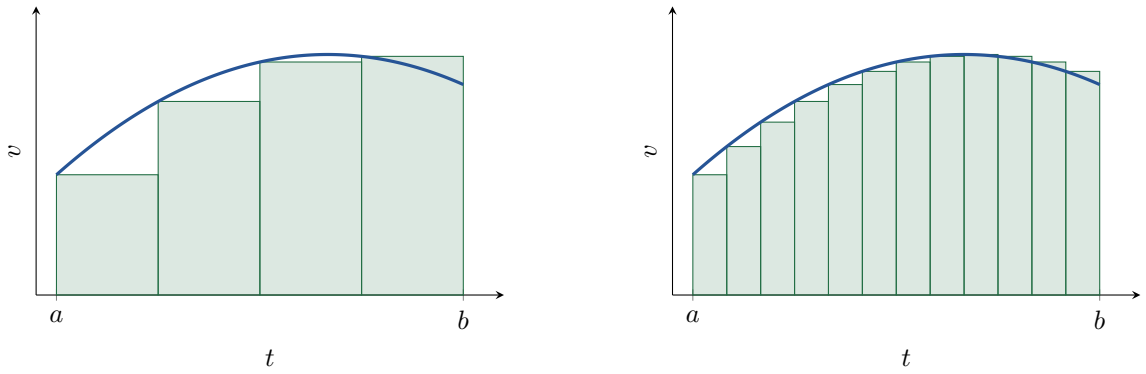
The left-hand side is the number we want and cannot compute. The right-hand side is a finite sum of products of numbers we know. And as the steps get shorter the approximation gets better — in a way worth doing carefully, because there are more errors to worry about as well as smaller ones. Suppose that once Δt is small enough, every one of the n individual errors is at most $\varepsilon\,\Delta t$ in size — one step size working at all n sample points at once. There are $n = (b-a)/\Delta t$ of them, so together they come to at most

$$n \cdot \varepsilon\,\Delta t = \varepsilon\,(b-a),$$

which we can make as small as we like by choosing ε small. Shrinking the steps does win the race against having more of them.

That "one step size working at all n points at once" is a real assumption and it is worth flagging rather than sliding past. Theorem 2.4 is a statement about one point at a time: it gives a good step size at *each t, and nothing says those choices can be made simultaneously. What rescues us here is that the sample points live in a closed bounded interval and v is continuous on all of it, which is enough — and in Section 3.4 the mean value theorem will remove the error term altogether rather than merely bounding it, which is tidier still.*

The uniformity just assumed is exactly the kind of thing that separates a convincing argument from a proof, and in Section 3.4 we will not need it.



(a) four steps (b) twelve steps

Figure 3.2: Here each rectangle is sampled at the *left* end of its step, so it has height $v(t_{i-1})$ and width Δt; any sample point inside the step would serve equally well, and the next section says so outright. The staircase is a crude stand-in for the curve on the left and a much better one on the right, and the mismatch — the little triangular slivers between staircase and curve — is exactly the accumulated $o(\Delta t)$ error. Total change and area under the rate graph are the same quantity, approached the same way.

Figure 3.2 is where the second question sneaks in. Each term $v(t_i)\Delta t$ is a height times a width, so it is the area of a thin rectangle standing under the velocity graph. The sum is the area of a staircase. Refining the steps improves the estimate of the distance travelled and simultaneously improves the fit of the staircase to the curve. The distance travelled *is* the area under the velocity graph — not by analogy, but because both are the limit of the same sum.

3.2 The Definite Integral

The sums in Section 3.1 deserve a name and a symbol, and the definition should not care whether we are talking about velocity or anything else.

3.2 Definition *Riemann sum and the definite integral*

Let f be defined on $[a,b]$. Divide $[a,b]$ into n equal subintervals of width $\Delta x = (b-a)/n$ with endpoints $x_0 = a, \dots, x_n = b$, and in each subinterval choose any **sample point** $x_i^* \in [x_{i-1}, x_i]$. The sum

$$\sum_{i=1}^{n} f(x_i^*)\,\Delta x$$

is a **Riemann sum** for f. If these sums approach the same limit as $n \to \infty$, no matter how the sample points are chosen, then f is **integrable** on $[a,b]$ and that limit is the **definite integral**

$$\int_a^b f(x)\,dx \;=\; \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\,\Delta x.$$

Every function we integrate over a closed bounded interval in this book will be integrable there. The reason is a fact we shall use and not prove: *every continuous function on a closed bounded interval is integrable* — and so is any function that is continuous apart from finitely many jumps, which covers the idealized collisions of Part II. (Proving it requires a careful look at what continuity does on a closed interval, and it would be a detour.)

Leibniz chose the symbol $\int$ as an elongated S, for summa. *The notation is a picture of its own definition: Δx relaxes into dx, the $\sum$ stretches into $\int$, and $f(x)\,dx$ is the typical little contribution being added. In Chapter 2 we learned to read $f'(x)\,dx$ as the differential df — so $\int_a^b df$ ought to be the total change in f. It is, and that is the Fundamental Theorem.*

Two features of the notation are worth saying out loud. The dx is not decoration: it records which variable is being chopped up, and it carries the units. If v is in m/s and dt in s, then $v\,dt$ is in metres and so is the integral — a habit that catches errors. And the variable itself is a dummy: $\int_a^b f(x)\,dx$ and $\int_a^b f(u)\,du$ are the same number, just as $\sum_i a_i$ and $\sum_j a_j$ are the same sum.

Because the sample values $f(x_i^*)$ may be negative, so may the terms.

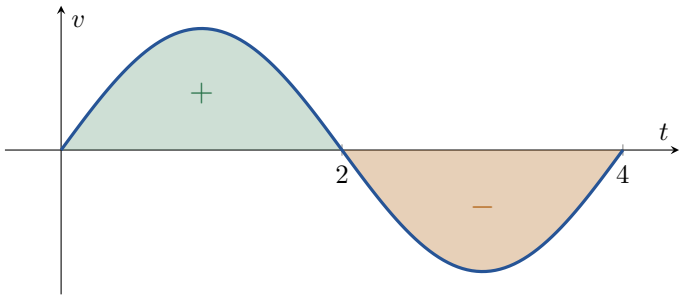


Figure 3.3: The integral measures *signed* area: regions below the axis count negatively. If this is a velocity graph, the object moves forward for the first two seconds and back for the next two, and $\int_0^4 v\,dt = 0$ — the displacement is zero even though the distance travelled is not. Displacement and distance are different questions, and the integral answers the first one.

Three properties follow from the definition, because each is the limit of the corresponding statement about finite sums. Part (b) needs one small extension first. We want a partition with a division point exactly at c, so that the sum for $[a,b]$ splits into the sums for $[a,c]$ and $[c,b]$; but c need not land on one of our equally spaced points. The repair is to allow unequal subintervals in Definition 3.2, requiring only that the widest of them shrinks to zero. That more general definition produces the same integral, and we take it as available from here on.

It is also convenient to adopt two conventions:

$$\int_a^a f = 0 \qquad \text{and} \qquad \int_b^a f = -\int_a^b f.$$

The first is the statement that a region of zero width has no area; the second makes reversed limits mean something, and both substitution and the logarithm of Chapter 4 will want it. With those in place, (b) holds for any three numbers a, b, c, not only for $a < c < b$.

We will use all three properties.

3.3 Proposition *basic properties of the integral*

Let f, g be integrable on the relevant intervals and c a constant. Then

(a) $\displaystyle\int_a^b [f+g] = \int_a^b f + \int_a^b g$ and $\displaystyle\int_a^b c\,f = c\int_a^b f$;

(b) $\displaystyle\int_a^b f = \int_a^c f + \int_c^b f$ for $a < c < b$;

(c) if $m \le f(x) \le M$ for all x in $[a,b]$, then $m\,(b-a) \;\le\; \displaystyle\int_a^b f \;\le\; M\,(b-a)$.

Part (c) deserves a picture in your head: the region under the graph contains a rectangle of height m and is contained in one of height M. It is the crudest possible estimate of an integral, and in Section 3.5 it is exactly the tool we need.

3.4 Example *free fall, by geometry*

A ball is dropped from rest, so $v(t) = 9.8\,t$ (Example 2.2). How far does it fall between $t = 0$ and $t = 1$?

Solution The graph of v on $[0,1]$ is a straight line from $(0,0)$ to $(1, 9.8)$, so the region beneath it is a triangle with base 1 and height 9.8:

$$\int_0^1 9.8\,t\,dt = \tfrac{1}{2}(1)(9.8) = 4.9 \text{ metres.}$$

Compare with $s(1) = 4.9(1)^2 = 4.9$ from Chapter 1: the same number. The area under the velocity graph is the distance fallen, exactly as Section 3.1 promised — and here we could get it without any limit at all, because the region happened to be a shape from elementary geometry.

3.5 Example *the same answer, the hard way*

Confirm $\int_0^1 9.8\,t\,dt = 4.9$ straight from Definition 3.2.

Solution Take n equal steps of width $\Delta t = 1/n$ and sample at right endpoints, $t_i^* = i/n$. Then

$$\sum_{i=1}^{n} 9.8\left(\frac{i}{n}\right)\frac{1}{n} = \frac{9.8}{n^2}\sum_{i=1}^{n} i = \frac{9.8}{n^2}\cdot\frac{n(n+1)}{2} = 4.9\cdot\frac{n+1}{n} = 4.9\left(1+\frac{1}{n}\right),$$

using the identity $1+2+\cdots+n = n(n+1)/2$. As $n\to\infty$ this tends to 4.9. Notice how much work that was for a problem we had already solved by looking at a triangle — and that we could only do it at all because the sum had a closed form. Sums of fifth powers have closed forms as well, and so do sums of sines, so $\int_0^1 t^5\,dt$ and $\int_0^1 \sin t\,dt$ are not strictly out of reach this way — but each demands its own piece of specialised algebra, found separately and good for nothing else. What we want is a method that does not require a new summation identity for every new integrand, and the next three sections build it.

3.3 A Detour We Need: the Mean Value Theorem

Section 3.1 left us with an approximation and a hope: that the errors wash out. There is a way to make the argument airtight, and it comes from a theorem about derivatives that we have not needed until now. It says something you already believe.

You drive 100 miles in two hours, so your average speed was 50 mph. Is it possible that you were never going exactly 50 at any instant of the trip? Think about what your speedometer would have had to do.

It is not possible. If you were always below 50 you could not have covered the distance, and if always above you would have overshot; and to get from below to above without passing through 50 your speed would have to jump — which is exactly what continuity forbids (Theorem 1.13). The average rate over an interval must be achieved as an instantaneous rate somewhere inside it.

3.6 Theorem *mean value theorem*

If f is continuous on $[a,b]$ and differentiable on (a,b), then there is at least one c in (a,b) with

$$f'(c) \;=\; \frac{f(b)-f(a)}{b-a}.$$

Proof sketch Measure the vertical gap between the curve and the chord: let

$$g(x) = f(x) - \left[f(a) + \frac{f(b)-f(a)}{b-a}(x-a)\right],$$

so $g(a) = g(b) = 0$. Being continuous on a closed interval, g attains a largest and a smallest value somewhere on $[a,b]$ — this is the extreme value theorem, which like the intermediate value theorem (Theorem 1.13) rests on the completeness of the real numbers, and which we take on trust here. If both are 0 then g is identically 0, f is the chord, and every c works. Otherwise one

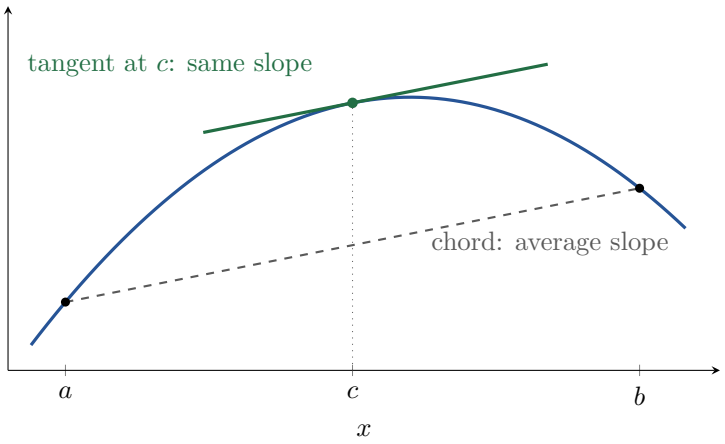


Figure 3.4: The mean value theorem. The dashed chord has the average slope across $[a, b]$. Slide a line of that slope up until it last touches the curve: where it touches, at c, it is tangent, so $f'(c)$ equals the average slope. The theorem promises such a c exists; it says nothing about where, and there may be several.

of them is attained at an interior point c — and there $g'(c) = 0$ by Fermat's rule (Theorem 2.18), since a nonzero slope would let us step one way or the other and do better. Differentiating the definition of g, $g'(c) = 0$ says precisely $f'(c) = \big(f(b) - f(a)\big)/(b - a)$. ■

Two consequences do work for us immediately.

3.7 Corollary *zero derivative means constant*

If $g'(x) = 0$ for every x in an interval, then g is constant on that interval. Consequently, if $F' = G'$ on an interval, then $F - G$ is constant there.

Proof Take any two points $x < y$ in the interval. By Theorem 3.6 applied to $[x, y]$, there is a c between them with $g(y) - g(x) = g'(c)(y - x) = 0$, so $g(y) = g(x)$. Since this holds for every pair, g is constant. For the second statement apply the first to $g = F - G$, whose derivative is $F' - G' = 0$. ■

That second sentence is the reason antiderivatives come with "$+ C$", and we were not entitled to say so until now.

3.4 The Fundamental Theorem, Part 2

Now we can finish the argument of Section 3.1, and the mean value theorem is exactly the tool that upgrades its "$\approx$" to "$=$".

Recall the setup: partition $[a, b]$ into n steps, and telescope the total change of some function F into a sum of little changes. That step was exact. The approximation was $\Delta F_i \approx F'(x_i)\Delta x$. But Theorem 3.6, applied to F on the i-th subinterval, says we can do better than "approximately": there is a particular point c_i inside that subinterval where the approximation is *exactly* right,

$$\Delta F_i \;=\; F(x_i) - F(x_{i-1}) \;=\; F'(c_i)\,\Delta x.$$

No error term at all — provided we sample at the right place, and the theorem guarantees that such a place exists even though it does not tell us where. Since Definition 3.2 lets us choose the sample points however we like, this is exactly the freedom we need.

3.8 Theorem *fundamental theorem of calculus, part 2*

Let f be continuous on $[a, b]$ and let F be any function with $F' = f$ there. Then

$$\int_a^b f(x)\,dx \;=\; F(b) - F(a).$$

Proof Partition $[a, b]$ into n equal subintervals of width Δx. Telescoping,

$$F(b) - F(a) = \sum_{i=1}^{n} [F(x_i) - F(x_{i-1})].$$

By Theorem 3.6 there is a point $c_i \in (x_{i-1}, x_i)$ with $F(x_i) - F(x_{i-1}) = F'(c_i)\,\Delta x = f(c_i)\,\Delta x$. Hence for *every* n,

$$F(b) - F(a) = \sum_{i=1}^{n} f(c_i)\,\Delta x,$$

and the right-hand side is a Riemann sum for f with sample points c_i. Since f is continuous it is integrable, so as $n \to \infty$ these sums converge to $\int_a^b f$. The left-hand side does not depend on n, so it must equal that limit. ■

Look at what that proof actually used: the total change is the sum of the little changes (telescoping, exact), and each little change is a rate times a step (the mean value theorem, also exact). The Fundamental Theorem is the bookkeeping identity of Section 3.1 with the approximation removed.

It is also a spectacular labour-saving device. To integrate, stop summing rectangles and instead go looking for a function whose derivative is the integrand. We write $F(b) - F(a)$ as $\big[F(x)\big]_a^b$.

3.9 Example *free fall, the easy way*

Recompute $\int_0^1 9.8\,t\,dt$.

Solution We need any F with $F'(t) = 9.8\,t$. The function $F(t) = 4.9\,t^2$ works, since that is exactly what Example 2.2 computed. So

$$\int_0^1 9.8\,t\,dt = \big[4.9\,t^2\big]_0^1 = 4.9 - 0 = 4.9.$$

One line, no limits, no triangles, no sums. And notice what F turned out to be: the position function $s(t)$ we started the book with. Differentiating s gave v; integrating v gave s back.

3.5 The Fundamental Theorem, Part 1

Part 2 says that if you can find an antiderivative, integration is easy. It leaves two questions open. Does an antiderivative always exist? And is there a reason, visible in the picture rather

than in the algebra, why areas and derivatives should have anything to do with each other?

One construction answers both. Fix the left endpoint and let the right one move: define the **accumulation function**

$$A(x) \;=\; \int_a^x f(t)\,dt,$$

the running total of *signed* area swept out so far. If f is a velocity, $A(x)$ is the displacement from the starting position up to time x — not an odometer reading, because an odometer never runs backwards. To get the distance actually travelled you would accumulate the speed $|f|$ instead.

Push the right edge out by a small Δx. The accumulated area grows by a thin sliver. What is the area of that sliver, roughly, and what does that make $A'(x)$?

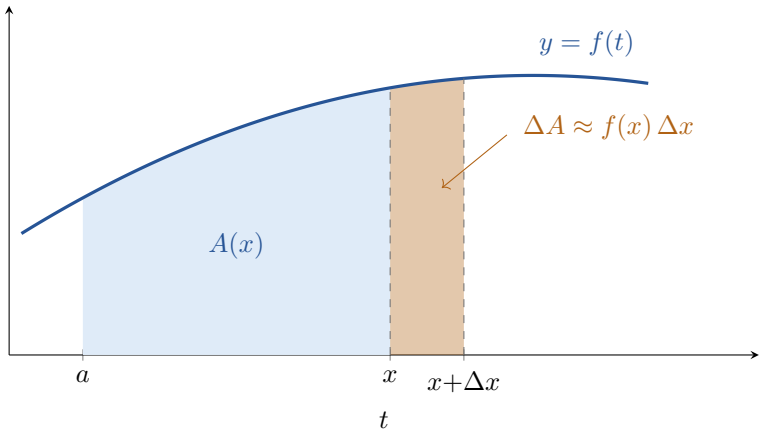


Figure 3.5: The accumulation function and the reason Part 1 is true. Widening the region by Δx adds a sliver which is very nearly a rectangle of height $f(x)$ and width Δx. So $\Delta A \approx f(x)\,\Delta x$, which says $A'(x) = f(x)$: the rate at which area accumulates is the height of the graph at the leading edge.

The sliver is almost a rectangle — height $f(x)$, width Δx — and the only thing spoiling it is the small amount by which the top of the curve is not flat over that short stretch. So $\Delta A \approx f(x)\,\Delta x$, which is the statement $A'(x) = f(x)$ in the language of Theorem 2.4. Differentiating an accumulation returns the thing being accumulated.

3.10 Theorem *fundamental theorem of calculus, part 1*

Let f be continuous on $[a, b]$ and define $A(x) = \int_a^x f(t)\,dt$ for x in $[a, b]$. Then A is differentiable and

$$A'(x) = f(x) \qquad \text{for every } x \text{ in } (a, b).$$

Proof Take $\Delta x > 0$ (the case $\Delta x < 0$ is the same argument with the inequalities read in the other direction). By Proposition 3.3(b),

$$\Delta A = A(x + \Delta x) - A(x) = \int_x^{x+\Delta x} f(t)\,dt,$$

so the sliver really is the integral over the short interval. Now let $\varepsilon > 0$. Because f is continuous at x, there is a $\delta > 0$ such that $|f(t) - f(x)| < \varepsilon$ whenever $|t - x| < \delta$ (Definition 1.12). So for

$0 < \Delta x < \delta$ the values of f on $[x, x+\Delta x]$ all lie between $f(x)-\varepsilon$ and $f(x)+\varepsilon$, and part (c) of Proposition 3.3 gives

$$(f(x)-\varepsilon)\Delta x \;\le\; \Delta A \;\le\; (f(x)+\varepsilon)\Delta x.$$

Dividing by $\Delta x > 0$,

$$\left|\frac{\Delta A}{\Delta x} - f(x)\right| \le \varepsilon.$$

Since ε was arbitrary, $\Delta A/\Delta x \to f(x)$, which is exactly $A'(x) = f(x)$. ■

Why there are two parts, and why both are obvious in hindsight

The two halves of the Fundamental Theorem are the same sentence read in the two possible directions.

Part 1 says: *accumulate a rate, then differentiate, and you get the rate back.* Start with a speedometer reading, build the odometer from it, and the odometer's rate of change is the speedometer again.

Part 2 says: *differentiate a quantity, then accumulate the result, and you get the total change in the quantity back.* Start with the odometer, read off the speedometer, and adding up what the speedometer said reproduces how far you went.

Each says that accumulation and differentiation undo each other; they differ only in which one you do first. That is what it means for two operations to be inverse, and the reason it took a genius to notice is that nobody expected the tangent problem and the area problem to be related at all.

Part 1 also settles the existence question left open above: for continuous f, an antiderivative always exists, because $A(x) = \int_a^x f$ is one. That is not an idle remark. Some perfectly innocent-looking integrands have antiderivatives that cannot be assembled out of powers, roots, exponentials, logarithms and trigonometric functions — $\int e^{-x^2}dx$ is the famous case — and Part 1 assures us the function exists anyway. It is a respectable function with a name (statisticians call a scaled version of it the error function) and a computer will evaluate it to any accuracy you like. It simply is not expressible in the vocabulary we started with.

3.6 Antiderivatives

3.11 Definition *antiderivative and indefinite integral*

F is an **antiderivative** of f on an interval if $F'(x) = f(x)$ throughout it. By Corollary 3.7 any two antiderivatives of f differ by a constant, so they are all captured by writing

$$\int f(x)\,dx \;=\; F(x) + C,$$

called the **indefinite integral** of f. Unlike $\int_a^b f(x)\,dx$, which is a number, this is a family of functions.

Every differentiation rule from Chapter 2 can now be read backwards.

3.12 Theorem *antiderivatives of powers and the basic functions*

For $n \neq -1$,

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C,$$

and

$$\int \cos x \, dx = \sin x + C, \qquad \int \sin x \, dx = -\cos x + C.$$

Antidifferentiation is also linear: $\int [c\,f + g] = c \int f + \int g$.

Proof Differentiate each proposed answer and check it reproduces the integrand. For the first, $\dfrac{d}{dx}\left[\dfrac{x^{n+1}}{n+1}\right] = x^n$ by Theorem 2.9; the restriction $n \neq -1$ is there because otherwise we would be dividing by zero. The other two are Theorem 2.23 and Corollary 2.24 read in reverse, and linearity is Theorem 2.9(b),(c) read the same way. ■

The missing case $n = -1$ is not an oversight, and no power of x will ever fix it: the antiderivative of $1/x$ is a genuinely new function, the natural logarithm. We will build it when we need it — and we will need it, since it governs radioactive decay, cooling, and the charging of a capacitor.

Notice that checking an antiderivative is always easy, because differentiation is mechanical: you differentiate your answer and see whether you get the integrand back. Integration is the harder direction — by hand it is a toolkit and practice rather than a procedure that always works — but verification is free, so use it.

3.13 Example *a few antiderivatives*

$$\int (3x^2 - 4x + 5)\, dx = x^3 - 2x^2 + 5x + C,$$

and, evaluating a definite integral with Theorem 3.8,

$$\int_0^{\pi} \sin x \, dx = \big[-\cos x\big]_0^{\pi} = -\cos \pi + \cos 0 = 1 + 1 = 2.$$

3.7 Substitution

The chain rule says $\dfrac{d}{dx}[(x^2+1)^5] = 5(x^2+1)^4(2x)$. So without doing any work at all, what is $\displaystyle\int 10x\,(x^2+1)^4\, dx$? Look for the chain rule hiding inside the integrand.

Every differentiation rule run backwards becomes an integration technique, and the chain rule — the most useful of them — becomes the most useful of these.

3.14 Theorem *substitution*

Let $u = g(x)$ be differentiable with g' continuous, and let f be continuous on the range of g. Then

$$\int f(g(x))\, g'(x)\, dx \;=\; \int f(u)\, du,$$

where the right side is evaluated and then u replaced by $g(x)$. For a definite integral the limits travel with the substitution:

$$\int_a^b f(g(x))\, g'(x)\, dx \;=\; \int_{g(a)}^{g(b)} f(u)\, du.$$

Proof Let F be an antiderivative of f, which exists by Theorem 3.10. By the chain rule (Theorem 2.14),

$$\frac{d}{dx}\Big[F(g(x))\Big] = F'(g(x))g'(x) = f(g(x))g'(x),$$

so $F(g(x))$ is an antiderivative of the left integrand, which is the first statement. For the second, apply Theorem 3.8 twice:

$$\int_a^b f(g(x))g'(x)\, dx = \Big[F(g(x))\Big]_a^b = F(g(b)) - F(g(a)) = \int_{g(a)}^{g(b)} f(u)\, du. \qquad \blacksquare$$

In practice nobody thinks about it that way. With differentials (Definition 2.5) the rule is pure bookkeeping: write $u = g(x)$, write $du = g'(x)\, dx$, and substitute both into the integral until no x remains. This is the second time the differential has paid for itself, and it is why the dx in $\int f(x)\, dx$ had better not be treated as decoration.

3.15 Example *undoing the chain rule*

Compute $\displaystyle\int 10x\,(x^2+1)^4\, dx$.

Solution Put $u = x^2 + 1$. Then $du = 2x\, dx$, so $10x\, dx = 5\, du$, and

$$\int 10x\,(x^2+1)^4\, dx = \int 5u^4\, du = u^5 + C = (x^2+1)^5 + C.$$

Check by differentiating: $5(x^2+1)^4(2x) = 10x(x^2+1)^4$, the integrand. ✓

3.16 Example *a definite integral by substitution*

Compute $\displaystyle\int_0^2 x\sqrt{x^2+1}\, dx$.

Solution Put $u = x^2 + 1$, so $du = 2x\, dx$ and $x\, dx = \tfrac{1}{2}\, du$. The limits travel too: $x = 0$ gives $u = 1$, and $x = 2$ gives $u = 5$. Hence

$$\int_0^2 x\sqrt{x^2+1}\, dx = \frac{1}{2}\int_1^5 u^{1/2}\, du = \frac{1}{2}\left[\frac{2}{3}u^{3/2}\right]_1^5 = \frac{1}{3}(5^{3/2} - 1) \approx 3.39.$$

Because the limits were converted, there is no need to change back to x at the end.

3.8 Integrals in Physics: Adding Up the Little Bits

Section 2.10 gave a recipe for recognising when to differentiate. Here is the matching recipe for integration, and it is the one to carry into Part II.

> Whenever a total is built out of contributions, each of which is (something) × (a small chunk), the total is an integral of that something over those chunks.

Displacement is velocity times a chunk of time, so displacement is $\int v\,dt$ — and distance travelled is $\int |v|\,dt$, as Figure 3.3 insisted. The pattern repeats everywhere: mass is density times a chunk of volume, so mass is $\int \rho\,dV$; charge is charge density times volume; the total force on a dam is pressure times a chunk of area. In each case the move is the same — chop the thing into pieces small enough that the varying quantity is effectively constant on each, multiply, and add.

The most important instance for us is work.

3.17 Example *work done by a varying force*

For a constant force F pushing an object a distance d along its line of action, the work done is $W = Fd$. What if the force changes as the object moves, as it does for a spring?

Solution Chop the path from $x = a$ to $x = b$ into steps of width Δx small enough that F barely changes across each one. On the i-th step the work is approximately $F(x_i)\,\Delta x$, and the total is

$$W \approx \sum_i F(x_i)\,\Delta x \;\longrightarrow\; W = \int_a^b F(x)\,dx.$$

For a spring, Hooke's law says the force needed to hold the spring at extension x is $F(x) = kx$. Stretching it from 0 to X therefore takes

$$W = \int_0^X kx\,dx = \left[\tfrac{1}{2}kx^2\right]_0^X = \tfrac{1}{2}kX^2,$$

which you may recognise as the potential energy stored in a stretched spring. It is not a formula to memorise; it is the area under the straight line $F = kx$, a triangle of base X and height kX.

Compare this with Example 3.4, where $\int_0^1 9.8t\,dt$ was also a triangle. Both are integrals of a quantity proportional to the variable, so both give "$\frac{1}{2}$ times the final value times the range". Seeing that the same shape answers a kinematics question and an energy question is the beginning of noticing that physics has far fewer ideas in it than formulas.

There is a converse habit worth building as well: when you meet a physical formula with a $\frac{1}{2}$ and a square in it — $\frac{1}{2}kX^2$, $\frac{1}{2}mv^2$, $\frac{1}{2}at^2$ — suspect that somebody integrated something linear. The $\frac{1}{2}$ is almost always the area of a triangle, which is to say $\int_0^X x\,dx = X^2/2$ in disguise.

3.9 Chapter Recap

We wanted to run Chapter 2 backwards: given a rate, recover the total change. The whole chapter followed from one line borrowed from Theorem 2.4, $\Delta f \approx f'(x)\,\Delta x$, used many times

over and added up. Two facts about that sum did the work: the little changes telescope to the total change exactly, and each term is a height times a width, hence an area. That is why the answer to a question about accumulation turned out to be the area under a graph, and it is why $\int$ is a stretched-out $\sum$.

Definition 3.2 made the limit of those sums precise. The mean value theorem (Theorem 3.6) — the formal version of "your average speed was achieved at some instant" — removed the approximation, turning the bookkeeping identity into the Fundamental Theorem (Theorems 3.10 and 3.8): accumulating a rate and differentiating a quantity are inverse operations, so an integral can be computed by finding an antiderivative and subtracting its values at the endpoints. Substitution (Theorem 3.14) then turned the chain rule around, and differentials made that manipulation literal rather than symbolic.

The first three chapters, together

Look back at what these three chapters were for. We built limits because instantaneous quantities cannot be measured over an interval and yet plainly exist. We built derivatives because that limit, applied to a ratio of changes, both defines those instantaneous quantities and — read the other way — promises that a small change is predictable by multiplication. We built integrals because physics hands us rates and wants totals.

Every one of those ideas was driven by a falling ball, and every one of them will now be used on problems where nothing falls at all.

Newton's second law, when we finally meet it, will turn out to be a statement about a second derivative. Where the force is known as a function of time, recovering the motion really is a matter of integrating twice, and this chapter is all the machinery required. Where the force depends on *where the object is* — a spring, a planet, a charge — something new is needed, because then the equation refers to the very function we are trying to find. That is Chapter 8, and the rest of Part I is the road to it.

Problem 3.1. Use a Riemann sum with right endpoints, as in Example 3.5, to compute $\int_0^2 3x\,dx$ directly from Definition 3.2. Then check your answer with Theorem 3.8, and again by computing the area of a triangle.

Problem 3.2. Find $\displaystyle\int (4x^3 - \sin x)\,dx$, and verify your answer by differentiating it.

Problem 3.3. Evaluate $\displaystyle\int_0^3 (t^2 - 4t + 5)\,dt$ using Theorem 3.8.

Problem 3.4. Use the substitution $u = 3x + 1$ to compute $\displaystyle\int (3x+1)^6\,dx$, and $u = \cos x$ to compute $\displaystyle\int \sin x\,\cos^3 x\,dx$.

Problem 3.5. A rocket's velocity is $v(t) = 6t^2$ m/s for the first 5 seconds of flight. Set up and evaluate an integral for the distance travelled, and explain in one sentence why that distance is an area.

Problem 3.6. An object moves with velocity $v(t) = 4 - 2t$ m/s for $0 \le t \le 4$. Compute $\int_0^4 v\,dt$ and interpret the answer. Then compute the total distance travelled, which is *not* the same number, and explain the difference using Figure 3.3.

Problem 3.7. A spring with $k = 200$ N/m is stretched from 0 to 0.1 m. Find the work done. Then find the work done stretching it from 0.1 m to 0.2 m, and explain why the second answer is three times the first even though the distance is the same.

Chapter 4

Exponentials and Logarithms

Twice already this book has written a cheque it could not cash.

In Chapter 2 we proved the power rule $\frac{d}{dx}[x^n] = nx^{n-1}$ for whole numbers n, then immediately used it with $n = -\frac{1}{2}$ to expand the relativistic energy, promising to justify that later. In Chapter 3 the reverse power rule came with an exclusion, $n \neq -1$, and a note that the antiderivative of $1/x$ is a function we had not yet built.

Both debts are paid by the same function, and it is worth seeing where it comes from, because the usual presentation — "let $e = 2.71828\ldots$, and take my word for it that this is special" — explains nothing at all. The honest route is the historical one, and it starts with a problem that has nothing to do with calculus.

4.1 A Problem in Arithmetic

In 1614 John Napier published logarithm tables to ease a problem that consumed astronomers' time: long calculations done by hand.[6]

The problem was multiplication. An astronomer reducing observations multiplied and divided seven- and eight-digit numbers all day long, by hand, and a single slip in the fourth digit silently ruined a night's work. Addition is easy and safe; multiplication is neither. What Napier wanted was a way to *trade one for the other*: a function L with the property

$$L(ab) = L(a) + L(b).$$

Given a table of such a function and its inverse, any multiplication becomes a look-up, an addition, and another look-up. Kepler seized on the tables the moment he saw them and used logarithms throughout the computation of his *Rudolphine Tables* of 1627. The slide rule is nothing but that identity carved into wood.

> Before reading on: can you think of *any* function that turns multiplication into addition? Try a few. What would its graph have to look like — what does the property $L(ab) = L(a)+L(b)$ say about how L responds when you double the input, then double it again?

Doubling repeatedly adds the same amount each time. That is the whole character of a logarithm: *equal multiplications produce equal additions*. We are writing the idea in its modern normalization; Napier's own tables used a different scale and starting point, and he built them not by blind arithmetic but from a kinematic model of two points moving along lines, one at constant speed and one slowing in proportion to the distance it had left — a model which, read with the eyes of Chapter 8, is already a differential equation. For some decades afterwards the logarithm was regarded chiefly as a computational aid.

Then someone measuring an area noticed that he had built one by accident.

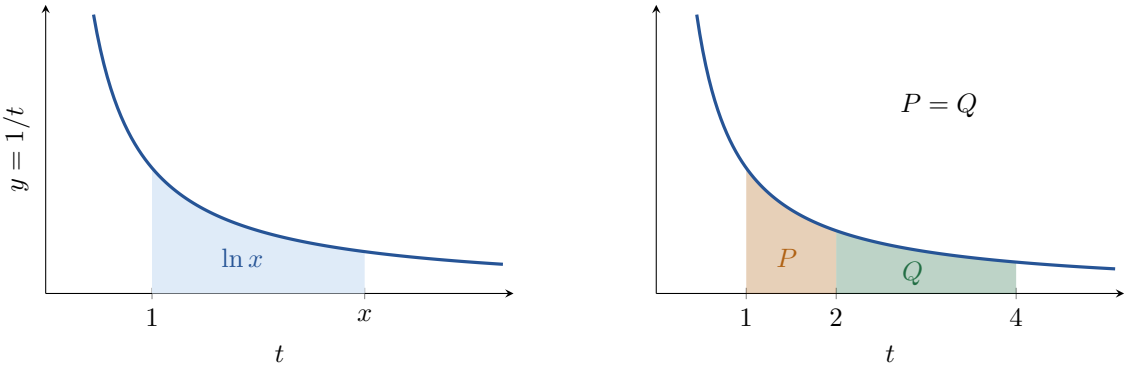
4.2 A Problem in Quadrature

Recall from Chapter 3 that quadrature — finding the area under a curve — was a major problem of the age. Cavalieri, Fermat and others found quadratures for many power curves. In our modern notation their results point toward the formula

$$\int_1^x t^n \, dt = \frac{x^{n+1} - 1}{n+1},$$

In modern terms this formula works for every real n except one, for $x > 0$. At $n = -1$ it divides by zero. The single curve that resisted was the plainest of all: the hyperbola $y = 1/t$.

In work published in 1647, Grégoire de Saint-Vincent established a striking property of hyperbolic areas. Alphonse Antonio de Sarasa connected this property with logarithms.[7] Look at the picture.



(a) the area *is* the logarithm

(b) twice as wide, half as tall

Figure 4.1: The hyperbola $y = 1/t$. On the right, the strip Q over $[2, 4]$ is twice as wide as the strip P over $[1, 2]$ — and because the curve is $1/t$, it is also exactly half as tall everywhere. The two effects cancel and the areas are equal. Stretching the interval by any factor leaves the area unchanged, which means that *multiplying* the endpoint *adds* a fixed amount of area. That is Napier's property, hiding under a curve.

This is worth pausing over, because it is the whole chapter in one picture. Take the region under $y = 1/t$ from 1 to 2, and now stretch it horizontally by a factor of 2: it lands on $[2, 4]$. Stretching doubles every width. But the height of the hyperbola at $2t$ is $1/(2t)$, exactly half the height at t. Doubled widths, halved heights, unchanged area. Any curve $y = C/t$ has this scaling property, for the same reason; fixing the height to be 1 at $t = 1$ picks out the particular one, $y = 1/t$, and it is the reciprocal power that matters, not the constant.

So define the area to be a function and see what we have.

4.1 Definition *the natural logarithm*

For $x > 0$,

$$\ln x \;=\; \int_1^x \frac{dt}{t},$$

the signed area under the hyperbola from 1 to x. (For $x < 1$ the integral runs backwards and $\ln x$ is negative; $\ln 1 = 0$.)

4.2 Theorem *the logarithm turns products into sums*

For all $a, b > 0$,

$$\ln(ab) = \ln a + \ln b.$$

Proof Split the area at a, using Proposition 3.3(b):

$$\ln(ab) = \int_1^{ab} \frac{dt}{t} = \int_1^{a} \frac{dt}{t} + \int_a^{ab} \frac{dt}{t} = \ln a + \int_a^{ab} \frac{dt}{t}.$$

It remains to show the second piece is $\ln b$, which is the stretching argument made algebraic. Substitute $t = au$, so $dt = a\,du$ (Theorem 3.14); when $t = a$, $u = 1$, and when $t = ab$, $u = b$. Then

$$\int_a^{ab} \frac{dt}{t} = \int_1^{b} \frac{a\,du}{au} = \int_1^{b} \frac{du}{u} = \ln b.$$ ■

Notice how the a cancelled: once in the width $dt = a\,du$, and once in the height $1/t = 1/(au)$. That cancellation is *Figure 4.1(b), and it is why no other power of t would have worked. For $y = t^n$ the stretch multiplies areas by a^{n+1}, which is a harmless constant only when $n = -1$.*

So the curve that refused to fit the quadrature formula turned out to be the curve that computes Napier's tables. A gap in a list of areas and a labour-saving device for astronomers led to the same mathematical structure. The connection joins two different problems without requiring them to have begun together.

4.3 The Derivative, for Free

Definition 4.1 says ln is an accumulation function — the running area under $1/t$ — and Chapter 3 told us exactly what happens when you differentiate one of those.

4.3 Theorem *derivative of the logarithm*

For $x > 0$,

$$\frac{d}{dx}[\ln x] = \frac{1}{x}.$$

Proof ln is the accumulation function of the continuous function $1/t$ on $(0, \infty)$, so this is Theorem 3.10 applied verbatim. ■

That is a remarkable return on investment: a two-line proof, resting entirely on machinery we already had. And it pays the first debt.

4.4 Corollary *the missing antiderivative*

$$\int \frac{dx}{x} = \ln|x| + C.$$

Proof For $x > 0$ this is Theorem 4.3 read backwards. For $x < 0$, write $|x| = -x$ and use the chain rule: $\dfrac{d}{dx}\ln(-x) = \dfrac{-1}{-x} = \dfrac{1}{x}$ as well. ■

The exception in Theorem 3.12 was therefore not a flaw in the formula but a signpost. Every power of x integrates to another power, except one, and that one integrates to something new — a function lying outside the reach of algebra altogether, since no combination of powers and roots has derivative $1/x$. (Logarithm *tables* had existed since 1614, of course. What was new was that the table and the area turned out to be the same object.)

4.4 Differentiating an Inverse

We now want to run the logarithm backwards, and to do that we need a small technique that is useful far beyond this chapter.

Suppose two quantities are tied together by an equation rather than by a formula solved for one of them — $x^2 + y^2 = 25$, say, or $\ln y = x$. We can still differentiate, provided we remember that y is a function of x and apply the chain rule every time y appears. This is called **implicit differentiation**, and it is nothing more than the chain rule taken seriously.

4.5 Example *a point moving on a circle*

A point moves along $x^2 + y^2 = 25$. Find $\dfrac{dy}{dx}$ when $x = 3$, $y > 0$.

Solution Differentiate both sides with respect to x. The term y^2 is a function of y, which is a function of x, so Theorem 2.14 gives $\dfrac{d}{dx}[y^2] = 2y\,\dfrac{dy}{dx}$:

$$2x + 2y\,\frac{dy}{dx} = 0 \qquad \Longrightarrow \qquad \frac{dy}{dx} = -\frac{x}{y}.$$

At $x = 3$ we have $y = 4$, so the slope is $-\frac{3}{4}$. We never solved for y; we differentiated the constraint itself. This is the standard move for *related rates* — two physical quantities tied by a geometric constraint, each changing in time — and we will use it constantly in Part II.

The same trick inverts any function whose derivative we know.

4.6 Proposition *derivative of an inverse function*

If f is differentiable with $f' \neq 0$, and g is its inverse, then

$$g'(x) = \frac{1}{f'(g(x))}.$$

Proof The tempting one-line proof is to differentiate $f(g(x)) = x$ with the chain rule and divide. That is circular: the chain rule may only be applied to g once we know g is differentiable, which is the larger half of what we are trying to prove. So argue from the ratio directly.

Work on an interval where f is one-to-one with $f' \neq 0$. Its inverse g is then one-to-one and continuous. Fix x, put $y = g(x)$, and let $\Delta y = g(x + \Delta x) - g(x)$. Continuity of g makes $\Delta y \to 0$ as $\Delta x \to 0$, and injectivity makes $\Delta y \neq 0$ whenever $\Delta x \neq 0$ — so we are allowed to divide by it.

Since $f(y) = x$ and $f(y + \Delta y) = x + \Delta x$, we have $\Delta x = f(y + \Delta y) - f(y)$, and therefore

$$\frac{\Delta y}{\Delta x} = \left(\frac{f(y + \Delta y) - f(y)}{\Delta y} \right)^{-1}.$$

As $\Delta x \to 0$, so does Δy, and the bracket tends to $f'(y) \neq 0$. Hence the left side has a limit, which is what "g is differentiable" means, and that limit is $1/f'(y) = 1/f'(g(x))$. ■

One fact is being used without proof: a continuous one-to-one function on an interval has a continuous inverse. It is believable — reflecting a gapless graph across $y = x$ gives a gapless graph — and, like the intermediate and extreme value theorems, it rests on the completeness of the real numbers. We assume it here.

Geometrically the formula is obvious, and worth seeing that way: the graph of the inverse is the graph of f reflected across the line $y = x$, and reflecting swaps rise with run. A slope of 3 becomes a slope of $\frac{1}{3}$. The content of the proof above is not the formula but the permission to write it down.

Two inverses we will need

The proposition earns its keep at once on the inverse trigonometric functions, which look forbidding and turn out to have two of the friendliest derivatives in the subject.

4.7 Proposition *derivatives of arcsine and arctangent*

For $-1 < x < 1$,

$$\frac{d}{dx}[\arcsin x] = \frac{1}{\sqrt{1 - x^2}},$$

and for every real x,

$$\frac{d}{dx}[\arctan x] = \frac{1}{1 + x^2}.$$

Consequently

$$\int \frac{dx}{\sqrt{1 - x^2}} = \arcsin x + C \quad \text{on } (-1, 1), \qquad \int \frac{dx}{1 + x^2} = \arctan x + C.$$

Proof Take $f = \sin$ restricted to $[-\pi/2, \pi/2]$, where it is one-to-one and $f' = \cos$ is positive on the interior; its inverse is arcsin, defined on $[-1, 1]$. Proposition 4.6, with $\theta = \arcsin x$, gives $\frac{d}{dx}[\arcsin x] = 1/\cos\theta$. Now $\sin\theta = x$ and θ lies in $[-\pi/2, \pi/2]$, where cosine is never negative, so $\cos\theta = +\sqrt{1 - x^2}$. The sign is forced by the interval we restricted to, and choosing it carelessly is the standard way to lose a minus sign here.

For arctangent, take $f = \tan = \sin / \cos$ on $(-\pi/2, \pi/2)$. The quotient rule (Theorem 2.12) gives

$$f'(\theta) = \frac{\cos^2\theta + \sin^2\theta}{\cos^2\theta} = \frac{1}{\cos^2\theta} = 1 + \tan^2\theta,$$

which at $\theta = \arctan x$ is $1 + x^2$; the proposition gives its reciprocal. ■

Look at what those two integrals are saying. A purely algebraic integrand, $1/(1+x^2)$, has an angle *for its antiderivative. That is not a curiosity — it is why angles appear unbidden in the middle of electrostatics and orbital mechanics, in calculations whose setup never mentioned a triangle. Chapter 6 will use both of these constantly.*

With those in hand we can return to the main thread.

4.5 The Number e

ln is increasing everywhere, since its derivative $1/x$ is positive for $x > 0$. It starts at $\ln 1 = 0$, and it is unbounded above: by Theorem 4.2, $\ln 2^n = n \ln 2$, which grows without limit. (Likewise $\ln 2^{-n} = -n \ln 2 \to -\infty$.) Being continuous, it takes every real value exactly once — Theorem 1.13 guarantees it hits everything in between.

In particular it hits 1, exactly once, and that one input deserves a name.

4.8 Definition *the number e and the exponential function*

e is the unique number with

$$\ln e = 1, \qquad \text{that is,} \qquad \int_1^e \frac{dt}{t} = 1.$$

Its value is $e = 2.718281828\ldots$. The inverse of ln is called the **exponential function** and written $\exp(x)$ or e^x.

This is an honest definition of e: the point you have to walk out to under the hyperbola before you have swept out one unit of area. Having chosen it, the other familiar descriptions of e — the limit of $(1+\frac{1}{n})^n$, the sum $1+1+\frac{1}{2}+\frac{1}{6}+\cdots$, the base a bank would use if it compounded interest continuously — become theorems about this number rather than definitions of it. One could equally start from any of those and prove its equivalence to this one; what is not honest is to start from the decimal $2.71828\ldots$ and claim to have explained anything.

Writing e^x for $\exp(x)$ is a small act of faith, since so far exp is just "the inverse of an area function" and has nothing obviously to do with raising anything to a power. It is justified by Theorem 4.2: applying ln to $\exp(a)\exp(b)$ gives $a + b$, so $\exp(a)\exp(b) = \exp(a + b)$, which is exactly the rule powers obey. The function multiplies when you add, so it behaves like a power, so we write it like one.

4.9 Theorem *the function that is its own derivative*

$$\frac{d}{dx}\left[e^x\right] = e^x.$$

Proof exp is the inverse of ln, and $\ln'(t) = 1/t$ is never zero on $(0, \infty)$. So Proposition 4.6 applies, telling us first that exp is differentiable at all — which we were not entitled to assume — and then that

$$\frac{d}{dx}\left[\exp x\right] = \frac{1}{\ln'(\exp x)} = \frac{1}{1/\exp x} = \exp x. \qquad \blacksquare$$

In practice nobody computes it that way; they differentiate implicitly. Put $y = e^x$, so that $\ln y = x$, and differentiate both sides with respect to x: the chain rule gives $\frac{1}{y}\frac{dy}{dx} = 1$, hence $\frac{dy}{dx} = y$. That is the same argument with the bookkeeping hidden, and it is legitimate only because the proposition has already guaranteed that $\frac{dy}{dx}$ exists.

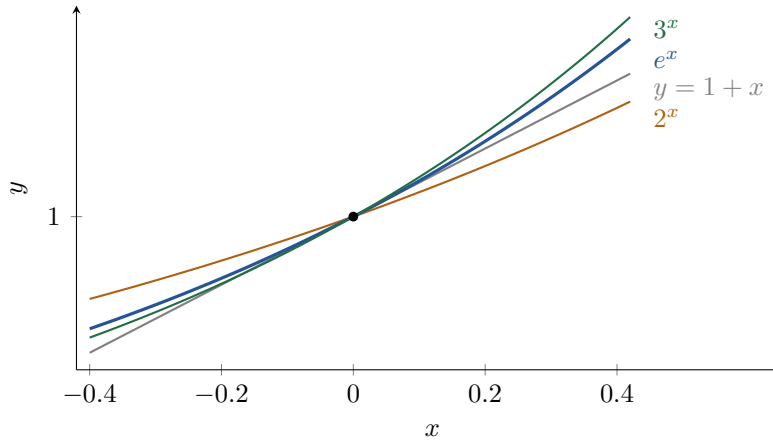


Figure 4.2: What makes e the natural base. Every exponential curve passes through $(0, 1)$, but they leave it at different angles. Here 2^x departs too shallowly and 3^x too steeply; only e^x leaves with slope exactly 1, hugging the line $y = 1 + x$. By Theorem 2.4 that says $e^{\Delta x} \approx 1 + \Delta x$ for small Δx — and a function whose slope equals its height at $x = 0$ turns out to have that property everywhere.

Theorem 4.9 is the most consequential sentence in this chapter. Read it as a statement about growth: *the rate at which e^x grows is equal to how big it already is.* Interest earns interest. A population's growth is proportional to its size. A radioactive sample decays at a rate proportional to how much is left. Every one of those sentences is an equation about a function and its own derivative, and this is the function that satisfies it.

4.10 Proposition *growth and decay*

For any constant k, the function $y = Ce^{kt}$ satisfies

$$\frac{dy}{dt} = k\,y,$$

and every solution of that equation has this form.

Proof The chain rule gives $\frac{d}{dt}[Ce^{kt}] = Ce^{kt} \cdot k = ky$. For the converse, suppose $\frac{dy}{dt} = ky$ and consider $u = y\,e^{-kt}$. By the product rule,

$$\frac{du}{dt} = \frac{dy}{dt}e^{-kt} - k\,y\,e^{-kt} = ky\,e^{-kt} - ky\,e^{-kt} = 0,$$

so u is constant by Corollary 3.7, say $u = C$, and $y = Ce^{kt}$. ∎

That converse matters more than it looks. It says e^{kt} is not merely *a* function with proportional growth — it is the *only* one. When we meet this equation in physics, and we will meet it

constantly, we will not have to search for solutions. There is exactly one family.

4.11 Example *half-life*

A radioactive sample decays at a rate proportional to the amount present, so $\dfrac{dN}{dt} = -\lambda N$ with $\lambda > 0$, giving $N(t) = N_0 e^{-\lambda t}$. How long until half is gone?

Solution Solve $N_0 e^{-\lambda t} = \frac{1}{2} N_0$, that is $e^{-\lambda t} = \frac{1}{2}$. Taking ln of both sides, $-\lambda t = \ln \frac{1}{2} = -\ln 2$, so

$$t_{1/2} = \frac{\ln 2}{\lambda} \approx \frac{0.693}{\lambda}.$$

Notice what is *not* in that answer: N_0. The time to halve does not depend on how much you started with, which is why "half-life" is a well-defined property of a substance rather than of a sample. That is the signature of exponential decay, and it is Napier's property again — equal multiplications take equal times.

4.6 Other Bases, and the Last Debt Paid

Everything else follows by writing it in terms of e.

4.12 Definition *general powers*

For $a > 0$ and any real x, define

$$a^x \;=\; e^{x \ln a}.$$

For $a > 0$ with $a \neq 1$, and for $x > 0$, define

$$\log_a x \;=\; \frac{\ln x}{\ln a}.$$

The restrictions are not pedantry. At $a = 1$ we would be dividing by $\ln 1 = 0$, which is the algebraic shadow of the fact that 1^x is stuck at 1 and has no inverse to speak of. And $\log_a x$ needs $x > 0$ because $\ln x$ does: the integral defining it starts at 1, and cannot cross the singularity at 0. The hyperbola has a negative branch too; on an interval entirely to the left of zero, $\ln|x|$ is an antiderivative of $1/x$.

For rational x this agrees with the meaning you already knew — $a^{1/2}$ really is the square root — and for irrational x it is a definition, because "multiply a by itself $\sqrt{2}$ times" has no independent meaning. Differentiating with the chain rule gives

$$\frac{d}{dx}\left[a^x\right] = a^x \ln a,$$

which explains Figure 4.2 completely: the slope at $x = 0$ is $\ln a$, and that equals 1 exactly when $a = e$.

And now the second debt.

4.13 Theorem *the power rule, at last, for every exponent*

For every real number n and every $x > 0$,

$$\frac{d}{dx}\left[x^n\right] = n\,x^{n-1}.$$

Proof Write $x^n = e^{n\ln x}$ by Definition 4.12 and differentiate with the chain rule:

$$\frac{d}{dx}\left[e^{n\ln x}\right] = e^{n\ln x}\cdot\frac{n}{x} = x^n\cdot\frac{n}{x} = n\,x^{n-1}. \qquad \blacksquare$$

The expansion in Section 2.10, which used $n = -\frac{1}{2}$, now has its justification. Our general proof uses the logarithm, hence integration and the Fundamental Theorem. Particular exponents admit other routes: rationalising the square-root difference quotient and using the quotient rule handles $n = -\frac{1}{2}$ in Chapter 2. We deferred that case to give one proof for every real exponent.

4.7 Chapter Recap

The arithmetic problem and the hyperbolic-area problem meet in $\ln x$. Its area definition gives the product law by substitution and the derivative $1/x$ by the Fundamental Theorem. Its inverse, e^x, equals its own derivative and describes proportional growth and decay. Both deferred debts are now paid: the reciprocal antiderivative and the power rule for every real exponent. Next we improve the straight-line approximations these rules make possible.

Problem 4.1. Using only Theorem 4.2 and $\ln 1 = 0$, show that $\ln(1/b) = -\ln b$ and that $\ln(a/b) = \ln a - \ln b$. Then explain in one sentence what the first of these says about Figure 4.1.

Problem 4.2. Differentiate: (a) $\ln(3x^2+1)$; (b) $x\ln x - x$; (c) e^{-x^2}; (d) 5^x. For (b), what does your answer tell you about $\displaystyle\int \ln x\,dx$?

Problem 4.3. A capacitor discharges so that the charge obeys $\dfrac{dQ}{dt} = -Q/(RC)$. Write down $Q(t)$ given $Q(0) = Q_0$, and find the time at which the charge has fallen to $1/e$ of its initial value. (This time is called the time constant, and you have just discovered why.)

Problem 4.4. Carbon-14 has a half-life of 5730 years. A bone contains 34% of the carbon-14 it had when the animal died. How old is it? (Use Example 4.11 to get λ first.)

Problem 4.5. Use implicit differentiation on $x^2 + y^2 = L^2$ to relate $\dfrac{dx}{dt}$ and $\dfrac{dy}{dt}$ for a ladder of length L sliding down a wall. If the foot slides out at a constant 0.5 m/s, how fast is the top falling when the foot is 3 m from a wall and the ladder is 5 m long? As the top approaches the ground, the model predicts an unbounded downward speed. Say precisely which assumption breaks down, and what that tells you about trusting a model at the edge of its range.

Problem 4.6. Show that $\dfrac{d}{dx}\left[\ln(kx)\right] = \dfrac{d}{dx}\left[\ln x\right]$ for any constant $k > 0$, and explain this using Theorem 4.2 rather than by differentiating.

Chapter 5

Series and Better Approximations

Chapter 2 built a machine that replaces a curve by a straight line, and Section 2.10 showed how much physics you can do with nothing else. But we also know the machine's weakness exactly. The error in the linear approximation is $o(\Delta x)$, and Section 2.11 identified what that error mostly *is*: about $\frac{1}{2}f''(a)(\Delta x)^2$.

That is a strange situation to leave alone. We know the leading error term. Why not add it to the approximation and be rid of it?

5.1 Why Stop at a Line?

This is not a modern tidying-up. It is how Newton actually worked.

The young Newton's great trick, worked out in the plague years, was to turn awkward functions into infinite polynomials and then handle them term by term. He knew, as everyone did, how to integrate x^n. The area under $y = \sqrt{1-x^2}$ — the upper half of a circle — was hardly unknown territory; circles and circular segments had been measured since antiquity. What was missing was a *method* that produced those areas the way the power quadratures did, mechanically and on demand. Newton used a binomial expansion and term-by-term integration to calculate a circular-segment area and hence approximate π. Here $\sqrt{1-x^2}$ is our modern unit-circle version of that idea, rather than a transcription of his calculation.[9] The series came first and the calculus grew up around it.

> The best straight line through $(a, f(a))$ is the one whose *value* and *slope* match f there. If you wanted the best parabola, what would you ask it to match? And having decided that, how many conditions does a cubic need?

Match as much as you can. A line has two coefficients, so it can match the value and the first derivative. A parabola has three, so it can also match the second derivative — which, by Section 2.11, is precisely the quantity that controls how fast the line was going wrong. A cubic can match three derivatives. Let us do this in general and see what the coefficients have to be.

5.2 Matching Derivatives

Suppose we want a polynomial in powers of $(x-a)$,

$$P(x) = c_0 + c_1(x-a) + c_2(x-a)^2 + \cdots + c_n(x-a)^n,$$

whose first n derivatives at a agree with those of f. The coefficients are forced, and finding them is easy because every term except one dies when you set $x = a$.

Put $x = a$ in P: every term with a factor of $(x-a)$ vanishes, leaving $P(a) = c_0$. So $c_0 = f(a)$. Differentiate once and put $x = a$ again: $P'(x) = c_1 + 2c_2(x-a) + \cdots$, so $P'(a) = c_1$, giving $c_1 = f'(a)$ — we have recovered the tangent line. Differentiate twice: the $(x-a)^2$ term becomes

$2c_2$, so $c_2 = f''(a)/2$. Each further differentiation of $(x-a)^k$ pulls down a factor k, then $k-1$, and so on, so after k differentiations the k-th term has become the constant $k!\,c_k$.

5.1 Definition *Taylor polynomial*

The **Taylor polynomial of order** n for f about the point a is

$$P_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}\,(x-a)^k = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

It is the unique polynomial of degree at most n whose value and first n derivatives at a match those of f.

The factorials are not decoration and not a convention — they are there to undo the factors that differentiation throws off. Each differentiation of $(x-a)^k$ drops the exponent by one and multiplies by the old exponent; doing it k times multiplies by $k(k-1)\cdots 1 = k!$. Dividing by $k!$ in the coefficient cancels that exactly, so that what survives at $x = a$ is $f^{(k)}(a)$ and nothing else.

Notice that P_1 is the linear approximation of Chapter 2, so this is a genuine extension rather than a new idea. And P_2 adds precisely the term $\frac{1}{2}f''(a)(\Delta x)^2$ that we identified as the leading error — so the parabola should be right where the line was wrong.

How wrong is P_n? The answer has the same shape as everything else in this book: the first term you did *not* take.

5.2 Theorem *Taylor's theorem with remainder*

If f has $n+1$ continuous derivatives on an interval containing a and x, then

$$f(x) = P_n(x) + R_n(x), \qquad R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}\,(x-a)^{n+1}$$

for some c between a and x.

Proof sketch For $n = 0$ the statement reads $f(x) = f(a) + f'(c)(x-a)$, which is exactly the mean value theorem (Theorem 3.6). The general case is proved by applying the same idea repeatedly to the difference $f - P_n$, which vanishes together with its first n derivatives at a; each application of the mean value theorem peels off one order. We take the result on trust. ∎

So the error behaves like $(x-a)^{n+1}$: each extra term you keep buys you one more power of smallness. Count those powers carefully, because it is easy to be off by one. The remainder bound for the *linear* approximation P_1 carries $(\Delta x)^2$; for the quadratic P_2 it carries $(\Delta x)^3$; for the cubic P_3, $(\Delta x)^4$. Halving the step therefore divides those bounds by 4, by 8 and by 16 respectively, which is why a parabola beats a line by a margin that is dramatic rather than marginal. (Those are the guaranteed bounds. The actual errors approach those ratios as the step tends to zero when the leading derivative at the expansion point is nonzero and the required derivatives are continuous. At a finite step the ratios need not be exact.)

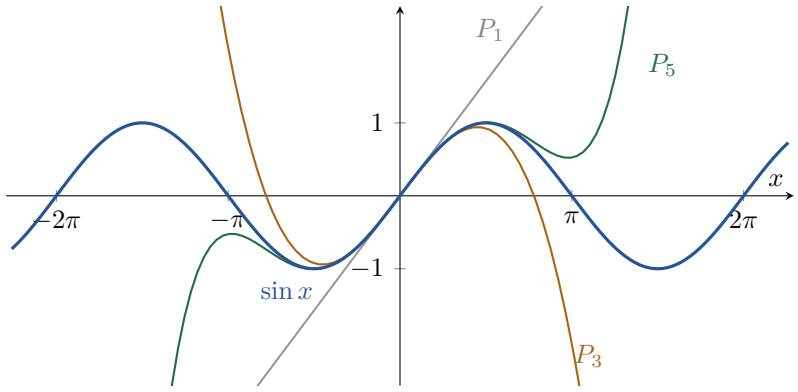


Figure 5.1: Taylor polynomials of $\sin x$ about the origin. $P_1 = x$ is the small-angle approximation of Chapter 1. Each higher polynomial clings to the sine curve further out before finally tearing away — and the place where it tears away moves steadily outwards. Keep going forever and it never tears away at all.

5.3 Infinite Series

Figure 5.1 suggests an obvious question: what if we never stop?

5.3 Definition *Taylor series*

The **Taylor series** of f about a is the infinite sum

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!}(x-a)^k,$$

formed from the derivatives of f at a, all of which are assumed to exist. When $a = 0$ it is often called a Maclaurin series.

Two different questions come attached to that definition, and running them together is the classic error. The series **converges** at x if its partial sums $P_n(x)$ have a finite limit. It **represents** f at x if that limit is $f(x)$. The second does not follow from the first. The standard warning is the function

$$f(x) = \begin{cases} e^{-1/x^2} & x \neq 0, \\ 0 & x = 0, \end{cases}$$

which is smooth everywhere and every one of whose derivatives at the origin works out to zero. Its Taylor series about 0 is $0 + 0x + 0x^2 + \cdots$: it converges perfectly, everywhere, to the wrong function. So whenever we write "$f(x) =$ " followed by a series, we are making a claim — that the remainder $R_n(x)$ of Theorem 5.2 tends to zero — and it is a claim to be checked, not assumed.

Three examples carry almost all the weight in physics. Each is a two-line computation once you notice the pattern in the derivatives.

5.4 Example *the exponential, sine and cosine series*

Expand e^x, $\sin x$ and $\cos x$ about 0.

Solution For $f(x) = e^x$, Theorem 4.9 says every derivative is e^x again, so $f^{(k)}(0) = 1$ for

every k and

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = \sum_{k=0}^{\infty} \frac{x^k}{k!}.$$

For $f(x) = \sin x$ the derivatives cycle $\cos, -\sin, -\cos, \sin$, so at 0 they run $1, 0, -1, 0, 1, \ldots$ — only odd powers survive:

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$$

Matching the derivatives of cosine at 0 in the same way — they run $1, 0, -1, 0, 1, \ldots$, so only even powers survive — gives

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$$

All three series really do represent their functions for every real x, and the check is short enough to do rather than assert. Fix x. Every derivative of e^x is e^x again, so on the interval between 0 and x we have $|f^{(n+1)}| \le e^{|x|}$, and Theorem 5.2 gives

$$|R_n(x)| \;\le\; e^{|x|}\,\frac{|x|^{n+1}}{(n+1)!};$$

for sine and cosine every derivative is bounded by 1, so the same bound holds without the factor $e^{|x|}$. Either way, dividing one bound by the one before it leaves a factor $|x|/(n+2)$, which falls below $\frac{1}{2}$ once n is large enough — and from that point on the bound at least halves at every step, so it tends to zero. The remainder is squeezed out and the equalities above are earned. Setting $x = 1$ in the first gives $e = 1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} + \cdots = 2.71828\ldots$, which is where that decimal expansion comes from.

Two things in that example deserve comment. First, $\cos x \approx 1 - x^2/2$ is the second-order correction promised in Section 2.10, where the first-order term vanished and we had to admit we could not yet say what replaced it. Second, the sine series begins with x — the small-angle approximation — and the next term tells you exactly what it costs: at $\theta = 0.1$ the correction is $\theta^3/6 \approx 1.7 \times 10^{-4}$, matching the table in Chapter 1 to the digit.

Not every series converges everywhere, and the simplest example is one we can check by hand.

5.5 Example *the geometric series, and a warning*

Expand $\dfrac{1}{1-x}$ about 0.

Solution Rather than differentiate repeatedly, use the telescoping trick of Chapter 3. For $x \neq 1$,

$$(1 - x)(1 + x + x^2 + \cdots + x^N) = 1 - x^{N+1},$$

since everything in the middle cancels. Hence

$$1 + x + \cdots + x^N = \frac{1 - x^{N+1}}{1 - x}.$$

If $|x| < 1$ then $x^{N+1} \to 0$ and the sum converges to $1/(1-x)$. If $|x| \ge 1$ it does not converge at all: the terms do not even shrink.

So a Taylor series need not represent its function everywhere; $1/(1-x)$ is perfectly well behaved at $x=3$, but its series about 0 is useless there. The general statement is this. A *power series* has a **radius of convergence** R, possibly zero and possibly infinite: it converges for $|x-a|<R$ and diverges for $|x-a|>R$. What happens at the two endpoints $|x-a|=R$ is not decided by R and must be tested separately — the logarithm series in this chapter's problems is exactly such a case, converging at one endpoint and diverging at the other. And convergence, as we have just seen, is still not the same thing as representing f.

Strictly inside the radius, though, the manipulations we want are legitimate: a power series may be differentiated and integrated term by term, and the result is a power series with the same radius. We will use that freely and not prove it, which is a debt worth naming since the next section and two of the problems lean on it.

In this book the series we use are applied to small quantities — small angles, small velocities compared with light, small displacements from equilibrium — so we will stay comfortably inside the safe region, but it is worth knowing the cliff is there.

5.4 Newton's Binomial Series

The expansion Newton needed for his circle is the general power, and it is the workhorse of physical approximation.

5.6 Theorem *binomial series*

For any real n and $|x|<1$,

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \cdots$$

Proof sketch Differentiate $f(x)=(1+x)^n$ repeatedly using Theorem 4.13: $f'=n(1+x)^{n-1}$, $f''=n(n-1)(1+x)^{n-2}$, and in general $f^{(k)}(0)=n(n-1)\cdots(n-k+1)$. Substituting into Definition 5.3 gives the stated coefficients. We take on trust here the remaining assertion: for $|x|<1$ this series converges *to* $(1+x)^n$, not merely to some function. ∎

For a nonnegative integer n the series stops, giving the ordinary binomial theorem used in Chapter 2. For every other real n it continues indefinitely: the extension to an infinite series is Newton's crucial idea.

5.7 Example *approximations worth memorising*

Keeping the constant term and two corrections in Theorem 5.6 gives, for small x,

$$\sqrt{1+x} \approx 1 + \tfrac{x}{2} - \tfrac{x^2}{8}, \qquad \frac{1}{\sqrt{1+x}} \approx 1 - \tfrac{x}{2} + \tfrac{3x^2}{8}, \qquad \frac{1}{1+x} \approx 1 - x + x^2.$$

The first-order parts are the ones Section 2.10 used; the quadratic parts tell you when to stop trusting them. For the relativistic factor $\gamma=(1-v^2/c^2)^{-1/2}$ the next term after $1+\frac{v^2}{2c^2}$ is $\frac{3v^4}{8c^4}$ — so the kinetic-energy formula $\frac12 mv^2$ is wrong by a relative amount of about $\frac34 v^2/c^2$, which at the Earth's orbital speed of 30 km/s is 7.5×10^{-9} — a little under one part in a hundred million.

5.5 Why Almost Every Valley Is a Spring

Here is the payoff, and it is the single most important idea in this chapter for everything that follows. It also comes with conditions, so we will state them first and then take them seriously afterwards.

Take a system described by a single coordinate x, with a potential energy $U(x)$ and a positive inertia, no friction and nothing driving it — a mass on a spring, a pendulum, two atoms in a molecule, a bead on a wire, a satellite near a stable orbit. Suppose U is smooth and has a **nondegenerate minimum** at x_0, meaning

$$U'(x_0) = 0 \qquad \text{and} \qquad U''(x_0) > 0.$$

Expand U about that point:

$$U(x) = U(x_0) + \underbrace{U'(x_0)}_{=0}(x - x_0) + \tfrac{1}{2}U''(x_0)(x - x_0)^2 + \cdots$$

The constant is just a choice of where to measure energy from and changes no physics. The linear term is *gone*, because we are at an equilibrium. So the first term that says anything is the quadratic one, and

$$U(x) \approx U(x_0) + \tfrac{1}{2}k\,(x - x_0)^2, \qquad k = U''(x_0).$$

That is the potential energy of a spring of stiffness k.

Now the conditions, because every one of them is doing work and the honest title of this section has the word *almost* in it.

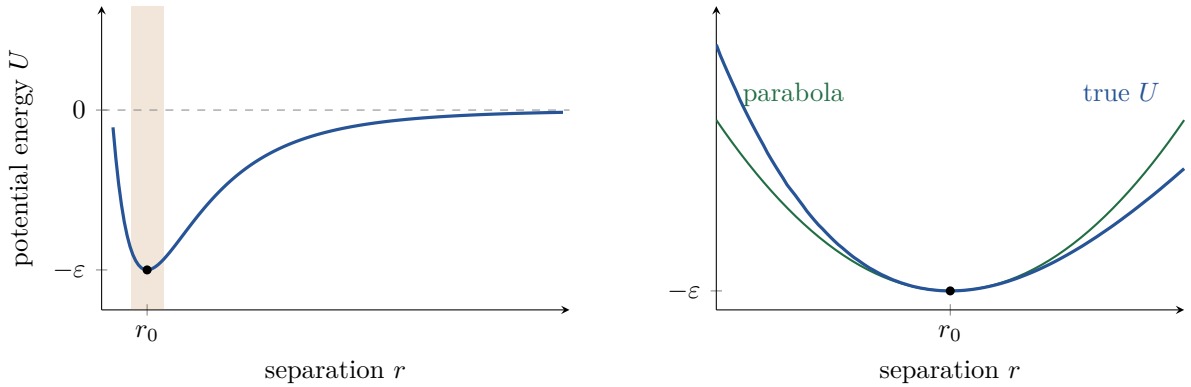
The assumption $U''(x_0) > 0$ is exactly that — an assumption, and not the definition of stability. The potential $U(x) = x^4$ has a perfectly stable minimum at the origin: push the bead off and it comes back. But $U''(0) = 0$, so the quadratic term is missing, the leading behaviour is quartic, and the resulting motion is not sinusoidal at all — its period depends on how far you pushed. Nondegenerate minima are overwhelmingly the common case, which is why the conclusion is so useful, but "stable" by itself does not deliver it.

The word *small* is load-bearing too. The parabola is the leading term and no more; Figure 5.2(b) shows the next correction already visible at modest displacements. And friction or an external drive changes the answer outright: a heavily damped system released from rest creeps back to equilibrium without ever oscillating, as Chapter 8 will show.

This is the same zooming picture as Figure 2.1, one order higher. There, magnifying a smooth curve made it look like a line; here, magnifying a smooth *minimum* makes it look like a parabola — because the linear term, which would otherwise dominate, has been killed off by the condition $U' = 0$.

Take the chain of reasoning seriously, because it explains an oddity of physics: why the same equation keeps appearing in unrelated places. A nondegenerate minimum, disturbed slightly, has an approximately quadratic potential; a quadratic potential gives a linear restoring force; and a linear restoring force gives sinusoidal motion. Springs, pendulums, atoms in a crystal, water in a U-tube, charge in an LC circuit — that they all obey the same equation is not a coincidence, it is Theorem 5.2 plus $U'(x_0) = 0$ plus $U''(x_0) > 0$.

The conclusion, with its conditions, is worth keeping: to leading order, an undamped one-coordinate system released from rest sufficiently close to a nondegenerate potential minimum behaves as a harmonic oscillator, with stiffness $U''(x_0)$. Chapter 8 derives the motion from the force law.



(a) the whole well

(b) the shaded strip, magnified

Figure 5.2: The potential energy of two atoms in a molecule, as a function of their separation — a steep repulsive wall, a well, and a long flat tail. It looks nothing like a parabola. But magnify the bottom of the well and the true curve and the fitted parabola become nearly indistinguishable, which is why a chemical bond, jostled gently, behaves approximately like a spring. The leading asymmetry here is cubic. Such anharmonicity helps explain ordinary thermal expansion: larger vibrations can have a shifted mean separation, although the behavior of a whole material needs more than this one-pair model.

5.8 Example *the pendulum is a spring*

A mass m hangs on a rigid rod of length L and swings through an angle θ. Its height above the lowest point is $L(1-\cos\theta)$, so

$$U(\theta) = mgL\,(1-\cos\theta).$$

Show that for small swings this is a spring.

Solution Use the cosine series from Example 5.4:

$$1-\cos\theta = \frac{\theta^2}{2!} - \frac{\theta^4}{4!} + \cdots$$

so

$$U(\theta) \approx \tfrac{1}{2}\,mgL\,\theta^2,$$

which is $\frac{1}{2}k\theta^2$ with stiffness $k = mgL$. The pendulum's potential is not a parabola — it is a cosine, and it flattens out and turns over if you swing hard enough. But near the bottom it is a parabola, and that is why a pendulum keeps time.
The error is the θ^4 term, relatively about $\theta^2/12$. At a swing of 0.1 radians that is one part in 1200, which is why a grandfather clock with a small swing is accurate and the same clock knocked into a wide swing runs slow.

5.6 A Preview: Euler's Formula

One computation is too beautiful to leave out, though we will not need it until oscillations in Part II.

Write i for a quantity with $i^2 = -1$. We have to be careful about what we are doing here: the

exponential of Chapter 4 was built as the inverse of an area function and is defined only for real inputs, so substituting a complex number into it is not something the earlier theorems license. What we do instead is *extend* it by decree, and take the series as the definition:

$$e^z = \sum_{k=0}^{\infty} \frac{z^k}{k!} \qquad \text{for any complex } z.$$

This agrees with the old exponential whenever z is real, so it is a genuine extension and not a different function. It converges for every z, and it converges *absolutely* — the series of magnitudes is $\sum |z|^k/k! = e^{|z|}$, which we already know is finite. Absolute convergence is what permits the two things we are about to do: regroup the terms into a real part and an imaginary part, and (in the problems) multiply two such series together to prove $e^{z+w} = e^z e^w$.

So put $z = i\theta$ and sort the terms. Since $i^2 = -1$, $i^3 = -i$, $i^4 = 1$, and so on,

$$\begin{aligned} e^{i\theta} &= 1 + i\theta + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \frac{(i\theta)^4}{4!} + \cdots \\ &= \Big(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots\Big) + i\Big(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots\Big). \end{aligned}$$

Those two bracketed series are exactly the ones from Example 5.4.

5.9 Theorem *Euler's formula*

$$e^{i\theta} = \cos\theta + i\sin\theta.$$

(Rearranging an infinite sum is not always legal. Here it is, because the series converges absolutely, which is precisely why that was worth checking before splitting the sum in two.)

The formula says that the exponential function, the one that describes growth, and the trigonometric functions, the ones that describe rotation, are the same function looked at along different directions. That is why e^x keeps turning up in the study of waves and alternating current, and it is the reason a physicist reaches for a complex exponential the moment anything oscillates.

5.7 Chapter Recap

We stopped settling for straight lines. Matching more derivatives at a point gives the Taylor polynomials (Definition 5.1), whose coefficients are forced to be $f^{(k)}(a)/k!$, and whose error is governed by the first term left out (Theorem 5.2) — the same pattern as every approximation in this book, one order deeper.

Letting the polynomials run forever gives series for e^x, sin and cos, the binomial series that started Newton off, and a permanent warning that a series need not be valid everywhere — and that converging and representing are two different things. The physical payoff was the small-oscillation result: sit at a nondegenerate minimum of the potential, where the linear term dies, and the surviving quadratic term turns the system, gently disturbed, into a spring.

Problem 5.1. Find the Taylor polynomial P_3 for $f(x) = e^x$ about $a = 0$ and use it to estimate $e^{0.2}$. Compare with the true value and check that the error is roughly the size Theorem 5.2 predicts.

Problem 5.2. Find the Taylor series of $\ln(1+x)$ about 0, but do it without appealing to termwise integration of an infinite series. Instead start from the *finite* identity

$$\frac{1}{1+t} = \sum_{k=0}^{N} (-t)^k + \frac{(-t)^{N+1}}{1+t},$$

which you can check by multiplying through by $1+t$ and telescoping as in Example 5.5. Integrate both sides from 0 to x and bound the leftover term to show the series converges to $\ln(1+x)$ for $|x| < 1$. Then handle the endpoints: at $x = 1$, show the leftover integral has size at most $1/(N+2)$, so the series works there too. What goes wrong at $x = -1$, and what goes wrong for $|x| > 1$?

Problem 5.3. You want $\sin(0.3)$ to an accuracy of 10^{-8}, using the series of Example 5.4. Use the remainder bound to decide, before computing anything, how many terms you need; then compute that many and check. Repeat for $\sin(3.0)$, and comment on why the same accuracy is so much more expensive there. (This is what "small" has to mean in practice: not that a number looks small in decimal, but that it is small compared with the scale the series is built on.)

Problem 5.4. A relativistic particle has $\gamma = (1 - v^2/c^2)^{-1/2}$. Keep the constant term and two corrections from the binomial series and show that

$$E = \gamma mc^2 \approx mc^2 + \tfrac{1}{2}mv^2 + \tfrac{3}{8}\frac{mv^4}{c^2}.$$

At what speed, as a fraction of c, does the third term reach one percent of the second?

Problem 5.5. A bead of mass $m > 0$ slides on a wire whose shape gives it potential energy

$$U(x) = U_0\Big[(x/\ell)^4 - 2(x/\ell)^2\Big],$$

where $U_0 > 0$ is an energy and $\ell > 0$ is a length. (Check first that the bracket is dimensionless, and say why $U_0(x^4 - 2x^2)$ would not have been.) Find the equilibria, classify each as stable or unstable with Theorem 2.20, and find the effective spring stiffness at each stable one. Then say what changes if $U_0 < 0$.

Problem 5.6. Two atoms have $U(r) = \varepsilon\big[(r_0/r)^{12} - 2(r_0/r)^6\big]$, the potential drawn in Figure 5.2. Verify that r_0 is the minimum, and show that the effective stiffness there is $k = 72\varepsilon/r_0^2$.

Problem 5.7. First establish the multiplication law $e^z e^w = e^{z+w}$ for complex z, w by multiplying the two defining series and collecting the terms of each total degree — you will find the binomial theorem doing the work, and absolute convergence is what makes the rearrangement legal. Then use Euler's formula to show $\cos\theta = \tfrac{1}{2}(e^{i\theta} + e^{-i\theta})$, and verify the identity $\cos^2\theta = \tfrac{1}{2}(1 + \cos 2\theta)$ by squaring. (Trigonometric identities become algebra once you have Theorem 5.9, which is most of why physicists like it.)

Chapter 6

The Craft of Integration

There is an asymmetry in this subject that nobody warns you about.

Differentiation is mechanical. Hand me any function built out of powers, exponentials, logarithms and trigonometric functions, and the rules of Chapter 2 will grind out its derivative without judgement or ingenuity. You could teach the procedure to a machine, and people did.

Integration, done by hand, is not like that. Two different things are worth separating here, because they are often run together. The first is that some perfectly innocent elementary functions — e^{-x^2} is the famous one — have no *elementary* antiderivative: no finite combination of powers, roots, exponentials, logarithms and trigonometric functions differentiates to them. That is a theorem, not an admission of defeat, and it says nothing against those integrals, which are still perfectly good functions that can be defined, named and computed to any accuracy you like. The second is that there is no short recipe a person can follow to integrate whatever lands in front of them. (Computer algebra systems do rather better than this suggests: algorithms exist that decide elementary integrability for large classes of functions. They are not something you would want to run in your head.)

What exists for us instead is a toolkit, and the skill is recognising which tool a given integral is asking for. Fortunately the toolkit is small, and every tool in it is a differentiation rule run backwards.

Differentiation rule	Integration technique
chain rule	substitution (Section 3.7)
product rule	integration by parts
—	partial fractions (algebra, not calculus)
trigonometric identities	trigonometric substitution

One consolation: checking is free. Whatever you produce, differentiate it. If you get the integrand back, you are right, and no amount of dubious manoeuvring along the way can make you wrong.

6.1 Integration by Parts

We can integrate e^x, and we can integrate x. What about xe^x?

Nothing we have will touch it. Substitution is no help — there is no inner function whose derivative is sitting there waiting. But try differentiating the thing you would guess the answer looks like:

$$\frac{d}{dx}\big[xe^x\big] = xe^x + e^x.$$

That is the integrand we want, *plus* one we already know how to integrate. So $\int xe^x dx$ ought to be $xe^x - \int e^x dx$, and a quick differentiation confirms it. The question is whether that lucky accident can be turned into a method, and it can, because the accident was just the product rule.

The product rule says

$$\frac{d}{dx}\big[uv\big] = u\,\frac{dv}{dx} + v\,\frac{du}{dx}.$$

Integrate both sides. The left-hand side integrates to uv, since that is what it is the derivative of, and rearranging gives the technique.

6.1 Theorem *integration by parts*

If u and v are differentiable with continuous derivatives, then

$$\int u\,dv \;=\; uv - \int v\,du,$$

and for a definite integral,

$$\int_a^b u\,dv = \Big[uv\Big]_a^b - \int_a^b v\,du.$$

In differential language — which is how everyone actually uses it — you split the integrand into a part you call u and a part you call dv, and the rule trades the integral you have for a different one. The art is choosing the split so that the new integral is easier than the old.

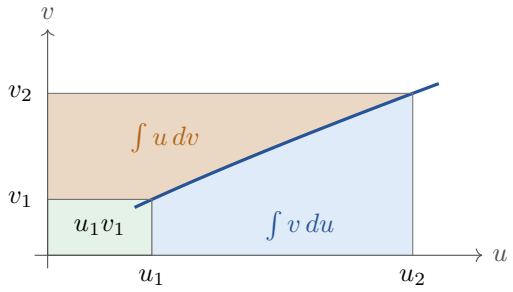


Figure 6.1: Integration by parts is a statement about two ways of slicing the same region. A curve runs from (u_1, v_1) to (u_2, v_2). Slicing vertically measures the blue area $\int v\,du$; slicing horizontally measures the orange area $\int u\,dv$. Together with the corner rectangle u_1v_1 they fill the large rectangle u_2v_2 exactly, which rearranged is $\int u\,dv = u_2v_2 - u_1v_1 - \int v\,du$.

6.2 Example *a polynomial times an exponential*

Compute $\displaystyle\int xe^x\,dx$.

Solution Take $u = x$ and $dv = e^x dx$, so that $du = dx$ and $v = e^x$. The point of that choice is that differentiating x makes it simpler while integrating e^x costs nothing. Then

$$\int xe^x\,dx = xe^x - \int e^x\,dx = xe^x - e^x + C = (x-1)e^x + C.$$

Check: $\dfrac{d}{dx}\big[(x-1)e^x\big] = e^x + (x-1)e^x = xe^x$. ✓
Had we chosen the other way round — $u = e^x$, $dv = x\,dx$ — we would have obtained $\frac{1}{2}x^2e^x - \frac{1}{2}\int x^2e^x dx$, which is worse than what we started with. Choosing badly is not fatal; it is just useless, and you notice immediately.

6.3 Example *integrating the logarithm*

Compute $\int \ln x \, dx$.

Solution There is only one factor, which looks like a problem until you notice you may always supply a second one. Take $u = \ln x$ and $dv = dx$, so $du = dx/x$ and $v = x$:

$$\int \ln x \, dx = x \ln x - \int x \cdot \frac{dx}{x} = x \ln x - \int dx = x \ln x - x + C.$$

This is the answer hinted at in the problems of Chapter 4, and the trick — taking $dv = dx$ when there is nothing else to take — works for any function whose derivative is simpler than itself.

6.4 Example *going round in a circle, productively*

Compute $I = \int e^x \sin x \, dx$.

Solution Integrating by parts with $u = \sin x$, $dv = e^x dx$ gives

$$I = e^x \sin x - \int e^x \cos x \, dx,$$

and doing it again on the new integral, with $u = \cos x$, gives

$$\int e^x \cos x \, dx = e^x \cos x + \int e^x \sin x \, dx = e^x \cos x + I.$$

Substituting back,

$$I = e^x \sin x - e^x \cos x - I \qquad \Longrightarrow \qquad I = \tfrac{1}{2} e^x (\sin x - \cos x) + C.$$

We came back to where we started and solved for the integral as though it were an unknown. This looks like a swindle the first time and a standard technique the second.

6.2 Rational Functions

A quotient of polynomials can always be integrated, and the method is algebra rather than calculus: break the fraction into pieces simple enough to recognise.

6.5 Example *partial fractions*

Compute $\int \dfrac{dx}{x^2 - 1}$.

Solution Factor the denominator and split:

$$\frac{1}{(x-1)(x+1)} = \frac{A}{x-1} + \frac{B}{x+1}.$$

Multiplying up gives $1 = A(x+1) + B(x-1)$. Setting $x = 1$ gives $A = \frac{1}{2}$; setting $x = -1$ gives $B = -\frac{1}{2}$. Hence, by Corollary 4.4,

$$\int \frac{dx}{x^2-1} = \tfrac{1}{2}\ln|x-1| - \tfrac{1}{2}\ln|x+1| + C = \tfrac{1}{2}\ln\left|\frac{x-1}{x+1}\right| + C.$$

This is not an idle exercise: the same integral appears when a falling body has air resistance proportional to v^2, and it appears again in logistic population growth, which is a problem in Chapter 8. (The worked falling-body example there uses drag linear in v, which is separable without partial fractions; the quadratic case is the one that needs this section.)

Example 6.5 is the easiest possible case, and on its own it does not equip you to act on the claim that *any* rational function can be integrated. Three more moves cover the common cases; a final extension is described below.

First, make the numerator smaller than the denominator. If it is not, divide. For instance

$$\frac{x^3}{x^2-1} = x + \frac{x}{x^2-1},$$

by long division, and the polynomial part integrates on sight. This step is not optional: partial fractions only exist for a proper fraction.

Second, repeated factors need repeated terms. A factor $(x-r)^2$ in the denominator contributes both $A/(x-r)$ and $B/(x-r)^2$, and in general $(x-r)^m$ contributes one term for each power up to m. The reason is a counting argument: a denominator of degree n needs n unknowns to match an arbitrary numerator of degree $n-1$, and only the full list supplies them. Thus

$$\frac{1}{x(x-1)^2} = \frac{1}{x} - \frac{1}{x-1} + \frac{1}{(x-1)^2},$$

whose integral is $\ln|x| - \ln|x-1| - 1/(x-1) + C$.

Third, quadratics that will not factor become arctangents. Over the real numbers some quadratics have no roots, and x^2+1 is the prototype. Such a factor contributes a term $(Ax+B)/(x^2+px+q)$, and completing the square turns it into something we met in Chapter 4:

$$\int \frac{dx}{x^2+2x+5} = \int \frac{dx}{(x+1)^2+4} = \tfrac{1}{2}\arctan\!\Big(\frac{x+1}{2}\Big) + C$$

by Proposition 4.7, after substituting $u = (x+1)/2$. An Ax in the numerator is handled separately, by arranging it to be a multiple of the derivative of the denominator, which makes that piece a logarithm.

6.6 Example *division and an irreducible quadratic*

Compute $\displaystyle\int \frac{2x^3+x}{x^2+1}\,dx$.

Solution The numerator has higher degree, so divide first:

$$\frac{2x^3+x}{x^2+1} = 2x - \frac{x}{x^2+1}.$$

The remaining fraction has an irreducible quadratic below, and its numerator is (half) the

derivative of that quadratic, so it is a logarithm:

$$\int \Big(2x - \frac{x}{x^2+1}\Big)dx = x^2 - \tfrac{1}{2}\ln(x^2+1) + C.$$

Had the numerator been a constant instead of x, the same denominator would have produced an arctangent. Which of the two you get depends entirely on whether the top is proportional to the derivative of the bottom, and looking for that is the first thing to do with any fraction.

There is one remaining case before this becomes a complete algorithm: a repeated irreducible quadratic contributes terms $(Ax+B)/(x^2+px+q)^j$ for each power j up to its multiplicity. After completing the square, the numerator's derivative part integrates directly and the remaining powers can be reduced by integration by parts. For example,

$$\int \frac{dx}{(1+x^2)^2} = \frac{x}{2(1+x^2)} + \frac{1}{2}\arctan x + C.$$

We leave that reduction beyond these examples. With it, rational integration is systematic: its answers contain rational functions, logarithms and arctangents.

6.3 Trigonometric Substitution

An integrand containing $\sqrt{a^2-x^2}$ is asking, in a rather broad way, to be put on a circle. Take $a>0$ and substitute $x = a\sin\theta$ with θ restricted to $[-\pi/2, \pi/2]$, which is exactly the range on which sin is one-to-one and on which $\cos\theta \geq 0$. Then

$$\sqrt{a^2-x^2} = \sqrt{a^2(1-\sin^2\theta)} = a\,|\cos\theta| = a\cos\theta,$$

and the square root disappears. The restriction is what earns the last equality: at $\theta=\pi$, for instance, $a\cos\theta$ would be $-a$ while the square root is $+a$, and the substitution would quietly flip a sign. The substitution is not a trick pulled from nowhere: it is the statement that x and $\sqrt{a^2-x^2}$ are the two legs of a right triangle with hypotenuse a, and θ is the angle.

6.7 Example *the area of a circle*

Compute $\displaystyle\int_{-a}^{a}\sqrt{a^2-x^2}\,dx$, the area of a semicircle of radius a.

Solution Put $x = a\sin\theta$, so $dx = a\cos\theta\,d\theta$, and the limits $x=\pm a$ become $\theta = \pm\pi/2$. Then

$$\int_{-a}^{a}\sqrt{a^2-x^2}\,dx = \int_{-\pi/2}^{\pi/2} a\cos\theta\cdot a\cos\theta\,d\theta = a^2\int_{-\pi/2}^{\pi/2}\cos^2\theta\,d\theta.$$

Using $\cos^2\theta = \frac{1}{2}(1+\cos 2\theta)$ — the identity from the problems of Chapter 5 —

$$a^2\int_{-\pi/2}^{\pi/2}\tfrac{1}{2}(1+\cos 2\theta)d\theta = \frac{a^2}{2}\Big[\theta + \tfrac{1}{2}\sin 2\theta\Big]_{-\pi/2}^{\pi/2} = \frac{a^2}{2}(\pi+0) = \frac{\pi a^2}{2}.$$

So the full circle has area πa^2 — which you knew, but had never derived. Newton approached circular-segment areas with his binomial series; here trigonometric substitution gives another route, and it is the reason quadrature of the circle mattered so much.

6.4 Improper Integrals

Every integral so far has run between two finite numbers, with a bounded integrand. Physics constantly asks for integrals that do neither.

How much work does it take to lift a satellite *completely away* from the Earth — not to some height, but out of the gravitational field altogether? The gravitational potential energy of a charge or a mass is conventionally measured from infinitely far away, which means an integral whose upper limit is not a number. We need to say what that means.

6.8 Definition *improper integrals*

If f is integrable on $[a, b]$ for every $b > a$, define

$$\int_a^{\infty} f(x)\,dx = \lim_{b\to\infty} \int_a^b f(x)\,dx,$$

and say the integral **converges** if this limit exists as a finite number and **diverges** otherwise. If instead f blows up at an endpoint, say at a, define

$$\int_a^b f(x)\,dx = \lim_{c\to a^+} \int_c^b f(x)\,dx.$$

The definition is doing something subtle and worth saying aloud: it refuses to treat infinity as a number. We never integrate "all the way to infinity". We integrate to b, which is an honest finite calculation, and then ask what happens to the answer as b grows. Infinity appears only inside a limit, where Chapter 1 taught us to handle it.

6.9 Example *two tails that look alike*

Do $\displaystyle\int_1^{\infty} \frac{dx}{x}$ and $\displaystyle\int_1^{\infty} \frac{dx}{x^2}$ converge?

Solution For the second,

$$\int_1^b \frac{dx}{x^2} = \Big[-\frac{1}{x}\Big]_1^b = 1 - \frac{1}{b} \longrightarrow 1,$$

so it converges, to exactly 1. For the first, by Corollary 4.4,

$$\int_1^b \frac{dx}{x} = \ln b \longrightarrow \infty,$$

so it diverges. The region under $1/x$ from 1 onwards has infinite area.

That dividing line is worth recording, since it settles most questions of this kind at a glance.

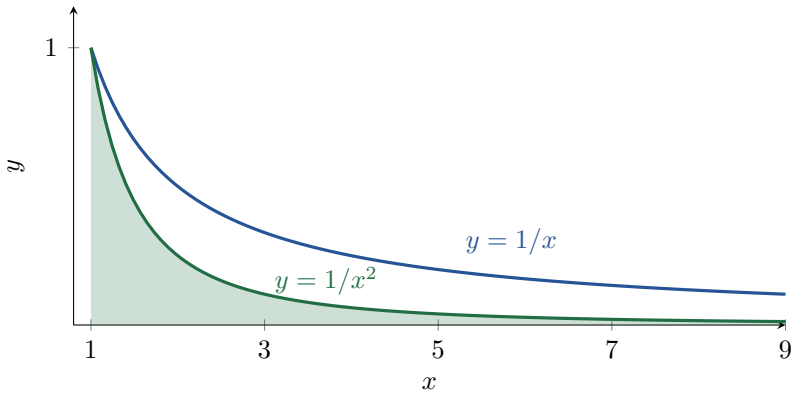


Figure 6.2: Both curves fall to zero, and to the eye they are much the same. But the complete region under $1/x^2$, continued forever to the right, has total area exactly 1 — of which only $8/9$ is shaded in this finite window. The corresponding region under $1/x$ has infinite area. Going to zero is not enough; a tail has to go to zero *fast enough*. Within the family $1/x^p$ the dividing line sits precisely at $p = 1$; other kinds of tail need their own test, and $1/(x \ln x)$, which is smaller than $1/x$ and still diverges, is the standard warning against reading too much into the picture.

6.10 Proposition *the p-test*

For $a > 0$,

$$\int_a^\infty \frac{dx}{x^p} \quad \text{converges if } p > 1, \quad \text{diverges if } p \le 1.$$

Near the origin the verdict reverses: $\displaystyle\int_0^a \frac{dx}{x^p}$ converges if $p < 1$ and diverges if $p \ge 1$.

Proof For $p \neq 1$, $\displaystyle\int_a^b x^{-p}dx = \frac{b^{1-p} - a^{1-p}}{1-p}$, and as $b \to \infty$ the term b^{1-p} tends to 0 when $p > 1$ and to ∞ when $p < 1$. The case $p = 1$ is Example 6.9. The statement near the origin is the same computation with the roles of the limits exchanged. ■

6.11 Example *escape velocity*

The gravitational force on a mass m at distance r from the centre of a planet of mass M is GMm/r^2, directed inwards. How much work is needed to carry the mass from the surface, $r = R$, out to infinity — and how fast must you throw it so that it never comes back?

Solution Work is the integral of force over distance (Example 3.17), and here the integral is improper:

$$W = \int_R^\infty \frac{GMm}{r^2}\,dr = GMm\Big[-\frac{1}{r}\Big]_R^\infty = \frac{GMm}{R},$$

which is *finite* — by the p-test, since $p = 2 > 1$. That single fact is why escape is possible at all. Had gravity fallen off as $1/r$ instead of $1/r^2$, the work would have been infinite and nothing could ever leave.

Supplying that work as kinetic energy, $\frac{1}{2}mv^2 = GMm/R$, gives

$$v_{\text{esc}} = \sqrt{\frac{2GM}{R}},$$

about 11.2 km/s for the Earth. Notice the mass m cancelled: a pebble and a spacecraft need the same speed.

The convergence of that integral is not a technicality, it is a fact about the universe. The exponent in the inverse-square law sits on the convergent side of Proposition 6.10, which is what allows gravitational potential energy to be measured from infinity and what makes "escape" a meaningful idea. In Part III we will find that the electric force obeys the same inverse-square law, and the same integral will give us the electric potential.

Improper integrals are also how we will define the potential of a charge, the total energy stored in a field, and the mass of an atmosphere that thins out forever without ever quite ending. In each case the question is the same: does the tail shrink fast enough?

Two traps

Definition 6.8 covers one bad endpoint at a time, and both ways of having two need a word, because the natural shortcut is wrong.

If the integrand blows up at an *interior* point c of $[a, b]$, the integral is defined only when *each side converges separately*:

$$\int_a^b f = \lim_{s \to c^-} \int_a^s f \; + \; \lim_{t \to c^+} \int_t^b f,$$

and if either limit fails, the whole thing is undefined. It is not enough for the two infinities to look as though they cancel. Consider

$$\int_{-1}^{1} \frac{dx}{x}.$$

It is tempting to say the answer is 0 by symmetry, and a calculator that uses a symmetric grid will happily tell you so. But $\int_t^1 dx/x = -\ln t \to \infty$, so the right-hand piece diverges on its own, and the integral does not exist. The symmetric cancellation is an artefact of how you chose to approach the singularity: approach it as $\int_{-1}^{-\varepsilon} + \int_{2\varepsilon}^{1}$ instead and the same reasoning gives $-\ln 2$. An expression whose value depends on how you sneak up on it is not a number, and in electrostatics — where $1/r$ singularities sit inside the region you are integrating over — this is the difference between a real answer and a plausible-looking wrong one.

If *both* limits are infinite, split at any convenient finite point and require both halves to converge:

$$\int_{-\infty}^{\infty} f = \int_{-\infty}^{0} f + \int_{0}^{\infty} f.$$

Where the split is made does not matter, by Proposition 3.3(b), but requiring both halves does: $\int_{-b}^{b} x\,dx$ is zero for every b, and yet $\int_{-\infty}^{\infty} x\,dx$ does not exist.

6.5 When There Is No Antiderivative

The chapter opened by admitting that some integrands have no elementary antiderivative. Others have one that is not worth finding, and in a laboratory the integrand is often not a formula at all but a column of measurements. In all three cases we still want the number.

Go back to the Riemann sums of Chapter 3 and improve them slightly. A rectangle approximates the graph over each step by a horizontal line, which is a zeroth-order approximation; using

a straight line through the two endpoints instead — a trapezoid — costs nothing and does much better.

6.12 Definition *the trapezoidal rule*

Divide $[a, b]$ into n steps of width $\Delta x = (b-a)/n$ with endpoints $x_0, \dots, x_n$. The **trapezoidal estimate** of $\int_a^b f$ is

$$T_n = \Delta x \left[\tfrac{1}{2}f(x_0) + f(x_1) + \cdots + f(x_{n-1}) + \tfrac{1}{2}f(x_n)\right].$$

Only the two end values get a half. That is the whole content of the formula: each interior point is the right edge of one trapezoid and the left edge of the next, so it is counted twice at half weight, while the two outermost points are counted once. If you ever forget the rule, redraw the trapezoids and the pattern reassembles itself.

The error is where the interest is. On a single step, replacing f by the chord through its endpoints is exactly the situation Theorem 5.2 describes, and the discrepancy is governed by f'': if $|f''| \le M$ on $[a, b]$ then one step contributes at most $M(\Delta x)^3/12$, and with $n = (b-a)/\Delta x$ steps,

$$\left|\int_a^b f - T_n\right| \le \frac{M\,(b-a)}{12}\,(\Delta x)^2.$$

So halving the step divides the error bound by four. That is a usable guarantee: it tells you in advance how fine a grid you need, and — more practically — it tells you what to expect when you refine. Compute T_n and T_{2n}; if the two agree to the accuracy you want, and the change from one to the other shrank by roughly a factor of four as predicted, you have a useful numerical check, but not a proof of accuracy. A smooth integrand can show a different rate because leading errors cancel, the grid is too coarse, or roundoff matters. Agreement between grids can also miss an oscillation between sample points. For a guarantee, use the derivative bound; treat refinement as supporting evidence.

6.13 Example *an integral with no formula*

Estimate $\displaystyle\int_0^1 e^{-x^2}\,dx$.

Solution There is no elementary antiderivative, so there is nothing to evaluate at the endpoints. With $n = 4$, $\Delta x = 0.25$, the values of e^{-x^2} at 0, 0.25, 0.5, 0.75, 1 are 1, 0.939413, 0.778801, 0.569783, 0.367879, giving

$$T_4 = 0.25\big[\tfrac{1}{2}(1) + 0.939413 + 0.778801 + 0.569783 + \tfrac{1}{2}(0.367879)\big] = 0.742984.$$

With $n = 8$ the same computation gives $T_8 = 0.745866$. The true value is $0.746824\ldots$, so T_4 is off by about 0.003840 and T_8 by 0.000959 — approximately a fourfold improvement in this example, consistent with the bound's dependence on the square of the step.

This is the same idea as Euler's method for differential equations in Chapter 8, and for the same reason: when a formula is unavailable, chop the problem small enough that the linear approximation is good, and add up. Every numerical method in physics is a descendant of that

one sentence.

6.6 Chapter Recap

Integration by hand is a toolkit rather than a recipe, and every tool in it is a differentiation rule read backwards: substitution from the chain rule, parts from the product rule (Theorem 6.1), and beneath both, the habit of checking your answer by differentiating it. The one exception is rational functions, where division, repeated factors and completed squares really do amount to a complete method, with logarithms and arctangents as the only outputs.

Improper integrals (Definition 6.8) extend the whole apparatus to infinite ranges and to unbounded integrands, by the only honest method available — doing a finite calculation and taking a limit, with each bad endpoint handled on its own. Whether the answer is finite depends on how fast the integrand decays, and within the power family the dividing line sits at $1/x$ (Proposition 6.10); that dividing line has consequences as large as whether a rocket can leave the Earth.

And when no antiderivative is available, Section 6.5 gives the fallback: approximate the graph by straight lines, add up the trapezoids, and use the $(\Delta x)^2$ error bound both to choose a grid and to check the answer by refining it.

Problem 6.1. Compute $\int x \sin x \, dx$ and $\int x^2 e^x \, dx$. (The second needs parts twice.) Check both by differentiating.

Problem 6.2. Compute $\displaystyle\int_0^1 x \ln x \, dx$ by first integrating from $c > 0$ to 1 and then letting $c \to 0^+$. Note carefully what is and is not wrong at the lower limit: the integrand is *undefined* at 0, because $\ln 0$ is, but it does not blow up — its limit there is 0. To finish you need $c^2 \ln c \to 0$; get it by writing $c = e^{-u}$ with $u \to \infty$ and using $e^{2u} \geq (2u)^2/2$, which follows from the exponential series. Finally, explain why defining the integrand to be 0 at $x = 0$ removes the difficulty entirely.

Problem 6.3. Integrate $\displaystyle\int \frac{x^3 + 2}{x^2 + 4}\, dx$ and $\displaystyle\int \frac{dx}{x(x-2)^2}$, using all three moves of Section 6.2. Check both by differentiating.

Problem 6.4. Estimate $\displaystyle\int_1^2 \frac{dx}{x}$ with the trapezoidal rule using $n = 2$ and $n = 4$, and compare with the exact answer $\ln 2$. Does the error shrink by the factor the bound predicts? Then use the bound to decide how many steps you would need for six-decimal accuracy, and say whether that is a sensible way to compute $\ln 2$.

Problem 6.5. Show that $\displaystyle\int_{-1}^1 \frac{dx}{x^2}$ diverges, even though the integrand is positive and the interval is finite. Then explain what is wrong with the calculation $\left[-1/x\right]_{-1}^{1} = -1 - 1 = -2$, and why a negative answer for a positive integrand should have been a warning.

Problem 6.6. Use partial fractions to compute $\displaystyle\int \frac{dx}{x(1-x)}$, which is the integral you need to solve the logistic growth equation.

Problem 6.7. Determine whether each converges: (a) $\displaystyle\int_1^\infty \frac{dx}{x^{3/2}}$; (b) $\displaystyle\int_1^\infty \frac{dx}{\sqrt{x}}$; (c) $\displaystyle\int_0^1 \frac{dx}{\sqrt{x}}$; (d) $\displaystyle\int_0^\infty e^{-x}\, dx$.

Problem 6.8. A rod of length L has mass per unit length λ. Using $x_{\mathrm{cm}} = \dfrac{1}{M}\displaystyle\int x\,dm$, find the centre of mass when λ is constant, and again when $\lambda = \lambda_0 x/L$ (a rod that gets heavier along its length).

Problem 6.9. Two point charges a distance r apart exert on each other a radial force $F_r = kq_1q_2/r^2$, counted positive when it pushes them apart. Show that the work done *by this force* as the separation grows from R to infinity is finite and equal to kq_1q_2/R. Then answer a different question: if you separate them slowly, so that they have no kinetic energy at either end, what work must *you* supply? Discuss both signs of q_1q_2 — in particular, which case releases energy on separation and which demands it — and identify the part of Proposition 6.10 you relied on.

Chapter 7

Curves in Motion

Every curve in this book so far has been the graph of a function: one height for each horizontal position. That is a serious restriction, and physics breaks it immediately. A planet's orbit closes on itself, so it has two heights above most horizontal positions. A ball thrown straight up passes through every height twice, at two different times. A point on the rim of a rolling wheel does something stranger still.

What all these have in common is that they are not really graphs at all. They are *paths*: things traced out by a moving point. And the natural way to describe a path is to say where the point is at each moment, which means two functions of time rather than one function of position.

7.1 Paths, Not Graphs

7.1 Definition *parametric curve*

A **parametric curve** in the plane is a pair of functions

$$x = x(t), \qquad y = y(t),$$

giving the position of a moving point at each value of the parameter t, usually time. The curve is the set of points visited; the parametrisation also records *when* each point is visited, and in which direction.

That extra information is the whole advantage. The unit circle as a set is $x^2 + y^2 = 1$, which tells you the shape and nothing else. Written parametrically as $x = \cos \omega t$, $y = \sin \omega t$, it also tells you the motion is anticlockwise, at angular rate ω, returning every $2\pi/\omega$ seconds.

The most celebrated curve of the seventeenth century was a path of exactly this kind, and it has a good claim to being the one that made mathematicians believe the new methods were worth something.

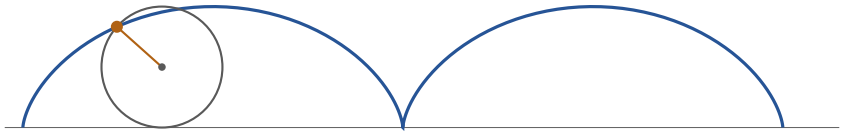


Figure 7.1: The cycloid: the path traced by a single point on the rim of a wheel rolling along a straight line. Parametrically, with the wheel of radius r having turned through an angle t, the point sits at $x = r(t - \sin t)$, $y = r(1 - \cos t)$ — the centre has moved forward by rt, and the point has swung back around it. As a graph the curve would be a nightmare; as a path it is two lines of algebra.

The cycloid caused so many priority quarrels among Roberval, Descartes, Torricelli and Pascal that it was nicknamed the Helen of geometry, after the woman who launched a thousand ships. It also has one useful property, found by Huygens in 1659: invert one cycloidal arch to make a bowl. In uniform gravity, a bead released from rest anywhere along either side reaches the bottom in the same time, if friction is neglected. He built a pendulum clock with cycloidal cheeks to exploit this, hoping to solve the longitude problem at sea.[10]

Once a curve is a path, the questions we ask about it change. We no longer want only its shape. We want the velocity of the point tracing it, the distance it travels, and the area it sweeps out — all of which are calculus questions, and all of which the parametrisation answers cleanly.

7.2 Velocity Along a Path

If x and y each change with time, then each has its own rate, and the pair of them is the velocity.

7.2 Definition *velocity and speed*

For a parametric curve, the **velocity components** are $\dot{x} = \dfrac{dx}{dt}$ and $\dot{y} = \dfrac{dy}{dt}$, and the **speed** is

$$|v| = \sqrt{\dot{x}^2 + \dot{y}^2}.$$

The speed formula is Pythagoras applied to a small step: in a short time Δt the point moves Δx across and Δy up, so the distance covered is $\sqrt{\Delta x^2 + \Delta y^2}$, and dividing by Δt gives the rate.

The slope of the path is also available, without solving for y as a function of x. By the chain rule, $\dfrac{dy}{dt} = \dfrac{dy}{dx}\dfrac{dx}{dt}$, so

7.3 Proposition *slope of a parametric curve*

Wherever $\dot{x} \neq 0$,

$$\frac{dy}{dx} = \frac{\dot{y}}{\dot{x}}.$$

7.4 Example *a thrown ball*

A ball is thrown with initial speed v_0 at angle α. Ignoring air resistance, its path is

$$x(t) = v_0 \cos\alpha \, t, \qquad y(t) = v_0 \sin\alpha \, t - \tfrac{1}{2} g t^2.$$

Find its speed, the time it is highest, and the shape of the path.

Solution Differentiating, $\dot{x} = v_0 \cos\alpha$ (constant — nothing pushes it horizontally) and $\dot{y} = v_0 \sin\alpha - gt$. The speed is $\sqrt{\dot{x}^2 + \dot{y}^2}$, smallest when $\dot{y} = 0$, which happens at $t = v_0 \sin\alpha / g$ — the top of the flight, where by Theorem 2.18 the vertical motion is momentarily stationary. For the shape, eliminate t: from the first equation $t = x/(v_0 \cos\alpha)$, and substituting into the second gives

$$y = x \tan\alpha - \frac{g}{2 v_0^2 \cos^2\alpha} x^2,$$

a parabola — which is Galileo's discovery, and now a two-line consequence of writing the motion as a path.

7.3 Arc Length

How far does the point actually travel? Not how far apart its endpoints are: how much road it covers.

Chop the path into short pieces. Over one piece the point moves dx across and dy up, and if the piece is short enough the path across it is indistinguishable from the straight chord — that is exactly local straightness from Chapter 2. So the little length is

$$ds = \sqrt{dx^2 + dy^2},$$

and the total length is the sum of them: an integral.

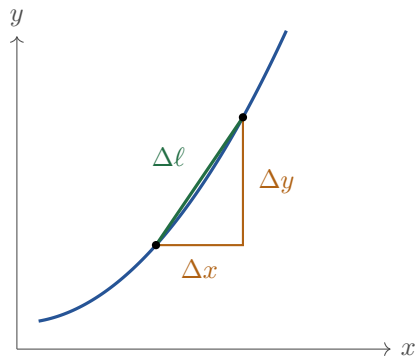


Figure 7.2: Leibniz's *characteristic triangle*, drawn large. The triangle as drawn joins two genuinely distinct points of the curve, so its legs are the finite increments Δx and Δy and its hypotenuse is the chord, $\Delta\ell = \sqrt{(\Delta x)^2 + (\Delta y)^2}$ — which approximates the arc rather than equalling it, and is always a little shorter. Shrink the step and the two become indistinguishable: in the tangent approximation $dx = \dot{x}\,dt$ and $dy = \dot{y}\,dt$, and for $dt > 0$ the length differential satisfies $ds^2 = dx^2 + dy^2$. This triangle was Leibniz's central image when he was inventing the calculus, and every formula in this section comes from dividing it through by something.

7.5 Theorem *arc length*

Let $x(t)$ and $y(t)$ have continuous first derivatives on $[a, b]$ (or on finitely many pieces of it). The total distance travelled for $a \le t \le b$, counting every traversal separately if part of the path is covered more than once, is

$$L = \int_a^b \sqrt{\dot{x}^2 + \dot{y}^2}\,dt = \int_a^b |v|\,dt.$$

For a graph $y = f(x)$, taking x itself as the parameter gives

$$L = \int_a^b \sqrt{1 + \big(f'(x)\big)^2}\,dx.$$

Proof sketch Divide $ds^2 = dx^2 + dy^2$ by dt^2 to get $\left(\dfrac{ds}{dt}\right)^2 = \dot{x}^2 + \dot{y}^2$, so the rate at which arc length accumulates is the speed — which is what "speed" ought to mean. Integrating that rate over the time of travel gives the total length, by Theorem 3.8. Making this rigorous means approximating the curve by inscribed polygons and passing to a limit, which is exactly the Riemann-sum argument of Chapter 3. ∎

7.6 Example *how long is one arch of a cycloid?*

Find the length of one arch of the cycloid $x = r(t - \sin t)$, $y = r(1 - \cos t)$, for $0 \le t \le 2\pi$.

Solution Differentiate: $\dot{x} = r(1 - \cos t)$ and $\dot{y} = r \sin t$. Then

$$\dot{x}^2 + \dot{y}^2 = r^2\big[(1 - \cos t)^2 + \sin^2 t\big] = r^2\big[1 - 2\cos t + \cos^2 t + \sin^2 t\big] = 2r^2(1 - \cos t).$$

Using $1 - \cos t = 2\sin^2(t/2)$, this is $4r^2 \sin^2(t/2)$, so the speed is $2r\,|\sin(t/2)|$, which is positive on $(0, 2\pi)$. Hence

$$L = \int_0^{2\pi} 2r \sin\frac{t}{2}\,dt = 2r\Big[-2\cos\frac{t}{2}\Big]_0^{2\pi} = 2r(2 + 2) = 8r.$$

The arch is exactly eight radii long — no π anywhere, which was established geometrically by Wren in 1658.[11] It is one of the first lengths of a curved path ever computed exactly, and a fair demonstration that the new methods could do things the Greeks could not.

7.4 Polar Coordinates

For anything that circles a centre — a planet, an electron, a rotating rigid body — describing position by how far out and in what direction is far more natural than by across and up.

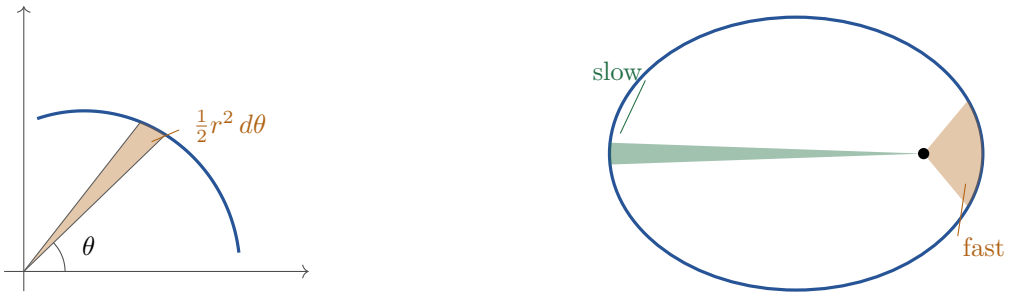
7.7 Definition *polar coordinates*

A point at distance $r \ge 0$ from the origin in the direction making angle θ with the positive x-axis has **polar coordinates** (r, θ), related to Cartesian coordinates by

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad r^2 = x^2 + y^2.$$

Going the other way, for $r > 0$ the angle is the one with $\cos\theta = x/r$ and $\sin\theta = y/r$, which fixes it up to multiples of 2π. At the origin the angle is arbitrary. A curve may then be given as $r = r(\theta)$.

Areas in polar coordinates are not rectangles but wedges, and the area element has to be worked out rather than assumed.



(a) the wedge (b) equal areas in equal times

Figure 7.3: Left: a thin wedge between θ and $\theta + d\theta$ is very nearly a triangle of base $r\,d\theta$ and height r, so its area is $\frac{1}{2}r^2\,d\theta$. Right: Kepler's second law. The two shaded slices have *equal area*, and the planet sweeps each out in the same time. Near the focus it must therefore cover a much wider angle, which is to say it moves faster — so the law is a statement about speed disguised as a statement about area.

7.8 Theorem *area in polar coordinates*

Let $r(\theta) \geq 0$ be continuous on $[\alpha, \beta]$ with $\alpha \leq \beta$. The area swept out by the radius from the origin to the curve $r = r(\theta)$, as θ runs from α to β — counting a region twice if the sweep passes over it twice — is

$$A = \int_{\alpha}^{\beta} \tfrac{1}{2}\, r(\theta)^2\, d\theta.$$

Proof sketch Chop the angle into small pieces $\Delta\theta$. Over one piece the radius barely changes, so the wedge is nearly a circular sector of radius r and angle $\Delta\theta$, whose area is $\frac{1}{2}r^2\Delta\theta$ — the same sector area used in Chapter 1's small-angle proof. Adding the wedges and refining gives a Riemann sum for the stated integral; the discrepancy between the true wedge and the circular sector is $o(\Delta\theta)$ and vanishes in the limit. ∎

7.9 Example *Kepler's second law as a rate*

Kepler found, from Tycho Brahe's observations and before any theory existed to explain it, that a line from the Sun to a planet sweeps out equal areas in equal times. Express this as a derivative.

Solution By Theorem 7.8 the swept area accumulates at the rate

$$\frac{dA}{dt} = \tfrac{1}{2}r^2\,\frac{d\theta}{dt},$$

by the Fundamental Theorem together with the chain rule. Kepler's statement is that this rate is *constant*:

$$r^2\dot{\theta} = \text{constant}.$$

In Part II we will recognise that combination as the angular momentum per unit mass, and Kepler's second law will turn out to be conservation of angular momentum — which in turn follows from the fact that gravity pulls straight toward the Sun. Kepler had the pattern in 1609; the explanation took Newton and eighty years.

7.5 Chapter Recap

Describing a curve as a path, by giving position as a function of time, costs one extra function and buys a great deal. Velocity becomes a pair of rates, speed becomes $\sqrt{\dot{x}^2 + \dot{y}^2}$, and the slope of the path is the ratio of the two rates rather than something requiring the curve to be solved for y.

Arc length came out of Leibniz's characteristic triangle (Figure 7.2), which is just Pythagoras applied to an infinitesimal step: $ds^2 = dx^2 + dy^2$, integrated. Polar coordinates replaced the rectangle with the wedge and gave us $\frac{1}{2}\int r^2 d\theta$, with which Kepler's second law becomes the statement that $r^2\dot{\theta}$ is constant.

Every one of these is the same move: take the little piece, work out its size to first order, and add the pieces up. It is Chapter 3 again, in coordinates better suited to things that go around.

Problem 7.1. A point moves with $x = 3\cos 2t$, $y = 3\sin 2t$. Describe the path, find the speed, and confirm it is constant. How long does one circuit take?

Problem 7.2. Find the length of the curve $y = x^{3/2}$ from $x = 0$ to $x = 4$ using the graph form of Theorem 7.5.

Problem 7.3. For the projectile of Example 7.4, find the range (the horizontal distance when it returns to $y = 0$) and use Section 2.8 to show that the range is greatest at $\alpha = 45°$.

Problem 7.4. Sketch the curve $r = 1 + \cos\theta$ (a cardioid) and find the area it encloses using Theorem 7.8. You will need $\cos^2\theta = \frac{1}{2}(1 + \cos 2\theta)$.

Problem 7.5. A planet at its closest approach is r_1 from the Sun and moving at speed v_1 perpendicular to the radius; at its furthest it is r_2 away. Use the constancy of $r^2\dot{\theta}$ from Example 7.9 to find its speed there.

Problem 7.6. Show that for the cycloid the speed of the tracing point is zero at the bottom of each arch. What is physically happening to that point on the wheel at the instant it touches the ground?

Chapter 8

Differential Equations

Chapter 3 ended by promising that where a force depends on position rather than on time, integrating twice is not enough and something new is needed. This is that something, and the gap it fills is the subject of the chapter.

Integrating twice works for a falling stone because the acceleration is a known constant. It works for anything whose acceleration is a known function of *time*: write down $a(t)$, integrate to get $v(t)$, integrate again to get $s(t)$, done. But look at what Newton's law actually says for a mass on a spring. The force depends on where the mass is:

$$m\,\ddot{x} = -k\,x.$$

There is nothing here to integrate. The right-hand side is not a known function of time — it involves x, which is the very thing we are trying to find. The equation relates a function to its own second derivative, and until we know the function we cannot evaluate the thing we would integrate.

This is not a difficulty to be got around. It is the normal situation in physics, and it is what Newton's second law *is*: not a formula that hands you the motion, but an equation the motion must satisfy.

8.1 What a Differential Equation Says

8.1 Definition *differential equation*

A **differential equation** is an equation relating an unknown function to its derivatives. Its **order** is the highest derivative appearing. A **solution** is a function which, substituted in, makes the equation true for all relevant values of the variable.

Here is an equation whose solutions we have already met once, in Chapter 4, and which is worth looking at again now that we know what to call it:

$$\frac{dy}{dx} = y.$$

It does not ask for a number. It asks: *which function is its own derivative?* You already know one. Are there others? How many?

The shift in what counts as an answer takes some getting used to. When you solve $x^2 = 9$ you produce numbers. When you solve $y' = y$ you produce functions, and there are infinitely many of them — $y = Ce^x$ for every constant C, as Proposition 4.10 established. To pin down one, you need extra information: where the solution starts.

8.2 Definition *initial conditions*

An **initial value problem** is a differential equation together with specified **initial data** at one point. For a first-order equation the data are usually the function's value there; for a second-order equation, its value and its first derivative — typically the position and the velocity at $t = 0$.

That count is worth understanding physically rather than memorising. Newton's law is second order, and the two numbers it asks for are exactly what you would have to tell someone to make the future of a mechanical system definite: where it is now, and how fast it is going. Nothing else.

But notice what the definition carefully does *not* say. It does not say that the data pick out exactly one solution. Whether they do is a theorem, with hypotheses, and it does not follow from the order of the equation. To see that it cannot, look at

$$\frac{dy}{dt} = 2\sqrt{|y|}, \qquad y(0) = 0.$$

One solution is $y(t) = 0$ for all t. Another is $y(t) = t^2$ for $t \geq 0$. Another sits at zero until any time T you care to name and then sets off as $(t - T)^2$. There are infinitely many solutions with the same initial data, and the equation is only first order. The culprit is that $\sqrt{|y|}$ has an infinitely steep graph at $y = 0$: continuous, but only just.

8.3 Theorem *existence and uniqueness, stated without proof*

Consider $\dfrac{dy}{dt} = f(t, y)$ with $y(t_0) = y_0$. If f is continuous near (t_0, y_0), then a solution exists on some interval around t_0. If in addition f does not vary too steeply in its second slot — precisely, if there is a constant K such that

$$|f(t, y_1) - f(t, y_2)| \;\leq\; K\,|y_1 - y_2|$$

for all nearby y_1, y_2, which holds whenever f has a bounded rate of change in y at fixed t — then that solution is unique. The same statement covers higher-order equations, once they are rewritten as systems of first-order ones.

Two warnings are hidden in that theorem. The first is the word "some": existence is promised only on *an* interval around t_0, possibly a short one. The equation $\dot{y} = y^2$ with $y(0) = 1$ has the perfectly unique solution $y = 1/(1 - t)$, which runs off to infinity at $t = 1$ and ceases to exist thereafter — not because the arithmetic broke down, but because the solution did. The second is that determinism in mechanics is not free. It rests on forces being smooth enough functions of position and velocity, and that is a physical claim about the world rather than a piece of logic.

So Newtonian determinism rests on regularity of the force law: position and velocity determine a unique motion for as long as that solution continues to exist. Theorem 8.3 gives this local guarantee, not existence for all future time. Our usual smooth force laws satisfy its uniqueness condition away from singularities; the exceptions and the square-root energy reductions require separate care.

Notice how much easier checking is than solving, exactly as with integration. To verify that $y = Ce^x$ solves $y' = y$, differentiate and compare — ten seconds, no ingenuity. Finding it in the first place needed Chapter 4. This asymmetry is why it is respectable, in this subject, to guess a solution and then verify it; a verified guess is a proof.

8.2 Seeing the Solutions Before Finding Them

A first-order equation $\dfrac{dy}{dt} = f(t, y)$ says something concrete and local: at the point (t, y), any solution passing through must have slope $f(t, y)$. So before solving anything, we can draw a little segment of that slope at each point of the plane, and any solution must thread its way through, tangent to the segments as it goes.

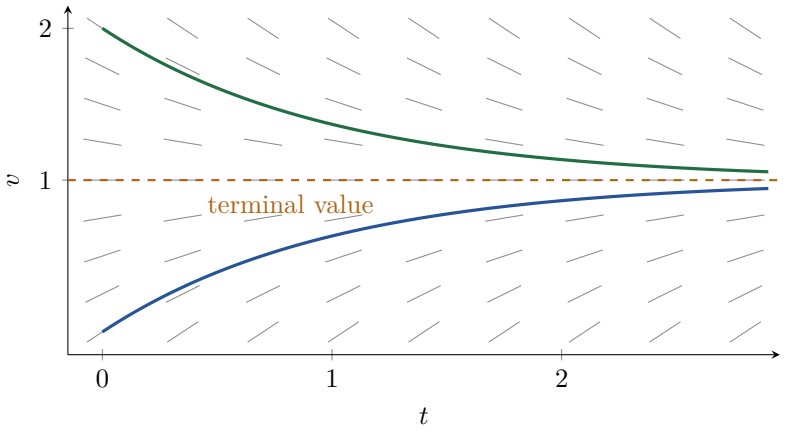


Figure 8.1: The slope field of $\dot{v} = 1 - v$. Each dash shows the slope a solution must have if it passes through that point: rising below $v = 1$, exactly horizontal on the line $v = 1$, falling above it, and steeper the further away it is. Two solutions are drawn, one starting below and one above. Neither can cross the dashed line, and both are dragged toward it — which you can read off the picture without solving anything. This is a falling body approaching terminal velocity, a cup of coffee cooling, and a capacitor charging, all at once.

This picture is worth more than it looks. Without doing any calculation we can already see that $v = 1$ is an equilibrium — the constant function $v(t) = 1$ is itself a solution, and it simply stays there — and that it is stable, in that every solution starting away from it moves toward it without reaching it in any finite time. Qualitative understanding first, formulas afterwards, is the right order — and for many equations in physics, the picture is all anyone ever gets, because no formula exists.

8.3 Separable Equations

The one general technique available to us works when the two variables can be pulled apart, and it is where the differentials of Chapter 2 finally justify all the fuss made over them.

8.4 Theorem *separation of variables*

If $\dfrac{dy}{dx} = f(x)\,g(y)$ with $g(y) \neq 0$, then the solutions satisfy

$$\int \frac{dy}{g(y)} = \int f(x)\,dx.$$

Proof Divide by $g(y)$ and integrate both sides with respect to x:

$$\int \frac{1}{g(y)}\frac{dy}{dx}\,dx = \int f(x)\,dx.$$

The left side is exactly what substitution (Theorem 3.14) turns into $\int dy/g(y)$, with $u = y(x)$ and $du = y'(x)\,dx$. ∎

In practice nobody writes it that way. One writes $dy/g(y) = f(x)\,dx$ — treating dy and dx as quantities to be shuffled, which Definition 2.5 entitles us to do — and integrates both sides. The theorem is the guarantee that this piece of apparent sleight of hand is legitimate.

The hypothesis $g(y) \neq 0$ is not a technicality, and dividing past it is how people lose solutions. Wherever $g(y_*) = 0$, the constant function $y \equiv y_*$ solves the equation exactly — it is an equilibrium — and the act of dividing by $g(y)$ deletes it from the answer. In $\dot y = y(1-y)$ the separation gives a family of S-shaped curves and quietly loses both $y = 0$ and $y = 1$, which are the two most important solutions in the problem. So: before separating, solve $g(y) = 0$ and write those constants down as solutions in their own right.

There is a standing rule that follows from that. Always check the zeros of g first, and only then divide.

8.5 Example *terminal velocity*

A falling body meets air resistance proportional to its speed, so

$$m\frac{dv}{dt} = mg - bv.$$

Find $v(t)$ for a body dropped from rest, and its eventual speed.

Solution Separate and integrate:

$$\int \frac{dv}{mg - bv} = \int \frac{dt}{m} \quad\Longrightarrow\quad -\frac{1}{b}\ln|mg - bv| = \frac{t}{m} + C,$$

using Corollary 4.4 and the chain rule on the left. Exponentiating and renaming the constant,

$$mg - bv = Ae^{-bt/m} \quad\Longrightarrow\quad v(t) = \frac{mg}{b}\Big(1 - e^{-bt/m}\Big),$$

where $v(0) = 0$ forced $A = mg$. As t grows the exponential dies and

$$v \longrightarrow v_{\mathrm{T}} = \frac{mg}{b},$$

the terminal velocity — which you could have read off the original equation without solving anything, by asking when $\dfrac{dv}{dt} = 0$. That is the equilibrium of Figure 8.1, and the solution curve there is this one in disguise.

The same equation, with different names, is a cooling cup of coffee ($\dot{T} = -k(T - T_{\text{room}})$), a charging capacitor, and a parachutist. Recognising that three problems are one problem is most of what a physics education is for.

8.4 First-Order Linear Equations

Separation is powerful, but it does not solve every equation. Some time-dependent coefficients and forcing terms leave an equation separable; others mix the variables in a way that separation cannot undo. Suppose the room the coffee sits in is itself warming through the morning, or the voltage driving a circuit is switched on and off. Then the equation looks like

$$\frac{dy}{dt} + p(t)\,y = q(t),$$

which is called **linear** because y and its derivative appear only to the first power and only separately. A general equation of this form need not separate, so we want a method that works even when it does not.

The left-hand side would be easy if it happened to be the derivative of a single product. The product rule gives $\dfrac{d}{dt}[\mu y] = \mu\,\dot{y} + \dot{\mu}\,y$. Compare that with $\dot{y} + py$. What would μ have to satisfy for the two to match, and can you solve for it?

Multiply the whole equation by an as-yet-unknown function $\mu(t)$:

$$\mu\,\dot{y} + \mu p\,y = \mu q.$$

For the left side to be $\dfrac{d}{dt}[\mu y] = \mu\dot{y} + \dot{\mu}y$, we need $\dot{\mu} = \mu p$ — which is separable, and whose solution we know from Chapter 4.

8.6 Theorem *the integrating factor*

Let p and q be continuous. Put

$$\mu(t) = \exp\left(\int p(t)\,dt\right),$$

which is positive and never zero. Then $\dfrac{dy}{dt} + p(t)y = q(t)$ becomes $\dfrac{d}{dt}[\mu y] = \mu q$, so

$$y(t) = \frac{1}{\mu(t)}\left[\int \mu(t)\,q(t)\,dt \;+\; C\right].$$

The trick looks like a rabbit out of a hat, and in a sense it is, but notice what made it possible: the product rule, run backwards, in exactly the way integration by parts was the product rule run backwards. Every method in this book is one of the four differentiation rules read in reverse. There are not very many ideas here; there are many costumes.

Note what the formula gives you: the constant C divided by μ is the solution of the equation with $q = 0$ — the system left to itself. It may decay, grow or remain constant, depending on p.

The other term is a particular solution forced by q. Every linear system in physics splits this way, into what it would do alone and what the outside world makes it do, and the two simply add. That is what *linear* buys you.

8.7 Example *a circuit that is switched on*

A coil of inductance L and resistance R is connected at $t = 0$ to a constant voltage V. The current obeys

$$L\frac{dI}{dt} + RI = V, \qquad I(0) = 0.$$

Find $I(t)$.

Solution Divide by L to put it in standard form, so $p = R/L$ and $q = V/L$. Then $\int p\,dt = Rt/L$ and $\mu = e^{Rt/L}$. So

$$\frac{d}{dt}\Big[e^{Rt/L}I\Big] = \frac{V}{L}e^{Rt/L} \implies e^{Rt/L}I = \frac{V}{R}e^{Rt/L} + C,$$

and $I(0) = 0$ forces $C = -V/R$. Hence

$$I(t) = \frac{V}{R}\Big(1 - e^{-Rt/L}\Big).$$

Compare Example 8.5: the current rises to V/R exactly as the parachutist's speed rises to mg/b, and for the same reason. The inductance plays the part of the mass.

Had V been a function of time rather than a constant — switched on and off, or oscillating — nothing in the method would change. Only the integral $\int \mu q\,dt$ would get harder, which is what Chapter 6 was for.

8.5 The Most Important Equation in Physics

Now the equation we could not integrate, and the reason Chapter 5 spent a section on potential wells.

8.8 Theorem *simple harmonic motion*

Let $\omega > 0$. Every solution of

$$\ddot{x} = -\omega^2 x$$

is

$$x(t) = A\cos\omega t + B\sin\omega t = C\cos(\omega t - \varphi),$$

where $C = \sqrt{A^2 + B^2}$ and, when $C > 0$, the phase φ is the angle with

$$\cos\varphi = A/C, \qquad \sin\varphi = B/C$$

(if $C = 0$ the motion is $x \equiv 0$ and the phase means nothing). The constants are fixed by the initial position and velocity: $A = x(0)$ and $B = \dot{x}(0)/\omega$.

Two small things people get wrong here. The phase is not *determined by* $\tan\varphi = B/A$*: that equation cannot tell* φ *from* $\varphi+\pi$*, so with* $A=-1, B=0$ *it happily accepts* $\varphi=0$ *and hands back the solution with its sign reversed. You need both the cosine and the sine to fix the quadrant. And* $\omega>0$ *is assumed throughout: at* $\omega=0$ *the equation is* $\ddot{x}=0$*, whose solutions are the straight lines* $x=A+Bt$*, and nothing oscillates at all.*

Proof That these are solutions is a verification: differentiating $A\cos\omega t + B\sin\omega t$ twice multiplies it by $-\omega^2$, by Theorem 2.23 and the chain rule. The second form is the first with the angle-addition identity applied backwards.

That there are no others needs an argument, and we can give a complete one here rather than appealing to a general uniqueness theorem. Let x be *any* solution, let $\tilde{x}$ be the one of the stated form with the same $x(0)$ and $\dot{x}(0)$, and put $z = x - \tilde{x}$. Then z also satisfies $\ddot{z} = -\omega^2 z$, and $z(0) = \dot{z}(0) = 0$. Consider

$$E(t) = \left(\dot{z}\right)^2 + \omega^2 z^2.$$

Differentiating, $\dot{E} = 2\dot{z}\ddot{z} + 2\omega^2 z\dot{z} = 2\dot{z}\left(\ddot{z} + \omega^2 z\right) = 0$, so E is constant by Corollary 3.7. Its value at $t=0$ is 0, so it is 0 always. But it is a sum of two squares with $\omega^2 > 0$, so both must vanish: $z \equiv 0$, and $x = \tilde{x}$. ■

That proof is worth a second look, because the quantity E was not pulled from nowhere: it is the energy, up to a factor of $m/2$. The argument reads as *if two solutions start out identical, their difference is a motion with no energy, and a motion with no energy is no motion at all.* Energy arguments settle uniqueness questions throughout physics, and this is the simplest one there is.

Three features of that solution deserve to be noticed, because between them they explain a great deal of the physical world.

The motion is **sinusoidal**. Not approximately, not usually — exactly, whenever the restoring force is proportional to the displacement. This is why sine waves are everywhere, and it is a fact about a differential equation rather than about springs.

The frequency ω is fixed by the system, while the amplitude C and phase φ are fixed by how you started it. In this harmonic model, pushing harder changes the amplitude and leaves the period alone: that property is **isochronism**, and it is what makes a clock possible. Be careful how far you carry it, though. It is exact for $\ddot{x} = -\omega^2 x$ and only approximate for a real pendulum, whose potential is a cosine rather than a parabola — which is why Chapter 5 could say that a wide swing runs slow. According to Viviani's later account, Galileo's interest in pendulums began with a swinging lamp in Pisa cathedral; that famous episode is uncertain. Galileo did study pendulum motion, but amplitude-independent timing is only an approximation for an ordinary pendulum.[12]

Two words for frequency get used interchangeably and should not be. The ω *in these formulas is the* ***angular frequency****, measured in radians per second, because it is the rate at which the angle inside the cosine advances. The* ***ordinary frequency*** f *is cycles per second, and since one cycle is* 2π *radians,*

$$f = \frac{\omega}{2\pi}, \qquad T = \frac{1}{f} = \frac{2\pi}{\omega}.$$

A pendulum with $\omega = 6$ *rad/s does not go round six times a second; it goes round* $6/2\pi \approx 0.95$ *times a second. Mixing the two up by a factor of* 2π *is the single most common arithmetic error in this part of physics.*

A second piece of bookkeeping deserves the same care. In $\omega = \sqrt{k/m}$, the k and the m must belong to the *same* coordinate. If you describe a pendulum by its angle θ rather than by an arc length, then the "stiffness" is $\dfrac{d^2U}{d\theta^2} = mgL$, and the quantity playing the part of the mass is not m but the rotational inertia mL^2 — which is why $\omega = \sqrt{mgL/mL^2} = \sqrt{g/L}$ comes out with the m cancelled. Use the inertia that goes with the coordinate you chose, and the dimensions will check.

And energy is conserved. Writing $k = m\omega^2$,

$$E = \tfrac{1}{2}m\dot{x}^2 + \tfrac{1}{2}kx^2, \qquad \frac{dE}{dt} = m\dot{x}\ddot{x} + kx\dot{x} = \dot{x}\big(m\ddot{x} + kx\big) = 0$$

by the equation itself. The energy that the motion takes out of the spring it puts back, forever.

8.9 Example *why almost every stable system oscillates*

A particle of mass $m > 0$ sits at a nondegenerate minimum x_0 of a smooth potential U, so $U'(x_0) = 0$ and $U''(x_0) > 0$, with no friction or driving. Displace it slightly and release it from rest. Find the leading small-amplitude approximation to its motion and frequency.

Solution The force is $F = -U'(x)$. By the small-oscillation expansion of Chapter 5, near x_0 we have $U(x) \approx U(x_0) + \tfrac{1}{2}k(x - x_0)^2$ with $k = U''(x_0) > 0$, so

$$F \approx -k\,(x - x_0),$$

and, to this order, Newton's law becomes $m\ddot{x} = -k(x - x_0)$. Writing $u = x - x_0$ turns this into $\ddot{u} = -(k/m)u$, which is Theorem 8.8 with

$$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{U''(x_0)}{m}}.$$

In this linear approximation the motion is sinusoidal, with a frequency set by the curvature at the bottom of the well and the mass. A general well adds corrections to the waveform and period at finite amplitude.

For a pendulum, Example 5.8 gave $k = mgL$ for the angular coordinate with the corresponding inertia mL^2, and $\omega = \sqrt{g/L}$ — period $2\pi\sqrt{L/g}$, independent of the mass and, to this order, of the amplitude.

Trace the chain once more, because it is the most valuable thing in Part I. A nondegenerate minimum of the potential means $U' = 0$ and $U'' > 0$ (Section 2.8). Taylor then says the potential is approximately a parabola near the bottom (Chapter 5). A parabolic potential means a linear restoring force, and a linear restoring force means Theorem 8.8. Therefore: an undamped system at a nondegenerate minimum, nudged gently, oscillates sinusoidally. *Atoms in a crystal, a ship rolling, a guitar string, charge in an LC circuit, a planet wobbling about a resonance. Four chapters of this book exist to justify that one sentence — and the qualifiers in it are not decoration, as $U = x^4$ and the overdamped door closer of the next section both show.*

8.6 Energy: Turning a Second-Order Equation into a First-Order One

Theorem 8.8 solved one second-order equation completely. Most potentials are not parabolas, and for those there is no formula — but there is a general method that gets you a great deal, and it is the one physics actually uses. It rests on the observation we have already made twice: energy is conserved.

Take $m\ddot{x} = -U'(x)$, a mass moving in any potential at all, with no friction. Multiply both sides by $\dot{x}$ and look at what appears:

$$m\dot{x}\ddot{x} + U'(x)\dot{x} = 0 \qquad \Longleftrightarrow \qquad \frac{d}{dt}\Big[\tfrac{1}{2}m\dot{x}^2 + U(x)\Big] = 0,$$

using the chain rule on $U(x(t))$. So the bracket is a constant of the motion, which we call the energy E. That single multiplication has traded a second-order equation for a first-order one:

$$\tfrac{1}{2}m\dot{x}^2 + U(x) = E \qquad \Longrightarrow \qquad \dot{x} = \pm\sqrt{\frac{2[E - U(x)]}{m}},$$

and the first-order equation is separable, which we know how to handle.

Three things can be read off before any integration.

The motion is confined to the region where $U(x) \le E$, since the square root must be real — and that is exactly the closed-interval reasoning of Proposition 2.21 showing up in mechanics. A particle does not go where it cannot afford to.

A point where $U(x) = E$ has zero speed if it is reached. It is a **simple turning point** if also $U'(x) \neq 0$: Newton's law then gives a nonzero acceleration back into the allowed region, and the sign of the velocity changes. A root with $U'(x) = 0$ needs separate analysis; it may be an equilibrium or an asymptotic barrier. For two simple turning points $x_1 < x_2$ with $U(x) < E$ throughout (x_1, x_2), and a smooth potential on the closed interval, the motion on each crossing is monotone, and the time to cross is

$$t = \int_{x_1}^{x_2} \frac{dx}{\sqrt{2[E - U(x)]/m}},$$

an ordinary integral, improper at each end because the denominator vanishes there, and usually convergent because it vanishes only like a square root.

And the motion between two turning points repeats, so the period is twice that crossing time.

8.10 Example *recovering the harmonic period, and losing it*

Use the energy method on $U = \frac{1}{2}kx^2$, and then on $U = \alpha x^4$.

Solution For the spring, release from rest at $x = A$, so $E = \frac{1}{2}kA^2$ and

$$\dot{x} = \pm\sqrt{\frac{k}{m}}\sqrt{A^2 - x^2} = \pm\omega\sqrt{A^2 - x^2}.$$

The quarter period is the time from $x = 0$ out to the turning point $x = A$:

$$\frac{T}{4} = \int_0^A \frac{dx}{\omega\sqrt{A^2 - x^2}} = \frac{1}{\omega}\Big[\arcsin\frac{x}{A}\Big]_0^A = \frac{\pi}{2\omega},$$

by Proposition 4.7. So $T = 2\pi/\omega$, which agrees with Theorem 8.8 — and notice that A cancelled, which is isochronism falling out of the arithmetic rather than being asserted.

Now the quartic well, $U = \alpha x^4$ with $\alpha > 0$: a perfectly stable minimum at the origin, but a degenerate one, with $U''(0) = 0$. Here $E = \alpha A^4$ and

$$\frac{T}{4} = \int_0^A \frac{dx}{\sqrt{2\alpha(A^4 - x^4)/m}}.$$

Substituting $x = Au$ pulls every A outside:

$$\frac{T}{4} = \frac{1}{A}\sqrt{\frac{m}{2\alpha}} \int_0^1 \frac{du}{\sqrt{1 - u^4}},$$

and the remaining integral is just a number. So $T \propto 1/A$: the wider the swing, the *shorter* the period. This system is stable, it oscillates, and it is not isochronous and not sinusoidal — which is precisely the counterexample Chapter 5 warned about, now with its period computed.

The integrand $1/\sqrt{1 - u^4}$ *has no elementary antiderivative; the definite integral is an example of an* elliptic integral. *Note that we did not need it: the* scaling $T \propto 1/A$ *came out before the integral was evaluated. Getting the dependence on the parameters without getting the number is very often all physics wants, and it is usually much cheaper.*

This energy method is the reason a potential-energy curve is such a useful picture. Draw $U(x)$, draw a horizontal line at height E, and you can read off where the particle may go, where it turns around, where its speed is large (where $E - U$ is large) and where it dawdles (near an accessible turning point) — all without solving anything.

8.7 Damping and Driving

Real oscillators lose energy. Adding a resistance proportional to velocity gives

$$m\ddot{x} + b\dot{x} + kx = 0.$$

Guessing $x = e^{rt}$ — legitimate, since a verified guess is a proof — and substituting gives $mr^2 + br + k = 0$, an ordinary quadratic. When the damping is light its roots are complex, and by Euler's formula (Theorem 5.9) the solution is an oscillation inside a decaying envelope:

$$x(t) = Ce^{-bt/2m}\cos(\omega' t - \varphi), \qquad \omega' = \sqrt{\frac{k}{m} - \frac{b^2}{4m^2}}.$$

Read that formula carefully, because two different things are happening and it is easy to run them together. The *amplitude* decays, at the rate $b/2m$; separately, the *angular frequency* ω' is lower than the undamped $\sqrt{k/m}$, so the oscillation is also slower. Both require $m, k > 0$ and $0 < b < 2\sqrt{mk}$, which is the underdamped range and what makes the roots complex.

At exactly $b = 2\sqrt{mk}$ — critical damping — the quadratic has a repeated root, the formula above degenerates, and the solution is instead $(A + Bt)e^{-bt/2m}$: there is no repeated oscillation. For fixed m and k, critical damping has the fastest asymptotic decay rate among the nonoscillatory regimes for generic initial data. With stronger damping, a slower exponential mode usually controls the late motion. The actual settling time and whether equilibrium is crossed once also

depend on the initial position and velocity. A door closer is designed to sit at or just past the critical value, which is why a well-adjusted door neither slams nor hangs open.

Push such a system periodically and something dramatic happens when the driving frequency approaches ω: the response grows very large. That is **resonance**, and it explains the opera singer and the wine glass, the radio tuner selecting one station out of thousands, and the collapse of badly designed bridges. We will do it properly in Part II, where it belongs; the equation is the same one.

8.8 When There Is No Formula

Most differential equations cannot be solved in closed form. This is not a failure of technique but a fact about the equations, and physics long ago stopped being embarrassed about it.

The way out is to go back to the meaning of the derivative. The equation $\dfrac{dy}{dt} = f(t, y)$ tells you the rate right now; local linearity (Theorem 2.4) says that over a short step the change is rate times step. So step forward, recompute the rate, and repeat.

8.11 Definition *Euler's method*

Given $\dfrac{dy}{dt} = f(t, y)$ with $y(t_0) = y_0$, choose a step $\Delta t > 0$ and compute

$$t_{n+1} = t_n + \Delta t, \qquad y_{n+1} = y_n + f(t_n, y_n)\,\Delta t.$$

Written that way the second line is literally Theorem 2.4: $\Delta y = (\text{rate}) \times \Delta t$, with the $o(\Delta t)$ thrown away. That is the whole method. Every refinement of it — and there are many, used in every simulation anyone runs — consists of being cleverer about which rate to use across the step, not of abandoning the idea.

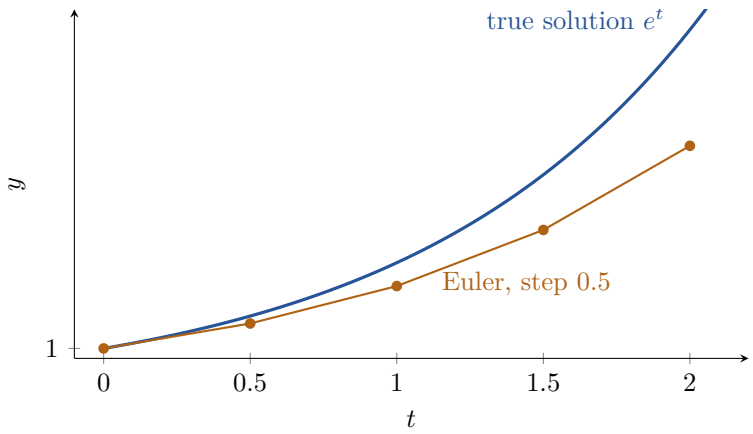


Figure 8.2: Euler's method on $\dot{y} = y$, $y(0) = 1$, with a deliberately crude step of 0.5. Each segment uses the slope at its left end and holds it for the whole step, so the polygon keeps falling behind a curve that is always steepening. Halving the step roughly halves the final error; the price of accuracy is arithmetic, which is what computers are for.

Compare Figure 8.2 with the staircases of Chapter 3. They are the same idea: chop the interval, hold the rate constant across each piece, add up the little changes. Riemann sums

estimate an integral that way; Euler's method estimates a solution that way. The error behaves the same too, under suitable assumptions, and it is worth being precise about them because the naive version of the argument is not quite right. Here is the honest statement. If the exact solution has a bounded second derivative, then one step taken *from a point on the exact solution* lands within a constant times $(\Delta t)^2$ of it; that much is Chapter 5's remainder bound. But Euler does not start each step on the exact solution — it starts wherever the previous steps left it — so one must also know that an error already made does not get amplified without limit as it is carried forward. The condition that controls this is the same Lipschitz condition that appeared in Theorem 8.3. Given both, the errors accumulate to order Δt over a fixed interval: halve the step and the final error roughly halves, because there are twice as many steps each contributing four times less.

Every trajectory NASA has ever computed, every weather forecast and every molecular simulation is a refinement of this one procedure. It is worth knowing that the practical machinery of modern physics rests on the definition of the derivative, used forwards.

8.9 Chapter Recap, and the End of Part I

Newton's second law is a differential equation: it does not hand you the motion, it constrains it. When the force depends only on time you can integrate twice, and Chapter 3 was enough. When the force depends on position — springs, gravity, essentially everything interesting — you need this chapter.

Slope fields let us see the solutions before finding them. Initial data do not by themselves guarantee a unique solution — that is Theorem 8.3, with hypotheses — so the determinism of Newtonian mechanics turns out to be a claim about forces, not about counting derivatives. Separation of variables solves the first-order equations of decay, cooling, charging and terminal velocity, provided you remember to collect the equilibrium solutions before dividing them away; integrating factors (Theorem 8.6) extend the reach to any linear first-order equation, including driven ones, and split every answer into what the system would do alone plus what the world does to it.

The second-order equation $\ddot{x} = -\omega^2 x$ has sinusoidal solutions whose frequency belongs to the system and whose amplitude and phase belong to the initial conditions; an energy argument shows there are no others. Combined with the small-oscillation result of Chapter 5, that one equation governs any undamped system sitting at a nondegenerate minimum and not disturbed too hard — a large class, but a described one. For potentials outside that class, Section 8.6 trades the second-order equation for a first-order one using conservation of energy, and turns the period into an integral between turning points. And when no formula exists at all, Euler's method steps forward using the derivative for exactly what it was defined to be.

That closes Part I. We began with a ball dropped from a tower and a question about what "how fast" could possibly mean at a single instant. We now have limits, derivatives, differentials, integrals, the Fundamental Theorem, the exponential and the logarithm, series, the techniques of integration, curves described as motion, and the equations that motion obeys. It is, near enough, the mathematics Newton had — and rather more of it than he had when he started.

Part II asks what a force is.

Problem 8.1. Verify by substitution that $y = Ce^{-3t}$ solves $\dot{y} = -3y$, and find the solution with $y(0) = 5$. What is $y(2)$?

Problem 8.2. Solve $\dfrac{dy}{dx} = xy$ by separation, given $y(0) = 2$. Before you divide, say what the zeros of the right-hand side are and which solution you would have lost.

Problem 8.3. Solve $\dfrac{dy}{dt} + 2y = e^{-t}$ with $y(0) = 0$ using Theorem 8.6. Identify which part of your answer is the system decaying on its own and which part is forced by the right-hand side, and describe what happens to each as t grows.

Problem 8.4. A pendulum of length $L > 0$ swings with amplitude $0 < \theta_0 < \pi$. For the comparison below, use $\sin(\theta/2) = \sin(\theta_0/2)\sin\varphi$, so $0 \le \varphi \le \pi/2$. Its energy is $E = mgL(1 - \cos\theta_0)$ and its potential is $U = mgL(1 - \cos\theta)$ with rotational inertia mL^2. Using Section 8.6, write the period as an integral over θ from 0 to θ_0. Do not try to evaluate it. Instead, use $1 - \cos\theta = 2\sin^2(\theta/2)$ to put it in a tidier form, and then argue from the integrand that the period is always *greater* than $2\pi\sqrt{L/g}$ and approaches that value as $\theta_0 \to 0$. Compare with what Chapter 5 said about a grandfather clock knocked into a wide swing.

Problem 8.5. Use $\frac{1}{2}m\dot{x}^2 + U(x) = E$ to show that a particle of mass $m > 0$ is never found where $U(x) > E$. Then use Newton's law $m\ddot{x} = -U'(x)$ to show that, if it reaches a point where $U(x) = E$ and $U'(x) \neq 0$, it reverses direction. Assume U is smooth. Then consider the inverted parabola $U = -\frac{1}{2}kx^2$ with $E = 0$, where the particle is heading toward the origin. Set up the time integral and show that it diverges — the particle approaches the top of the hill but never arrives. Why does this not contradict Theorem 8.3?

Problem 8.6. A cup of coffee at 90°C sits in a 20°C room and obeys $\dot{T} = -k(T - 20)$. After 5 minutes it is at 70°C. Find k, and find when it reaches 40°C.

Problem 8.7. A mass $m = 0.5$ kg on a spring with $k = 18$ N/m is pulled 0.1 m from equilibrium and released from rest. Write $x(t)$, and find the period, the maximum speed, and the total energy. At what displacement is the kinetic energy equal to the potential energy?

Problem 8.8. Sketch the slope field of $\dot{y} = y(1 - y)$ for $-0.5 < y < 1.5$ by evaluating the slope at a few points. Identify *both* equilibria and classify the stability of each — note that judging $y = 0$ requires looking at values on both sides of it. Then say which part of the picture survives if y is a population and only $y \ge 0$ is meaningful. (This is the logistic equation; solving it needs the partial fractions of Chapter 6.)

Problem 8.9. Apply Euler's method to $\dot{y} = y$ with $y(0) = 1$ and $\Delta t = 0.25$ over $[0, 1]$. Compare with $e \approx 2.71828$, then repeat with $\Delta t = 0.125$ and check that the error roughly halves.

Problem 8.10. A particle of mass $m > 0$ moves in the potential $U(x) = U_0[(x/\ell)^4 - 2(x/\ell)^2]$, with $U_0, \ell > 0$, from the problems of Chapter 5. Using Example 8.9, find the angular frequency, the ordinary frequency and the period of small oscillations about each stable equilibrium. Check that both frequencies have dimensions of inverse time and that the period has dimensions of time.

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